

Calculus II: Complete Lesson Notes

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Course Overview

These notes follow the topics in the coordinator’s note-taking guides, with complete explanations and worked examples. Lesson numbers divide the material into teaching units and do not prescribe one class meeting per lesson.

Lessons	Topics	Guide sections
1–6	Integrals and the Fundamental Theorem	5.1–5.5
7–12	Area, volume, work, average value	6.1–6.5
13–20	Integration methods and improper integrals	7.1–7.4, 7.8
21–25	Arc length, surface area, physical applications	8.1–8.3, 10.2
26–33	Sequences, series, convergence tests	11 introduction, 11.1–11.6
34–37	Power series, Taylor series, applications	11.8–11.10

The integration-strategy and convergence-strategy lessons consolidate the preceding methods. Optional error estimates and the smooth-function example are marked as extensions. Binomial series are included to complete the coordinator’s Taylor-series table.

Lesson 1: Approximating Areas Under Curves

Goals of this lesson.

- Approximate the area under a curve $y = f(x)$ on an interval $[a, b]$.
- Construct and evaluate Riemann sums (left-endpoint, right-endpoint, and midpoint approximations).
- Understand and manipulate sigma (Σ) notation for sums.

1.1 The Area Problem

Problem 1.1. How do we find the area under a continuous curve $y = f(x)$ bounded by the lines $x = a$, $x = b$, and the x -axis (where $f(x) \geq 0$)?

Example 1.2. Consider the region bounded by $y = \sin x$ on the interval $[0, \frac{\pi}{2}]$ and the x -axis.

This is **not** an elementary geometric shape (like a rectangle, triangle, or trapezoid), so classical formulas cannot compute the exact area directly.

Main idea: approximate with rectangles

One natural way to estimate the area is to use a **Riemann sum**:

1. Divide the interval $[a, b]$ into n smaller subintervals (a *partition*).
2. Build a rectangle over each subinterval whose height is the function value at a chosen sample point.
3. Add up the areas of all the rectangles.

1.2 Partitioning the Interval and Constructing Approximations

Let $f(x) = \sin x$ on $[0, \frac{\pi}{2}]$ and divide the interval into $n = 4$ equal subintervals.

The width of each subinterval is:

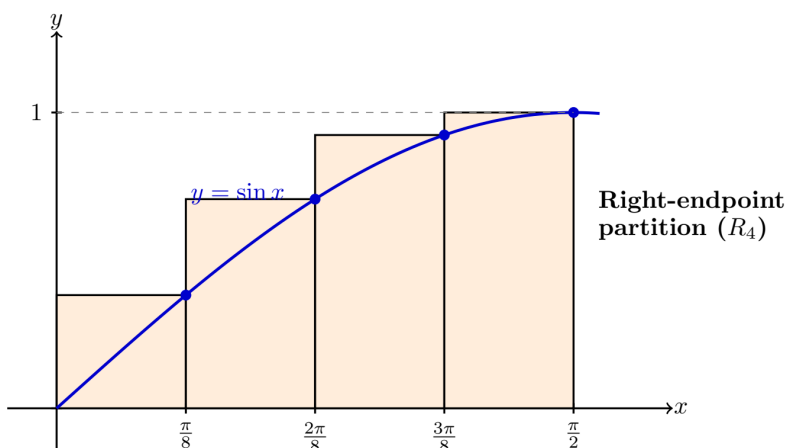
$$\Delta x = \frac{b - a}{n} = \frac{\frac{\pi}{2} - 0}{4} = \frac{\pi}{8}.$$

The partition points are:

$$x_0 = 0, \quad x_1 = \frac{\pi}{8}, \quad x_2 = \frac{2\pi}{8} = \frac{\pi}{4}, \quad x_3 = \frac{3\pi}{8}, \quad x_4 = \frac{4\pi}{8} = \frac{\pi}{2}.$$

This generates 4 subintervals:

$$\left[0, \frac{\pi}{8}\right], \quad \left[\frac{\pi}{8}, \frac{\pi}{4}\right], \quad \left[\frac{\pi}{4}, \frac{3\pi}{8}\right], \quad \text{and} \quad \left[\frac{3\pi}{8}, \frac{\pi}{2}\right].$$

Figure 1: Right-endpoint approximation (R_4) for $y = \sin x$ on $[0, \pi/2]$.

1.3 Sampling the Rectangle Heights

To determine the height of each rectangle, typical choices for sample points x_i^* within each subinterval $[x_{i-1}, x_i]$ include:

- **Right endpoint:** $x_i^* = x_i$
- **Left endpoint:** $x_i^* = x_{i-1}$
- **Midpoint:** $x_i^* = \bar{x}_i = \frac{x_{i-1} + x_i}{2}$

1. Right-Endpoint Approximation (R_4)

Using the right endpoints $x_1 = \frac{\pi}{8}, x_2 = \frac{2\pi}{8}, x_3 = \frac{3\pi}{8}, x_4 = \frac{4\pi}{8}$:

$$\begin{aligned}
 R_4 &= \sum_{i=1}^4 f(x_i) \Delta x \\
 &= \left(\frac{\pi}{8}\right) \sin\left(\frac{\pi}{8}\right) + \left(\frac{\pi}{8}\right) \sin\left(\frac{2\pi}{8}\right) + \left(\frac{\pi}{8}\right) \sin\left(\frac{3\pi}{8}\right) + \left(\frac{\pi}{8}\right) \sin\left(\frac{4\pi}{8}\right) \\
 &= \frac{\pi}{8} \left[\sin\left(\frac{\pi}{8}\right) + \sin\left(\frac{\pi}{4}\right) + \sin\left(\frac{3\pi}{8}\right) + \sin\left(\frac{\pi}{2}\right) \right] \\
 &\approx \frac{\pi}{8} [0.3827 + 0.7071 + 0.9239 + 1.0000] \approx 1.183.
 \end{aligned}$$

From the graph in [Figure 1](#), each rectangle extends above the curve, so R_4 is an **overestimate**.

2. Left-Endpoint Approximation (L_4)

Using the left endpoints $x_0 = 0, x_1 = \frac{\pi}{8}, x_2 = \frac{2\pi}{8}, x_3 = \frac{3\pi}{8}$:

$$\begin{aligned} L_4 &= \sum_{i=1}^4 f(x_{i-1}) \Delta x \\ &= \frac{\pi}{8} \left[\sin(0) + \sin\left(\frac{\pi}{8}\right) + \sin\left(\frac{\pi}{4}\right) + \sin\left(\frac{3\pi}{8}\right) \right] \\ &\approx \frac{\pi}{8} [0.0000 + 0.3827 + 0.7071 + 0.9239] \approx 0.791. \end{aligned}$$

Here each rectangle sits below the curve, so L_4 is an **underestimate**. The true area lies somewhere between 0.791 and 1.183.

Monotonicity and Over/Underestimates

- If $f(x)$ is **increasing** on $[a, b]$, then left endpoints underestimate ($L_n \leq \text{Area}$) and right endpoints overestimate ($R_n \geq \text{Area}$).
- If $f(x)$ is **decreasing** on $[a, b]$, then left endpoints overestimate ($L_n \geq \text{Area}$) and right endpoints underestimate ($R_n \leq \text{Area}$).

1.4 General Riemann Sum Formulation

For a continuous function, these approximations converge to the area as $n \rightarrow \infty$. Increasing n need not improve every individual approximation monotonically. For any choice of sample points $x_i^* \in [x_{i-1}, x_i]$:

Riemann sum

The area A under $y = f(x)$ on $[a, b]$ is approximated by

$$A \approx f(x_1^*) \Delta x + f(x_2^*) \Delta x + \cdots + f(x_n^*) \Delta x = \sum_{i=1}^n f(x_i^*) \Delta x,$$

where

$$\Delta x = \frac{b-a}{n} \quad \text{and} \quad x_i = a + i \Delta x.$$

1.5 Summation (Sigma Notation)

Definition 1.1 (Sigma Notation)

Sigma (Σ) notation is a compact way to write the sum of terms indexed by an integer k :

$$\sum_{k=1}^n a_k = a_1 + a_2 + a_3 + \cdots + a_n.$$

Here, k is the *index of summation*, 1 is the *lower limit*, and n is the *upper limit*.

Example 1.3. Expand and evaluate the sum

$$\sum_{k=1}^5 k^2.$$

Solution. Expanding term-by-term for $k = 1, 2, 3, 4, 5$:

$$\begin{aligned} \sum_{k=1}^5 k^2 &= \underbrace{1^2}_{k=1} + \underbrace{2^2}_{k=2} + \underbrace{3^2}_{k=3} + \underbrace{4^2}_{k=4} + \underbrace{5^2}_{k=5} \\ &= 1 + 4 + 9 + 16 + 25 = 55. \end{aligned}$$

□

Example 1.4. Write the following sum in sigma notation, starting at index $k = 1$:

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}.$$

Solution. Look at each term and how it depends on k :

$$\begin{aligned} k = 1 &\implies \frac{1}{3} = \frac{1}{1+2}, \\ k = 2 &\implies \frac{1}{4} = \frac{1}{2+2}, \\ k = 3 &\implies \frac{1}{5} = \frac{1}{3+2}, \\ k = 4 &\implies \frac{1}{6} = \frac{1}{4+2}, \\ k = 5 &\implies \frac{1}{7} = \frac{1}{5+2}. \end{aligned}$$

The general k -th term is $\frac{1}{k+2}$. Therefore:

$$\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} = \sum_{k=1}^5 \frac{1}{k+2}.$$

□

Useful finite sums

$$\sum_{i=1}^n 1 = n, \quad \sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Constants factor out of sums, and sums distribute over addition.

Example 1.5 (Midpoint Approximation). For the sine region above, compute M_4 .

Solution. The midpoints are $\pi/16, 3\pi/16, 5\pi/16, 7\pi/16$, so

$$M_4 = \frac{\pi}{8} \left(\sin \frac{\pi}{16} + \sin \frac{3\pi}{16} + \sin \frac{5\pi}{16} + \sin \frac{7\pi}{16} \right) \approx 1.026.$$

□

Lesson 2: The Definite Integral

Goals of this lesson.

- Define the definite integral as the limit of a Riemann sum.
- Identify the parts of the integral notation.
- Evaluate definite integrals geometrically using net signed area.
- Apply the algebraic and interval properties of definite integrals.

2.1 Riemann Sums and the Definite Integral

In [Lesson 1](#), the approximations improved as the number of rectangles grew. The definite integral is what we get when we take this to the limit.

Definition 2.1 (The Definite Integral)

Let f be a continuous function defined on $[a, b]$. Divide $[a, b]$ into n subintervals of equal width

$$\Delta x = \frac{b - a}{n}.$$

Let $x_0, x_1, x_2, \dots, x_n$ be the endpoints of these subintervals, and let x_i^* be any sample point in the i -th subinterval $[x_{i-1}, x_i]$. The **definite integral of f from a to b** is

$$\int_a^b f(x) dx := \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

Anatomy of Integral Notation

In the expression $\int_a^b f(x) dx$:

- $\int$ is the **integral sign** (an elongated “S” for summation).
- a is the **lower limit of integration**, and b is the **upper limit of integration**.
- $f(x)$ is the **integrand**.
- dx is the **differential**, which indicates that the variable of integration is x .
- The value of $\int_a^b f(x) dx$ is a **number**, not a function of x .

Example 2.1. Express the following limit of a Riemann sum as a definite integral on the interval $[0, \pi]$:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n ((x_i^*)^3 + x_i^* \sin(x_i^*)) \Delta x.$$

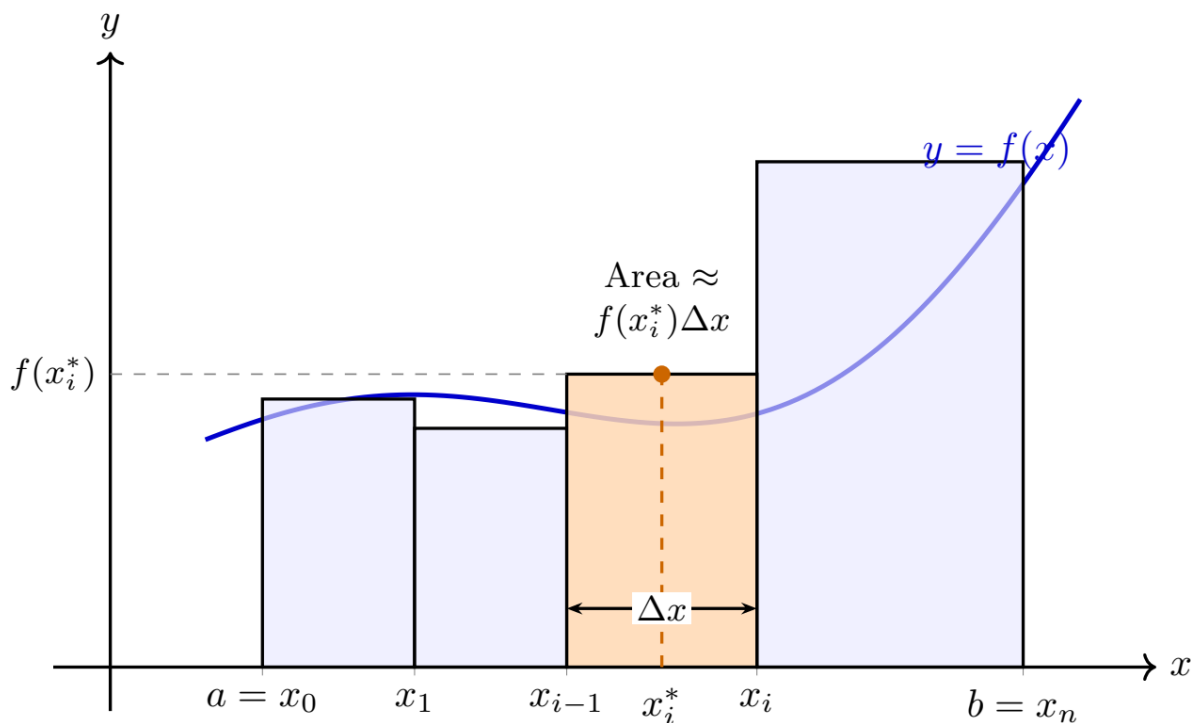


Figure 2: The definite integral as the limit of a Riemann sum on $[a, b]$.

Solution. Match each part of the sum to a part of the integral:

- Limits: $a = 0$ and $b = \pi$.
- Integrand: replacing the sample points x_i^* with the variable x gives $f(x) = x^3 + x \sin x$.
- Width: Δx becomes dx .

Thus:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n ((x_i^*)^3 + x_i^* \sin(x_i^*)) \Delta x = \int_0^\pi (x^3 + x \sin x) dx.$$

□

Why the Limit Is Well Defined

For continuous f , the Riemann-sum limit is independent of the sample points. More generally, a bounded function is Riemann integrable if all sums over partitions $a = x_0 < \cdots < x_n = b$ approach the same value as the mesh $\max_i (x_i - x_{i-1})$ tends to zero. In that version, each width is $\Delta x_i = x_i - x_{i-1}$.

Example 2.2 (An Integral from Its Definition). Compute $\int_0^2 (x^2 + 1) dx$ using right-endpoint sums.

Solution. Here $\Delta x = 2/n$ and $x_i = 2i/n$, so

$$\begin{aligned}\int_0^2 (x^2 + 1) dx &= \lim_{n \rightarrow \infty} \left(\frac{8}{n^3} \sum_{i=1}^n i^2 + \frac{2}{n} \sum_{i=1}^n 1 \right) \\ &= \lim_{n \rightarrow \infty} \left(\frac{4(n+1)(2n+1)}{3n^2} + 2 \right) = \frac{14}{3}.\end{aligned}$$

□

2.2 Evaluating Definite Integrals Using Areas

When f takes negative values, the terms $f(x_i^*) \Delta x$ in the Riemann sum are negative, so regions below the axis count negatively.

Net signed area

$$\int_a^b f(x) dx = (\text{Area above the } x\text{-axis}) - (\text{Area below the } x\text{-axis}) = A_{\text{top}} - A_{\text{bottom}}.$$

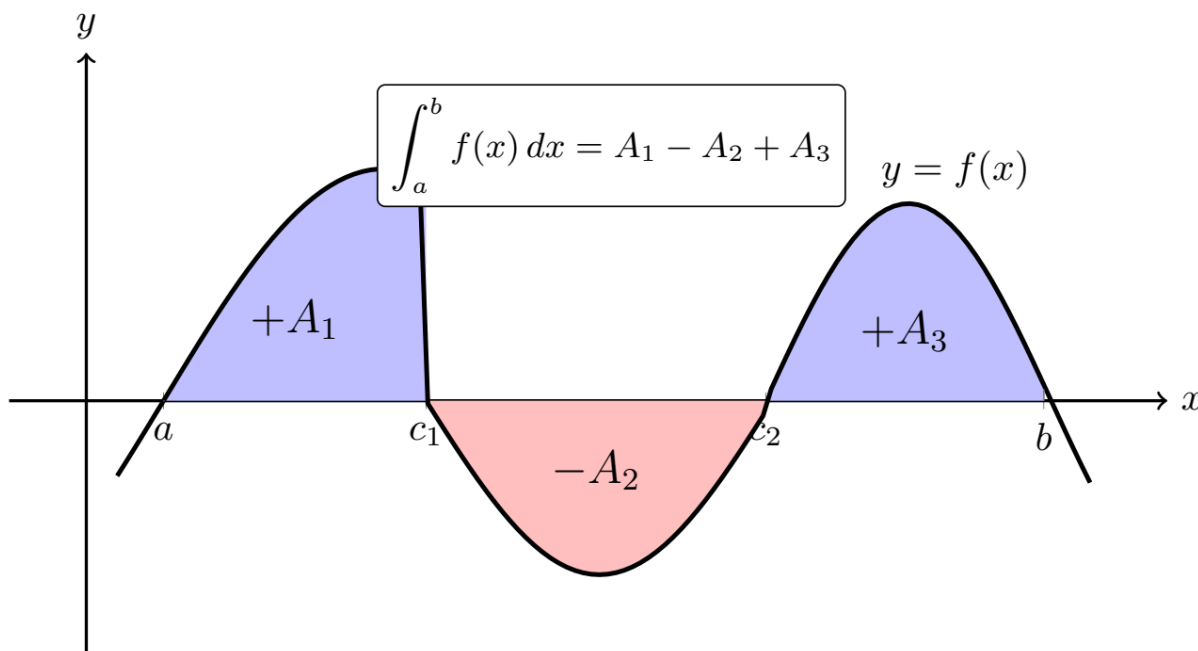


Figure 3: Geometric interpretation of the definite integral as net signed area.

Problem 2.3. Evaluate $\int_0^3 (x - 1) dx$ by interpreting it in terms of areas.

Solution. The integrand is the line $y = x - 1$. Sketching it on $[0, 3]$ as in Figure 4:

- On $[0, 1]$, the region lies **below** the x -axis and is a right triangle with base 1 and height 1:

$$A_1 = \frac{1}{2}(1)(1) = \frac{1}{2}.$$

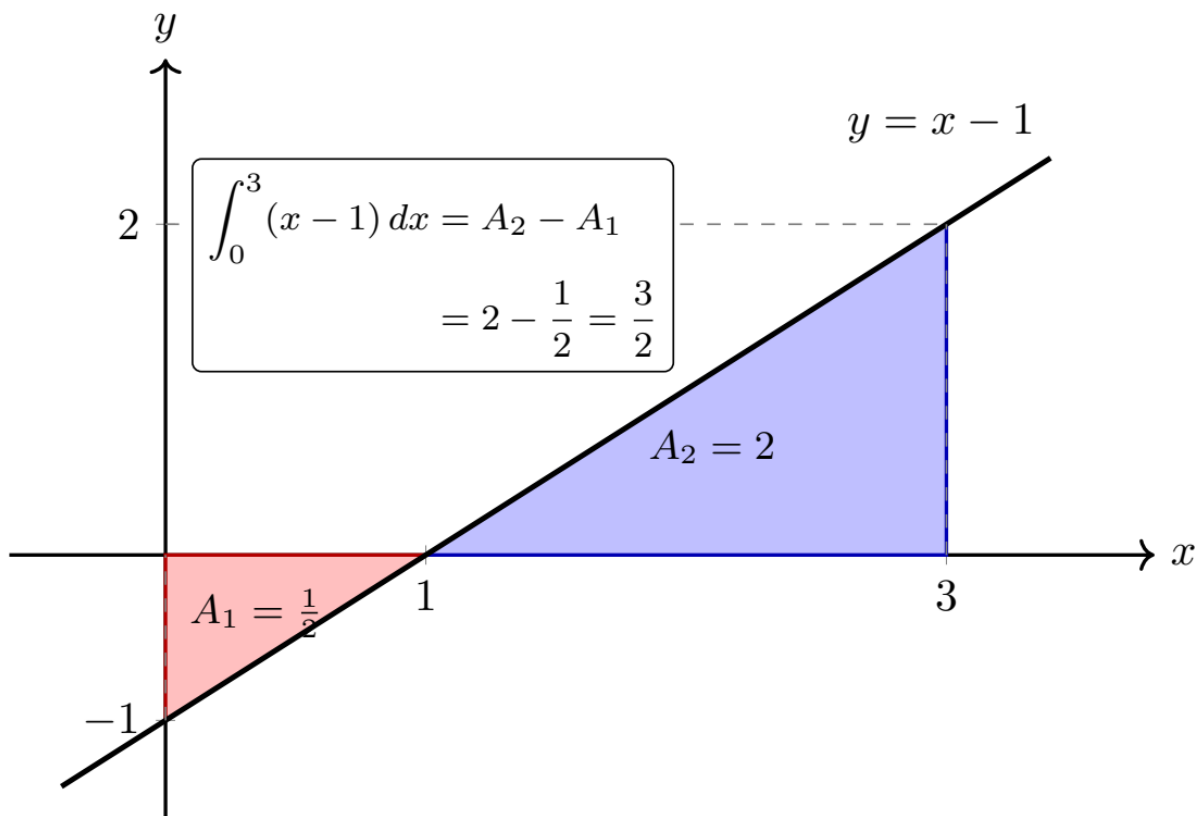


Figure 4: Evaluating $\int_0^3 (x - 1) dx$ using geometric areas.

- On $[1, 3]$, the region lies **above** the x -axis and is a right triangle with base 2 and height 2:

$$A_2 = \frac{1}{2}(2)(2) = 2.$$

Subtracting the area below the axis from the area above:

$$\int_0^3 (x - 1) dx = A_2 - A_1 = 2 - \frac{1}{2} = \frac{3}{2}.$$

□

Problem 2.4. Evaluate $\int_{-2}^2 \sqrt{4 - x^2} dx$ by interpreting it geometrically.

Solution. Let $y = \sqrt{4 - x^2}$. Squaring both sides gives

$$y^2 = 4 - x^2 \implies x^2 + y^2 = 4 \quad (y \geq 0).$$

This is the upper half of the circle centered at $(0, 0)$ with radius $r = 2$ (Figure 5).

The region lies entirely above the x -axis, so the integral is the area of the semicircle:

$$\int_{-2}^2 \sqrt{4 - x^2} dx = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi(2)^2 = 2\pi.$$

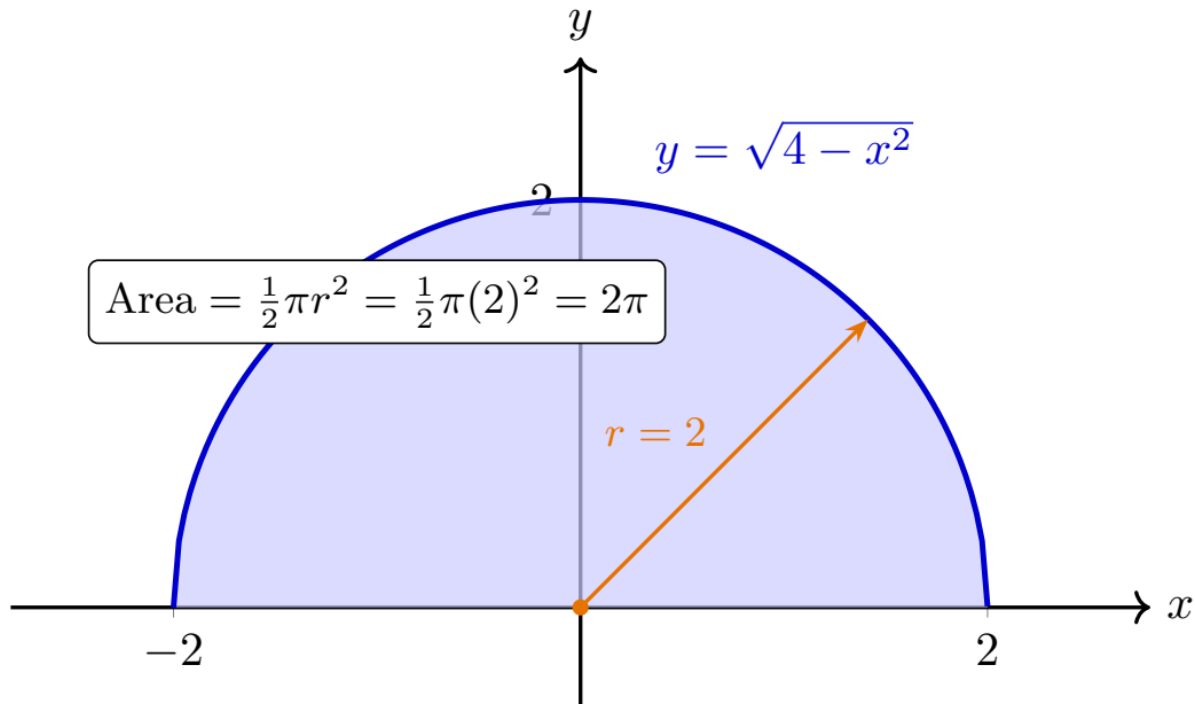


Figure 5: Evaluating $\int_{-2}^2 \sqrt{4-x^2} dx$ as the area of a semicircle.

□

2.3 Properties of the Definite Integral

Definition 2.2 (Basic Interval Conventions)

1. $\int_a^a f(x) dx = 0$ (an interval of width zero contains zero area)
2. $\int_b^a f(x) dx = -\int_a^b f(x) dx$ (reversing the direction negates the integral)

Theorem 2.1 (Algebraic and Interval Properties of Integrals)

Let f and g be continuous on an interval containing every endpoint used below, and let c be any constant.

1. **Constant Rule:**

$$\int_a^b c dx = c(b-a)$$

2. Constant Multiple Rule:

$$\int_a^b c f(x) dx = c \int_a^b f(x) dx$$

3. Sum and Difference Rule:

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

4. Interval Additivity Rule: For any number c between (or beyond) a and b :

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

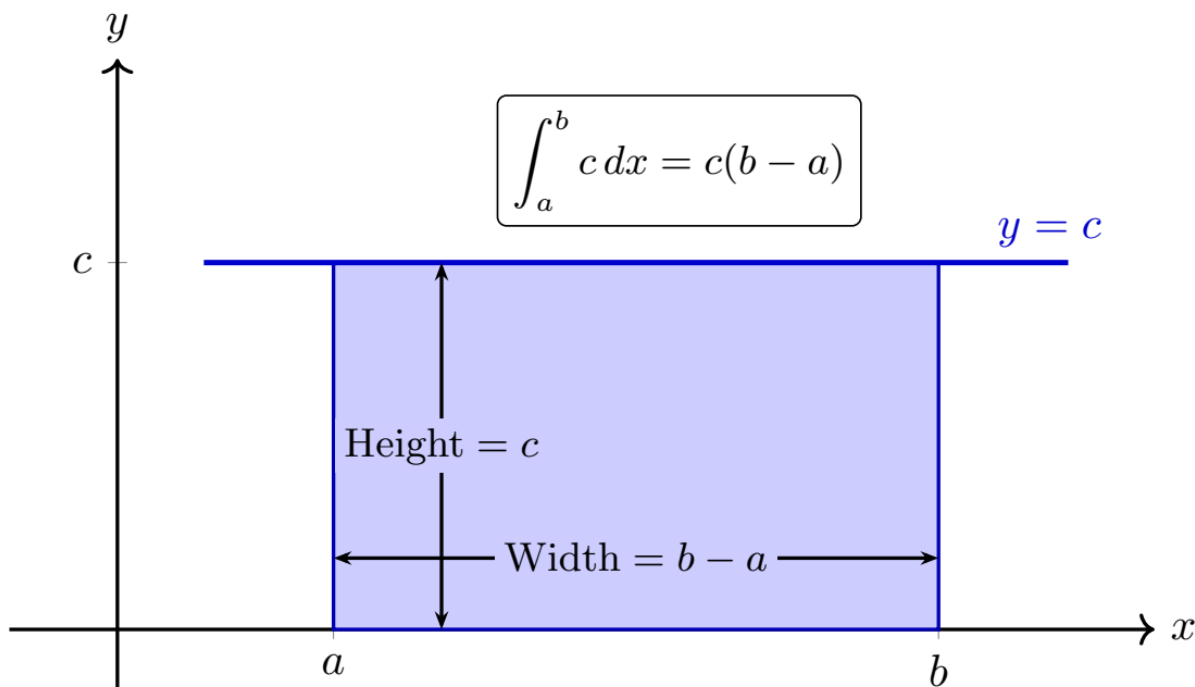


Figure 6: Geometric interpretation of the Constant Rule: $\int_a^b c dx = c(b - a)$.

Example 2.5. Suppose $\int_0^{10} f(x) dx = 17$ and $\int_0^8 f(x) dx = 12$. Find $\int_8^{10} f(x) dx$.

Solution. By interval additivity (Figure 7):

$$\int_0^{10} f(x) dx = \int_0^8 f(x) dx + \int_8^{10} f(x) dx.$$

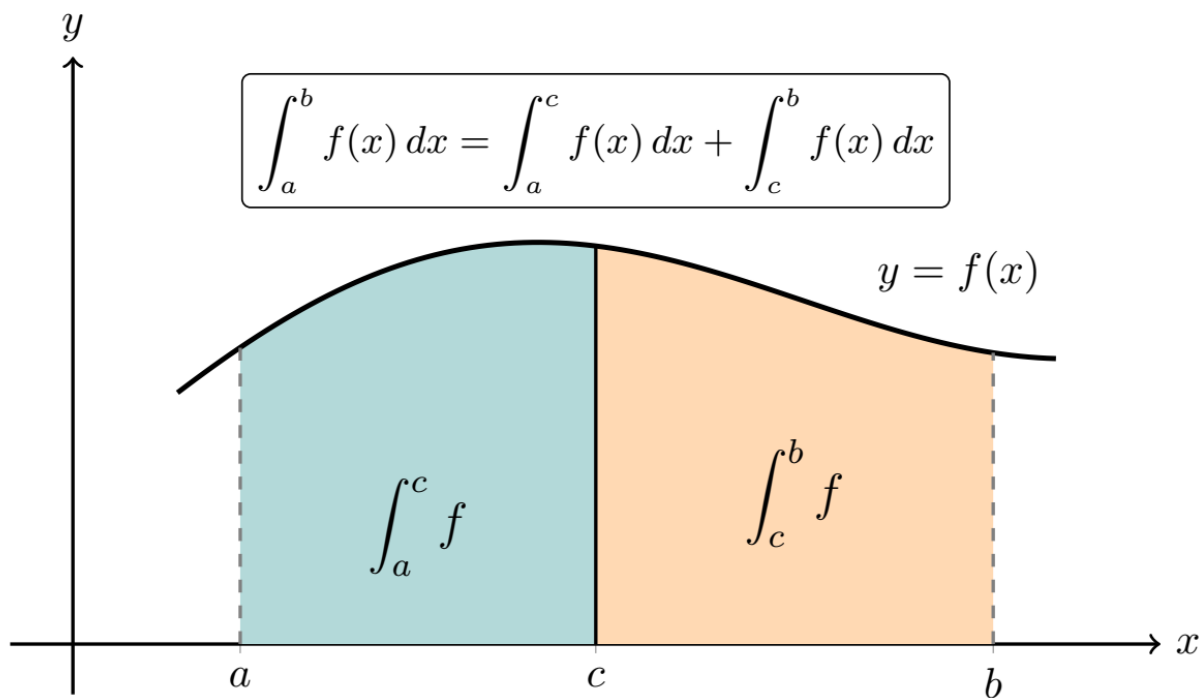


Figure 7: Interval Additivity Property: $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$.

Substituting the given values:

$$17 = 12 + \int_8^{10} f(x) dx$$

$$\Rightarrow \int_8^{10} f(x) dx = 17 - 12 = 5.$$

□

Lesson 3: The Fundamental Theorem of Calculus, Part 1

Goals of this lesson.

- Define and interpret the **area accumulation function** $g(x) = \int_a^x f(t) dt$.
- Understand the geometric intuition and formal statement of the **Fundamental Theorem of Calculus, Part 1 (FTC 1)**.
- Differentiate integral functions using FTC 1 together with interval properties and the **Chain Rule**.

3.1 The Area Accumulation Function

Recall that if $f(x) \geq 0$ is continuous on $[a, b]$, the definite integral

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

is the exact area under the curve $y = f(x)$ from $x = a$ to $x = b$.

Now let the right endpoint move. If we replace b by a variable x , the area becomes a *function* of x :

Definition 3.1 (Area Accumulation Function)

Let f be a continuous function on $[a, b]$. The **area accumulation function** is

$$g(x) := \int_a^x f(t) dt, \quad a \leq x \leq b.$$

Dummy Variable of Integration

Here t is a **dummy variable** (the variable of integration), while x is the independent variable that sets the right edge of the region:

- At the start of the interval: $g(a) = \int_a^a f(t) dt = 0$.
- At the end of the interval: $g(b) = \int_a^b f(t) dt$, the total (signed) area.

3.2 How Fast Does the Area Grow?

To see how the accumulated area $g(x)$ changes with x , consider its derivative from the limit definition:

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}.$$

By interval additivity,

$$g(x+h) - g(x) = \int_a^{x+h} f(t) dt - \int_a^x f(t) dt = \int_x^{x+h} f(t) dt.$$

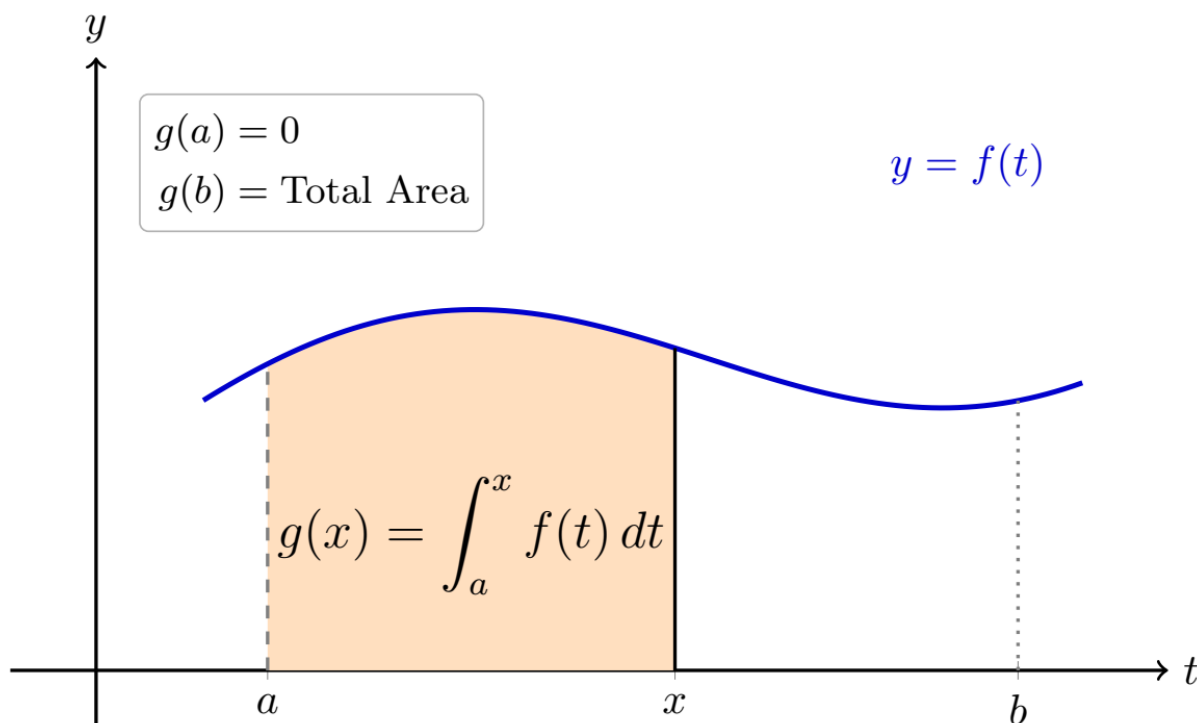


Figure 8: The area accumulation function $g(x) = \int_a^x f(t) dt$.

For small h , the region under $f(t)$ from x to $x+h$ is approximately a rectangle with base h and height $f(x)$ (Figure 10):

$$g(x+h) - g(x) = \int_x^{x+h} f(t) dt \approx f(x) \cdot h.$$

To justify the limit for either sign of h , continuity gives

$$\left| \frac{g(x+h) - g(x)}{h} - f(x) \right| \leq \sup_{t \text{ between } x \text{ and } x+h} |f(t) - f(x)| \rightarrow 0.$$

Thus $g'(x) = f(x)$. The rectangle argument explains the result, and this bound proves it.

Main idea

The rate at which area accumulates is the height of the curve. In other words, **differentiation undoes integration**: the derivative of the area function is the original integrand $f(x)$.

3.3 The Fundamental Theorem of Calculus, Part 1

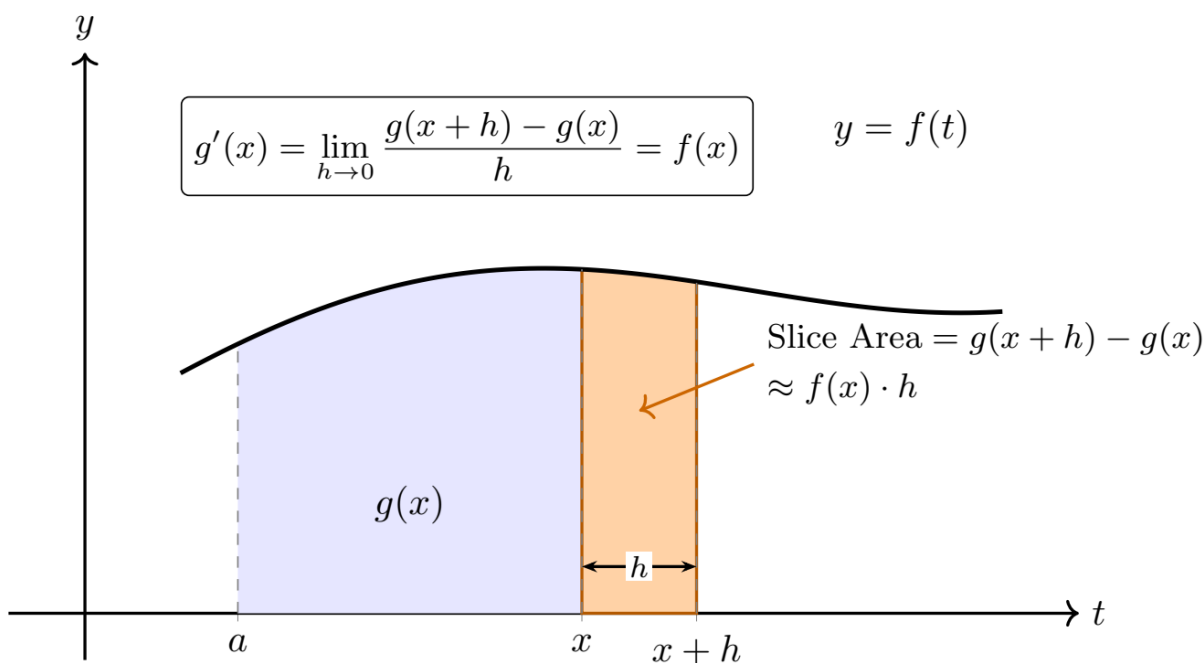


Figure 9: Geometric derivation of FTC 1: the difference $g(x+h) - g(x)$ is approximately a thin vertical rectangular slice.

Theorem 3.1 (Fundamental Theorem of Calculus, Part 1)

If f is continuous on $[a, b]$, then the function g defined by

$$g(x) = \int_a^x f(t) dt, \quad a \leq x \leq b$$

is continuous on $[a, b]$, differentiable on (a, b) , and

$$g'(x) = \frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x).$$

3.4 Examples and the Chain Rule

Example 3.1. Let $g(x) = \int_0^x \cos(\pi t) dt$. Find $g'(x)$ and evaluate $g'(1)$.

Solution. Applying FTC 1 directly with $f(t) = \cos(\pi t)$:

$$g'(x) = \frac{d}{dx} \left[\int_0^x \cos(\pi t) dt \right] = \cos(\pi x).$$

Evaluating at $x = 1$:

$$g'(1) = \cos(\pi) = -1.$$

□

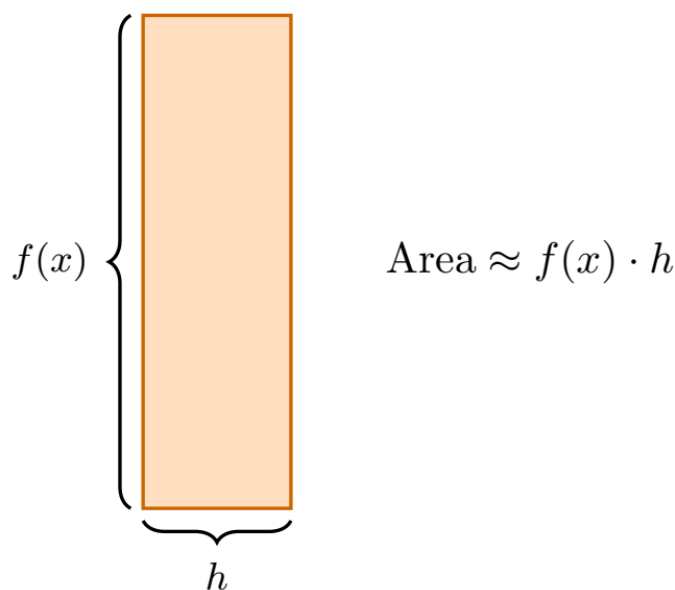


Figure 10: Geometric derivation of FTC 1: a thin vertical slice of width h and height $\approx f(x)$.

Example 3.2. Find the rate of change with respect to x of the area under $f(t) = t + 3$ from $t = -3$ to $t = x$.

Solution. The area function is $A(x) = \int_{-3}^x (t + 3) dt$. Its rate of change is

$$\frac{dA}{dx} = \frac{d}{dx} \left[\int_{-3}^x (t + 3) dt \right] = x + 3.$$

□

Example 3.3 (Variable Lower Limit). For $x > 0$, find $\frac{d}{dx} \left[\int_x^2 \ln(\ln(1 + t)) dt \right]$.

Solution. FTC 1 applies when the variable is in the **upper limit**. Reversing the limits introduces a negative sign:

$$\int_x^2 \ln(\ln(1 + t)) dt = - \int_2^x \ln(\ln(1 + t)) dt.$$

Applying FTC 1:

$$\frac{d}{dx} \left[- \int_2^x \ln(\ln(1 + t)) dt \right] = - \ln(\ln(1 + x)).$$

□

Example 3.4 (Applying the Chain Rule). Find $\frac{d}{dx} \left[\int_{-2}^{e^{2x}} \sin^3(5t) dt \right]$.

Solution. The upper limit is a function $u(x) = e^{2x}$, so we combine FTC 1 with the **Chain Rule**:

$$\frac{d}{dx} \left[\int_a^{u(x)} f(t) dt \right] = f(u(x)) \cdot u'(x).$$

Here $f(t) = \sin^3(5t)$, $u(x) = e^{2x}$, and $u'(x) = 2e^{2x}$. Therefore

$$\begin{aligned}\frac{d}{dx} \left[\int_{-2}^{e^{2x}} \sin^3(5t) dt \right] &= \sin^3(5e^{2x}) \cdot \frac{d}{dx}(e^{2x}) \\ &= 2e^{2x} \sin^3(5e^{2x}).\end{aligned}$$

□

Lesson 4: The Fundamental Theorem of Calculus, Part 2

Goals of this lesson.

- Derive the Fundamental Theorem of Calculus, Part 2 (FTC 2).
- Compute basic antiderivatives and evaluate definite integrals algebraically.

4.1 Connecting the Area Function to Antiderivatives

Computing areas as limits of Riemann sums is slow. FTC 1 gives us a shortcut.

Let $y = f(x)$ be a continuous curve, and let $g(x) = \int_a^x f(t) dt$ be its area accumulation function. By FTC 1 (Lesson 3),

$$g'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x).$$

So the area function $\int_a^x f(t) dt$ is an *antiderivative* of $f(x)$.

Now let $F(x)$ be *any* antiderivative of $f(x)$ (meaning $F'(x) = f(x)$). Two antiderivatives of the same function differ only by a constant, so

$$\int_a^x f(t) dt = F(x) + C.$$

To find C , evaluate at $x = a$:

$$\begin{aligned} \int_a^a f(t) dt &= F(a) + C \\ 0 &= F(a) + C \implies C = -F(a). \end{aligned}$$

Substituting back gives

$$\int_a^x f(t) dt = F(x) - F(a).$$

Finally, setting $x = b$ gives the definite integral over $[a, b]$.

4.2 The Fundamental Theorem of Calculus, Part 2

Theorem 4.1 (FTC 2)

Let $f(x)$ be a continuous function on $[a, b]$, and let $F(x)$ be any antiderivative of $f(x)$. Then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Main idea

To compute an area, you no longer need rectangles: find an antiderivative, and subtract its values at the endpoints.

Notation

We usually write the evaluation step with a vertical bar or brackets:

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a).$$

4.3 Evaluating Definite Integrals

Example 4.1. Evaluate $\int_0^2 2x dx$.

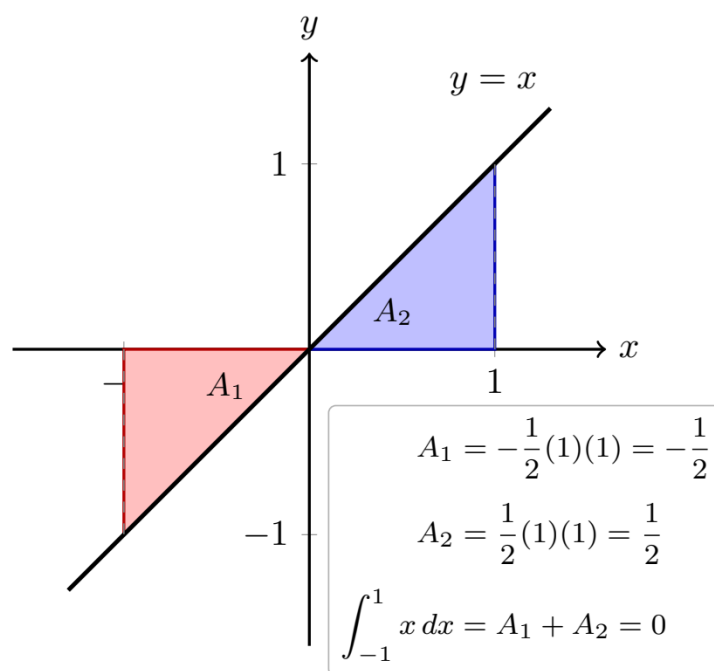


Figure 11: Triangular area under $y = 2x$ on $[0, 2]$.

Solution. Pick any antiderivative. If we include the constant of integration, $F(x) = x^2 + C$. Applying FTC 2:

$$\int_0^2 2x dx = (2^2 + C) - (0^2 + C) = 4 + C - C = 4.$$

The $+C$ always cancels when evaluating definite integrals, so from now on we simply choose $C = 0$.

Geometric check: This integral is the area of the triangle under $y = 2x$ on $[0, 2]$ (Figure 11), which is $\frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}(2)(4) = 4$. $\square$

Example 4.2. Evaluate $\int_{-1}^2 (x^2 - 2x + 7) dx$.

Solution. Find the antiderivative term-by-term and evaluate from -1 to 2 :

$$\begin{aligned}\int_{-1}^2 (x^2 - 2x + 7) dx &= \left[\frac{x^3}{3} - x^2 + 7x \right]_{-1}^2 \\ &= \left(\frac{2^3}{3} - 2^2 + 7(2) \right) - \left(\frac{(-1)^3}{3} - (-1)^2 + 7(-1) \right) \\ &= \left(\frac{8}{3} - 4 + 14 \right) - \left(-\frac{1}{3} - 1 - 7 \right) \\ &= \left(\frac{8}{3} + 10 \right) - \left(-\frac{25}{3} \right) = \frac{8}{3} + \frac{30}{3} + \frac{25}{3} = \frac{63}{3} = 21.\end{aligned}$$

□

Example 4.3. Evaluate $\int_0^{3\pi/4} \cos \theta d\theta$.

Solution. An antiderivative of $\cos \theta$ is $\sin \theta$:

$$\int_0^{3\pi/4} \cos \theta d\theta = \sin \theta \Big|_0^{3\pi/4} = \sin \left(\frac{3\pi}{4} \right) - \sin(0) = \frac{\sqrt{2}}{2}.$$

□

4.4 Two Essential Antiderivative Rules

Antiderivative rules

- **Power Rule:** An antiderivative of x^a is $\frac{x^{a+1}}{a+1}$ (for $a \neq -1$).
- **Logarithm Rule:** An antiderivative of $\frac{1}{x}$ is $\ln|x|$. (This covers the missing case $a = -1$.)

Example 4.4. Evaluate $\int_0^9 \sqrt{x} dx$.

Solution. Rewrite the radical as a fractional exponent and apply the Power Rule:

$$\begin{aligned}\int_0^9 x^{1/2} dx &= \left[\frac{x^{3/2}}{3/2} \right]_0^9 = \left[\frac{2}{3} x^{3/2} \right]_0^9 \\ &= \frac{2}{3} (9^{3/2}) - 0 = \frac{2}{3} (27) = 18.\end{aligned}$$

□

Example 4.5. Evaluate $\int_k^{k+1} \frac{dx}{x}$ (assuming $k > 0$).

Solution. Apply the Logarithm Rule:

$$\int_k^{k+1} \frac{1}{x} dx = \ln|x| \Big|_k^{k+1} = \ln(k+1) - \ln(k) = \ln \left(\frac{k+1}{k} \right).$$

□

Lesson 5: Indefinite Integrals and the Net Change Theorem

Goals of this lesson.

- Distinguish between definite and indefinite integrals.
- State and apply the **Net Change Theorem** to physical, biological, and real-world models.
- Distinguish between **displacement** and **total distance traveled** for a moving particle.

5.1 Indefinite Integrals vs. Definite Integrals

FTC 2 turned every definite integral into an antiderivative problem, so antiderivatives deserve their own notation.

Definition 5.1 (Indefinite Integral)

The notation $\int f(x) dx$ is called an **indefinite integral** and represents the most general antiderivative of $f(x)$:

$$\int f(x) dx = F(x) + C \quad \Longleftrightarrow \quad F'(x) = f(x).$$

Key Distinction

- A **definite integral** $\int_a^b f(x) dx$ is a **number** (the net signed area).
- An **indefinite integral** $\int f(x) dx$ is a **family of functions**, one for each value of C .

Domains and Constants

Antiderivative formulas hold on intervals where their integrands are defined. On a disconnected domain, the integration constant may differ between components. Real powers and logarithms also require their usual real domains. In particular, $\ln x$ and $x^x = e^{x \ln x}$ require $x > 0$.

5.2 Table of Standard Indefinite Integrals

Every entry below is a derivative rule read backwards. You can check any of them by differentiating the right-hand side.

Table of standard indefinite integrals

Basic & Algebraic

$$\int 0 \, dx = C$$

$$\int k \, dx = kx + C$$

$$\int x^a \, dx = \frac{x^{a+1}}{a+1} + C \quad (a \neq -1)$$

Trigonometric

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C$$

$$\int \csc x \cot x \, dx = -\csc x + C$$

Exponential & Logarithmic

$$\int e^x \, dx = e^x + C$$

$$\int b^x \, dx = \frac{b^x}{\ln b} + C \quad (b > 0, b \neq 1)$$

$$\int \frac{1}{x} \, dx = \ln |x| + C$$

Inverse Trigonometric

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \sin^{-1} x + C$$

$$\int \frac{1}{1+x^2} \, dx = \tan^{-1} x + C$$

$$\int \frac{1}{|x|\sqrt{x^2-1}} \, dx = \sec^{-1} x + C$$

Example 5.1 (Algebraic Simplification Before Integration). Evaluate $\int \frac{x^4 - 2\sqrt{x}}{x} \, dx$.

Solution. There is no quotient rule for integrals, so first split the integrand into power-rule terms:

$$\begin{aligned} \int \left(\frac{x^4}{x} - \frac{2x^{1/2}}{x} \right) dx &= \int \left(x^3 - 2x^{-1/2} \right) dx \\ &= \frac{x^4}{4} - 2 \left(\frac{x^{1/2}}{1/2} \right) + C \\ &= \frac{1}{4}x^4 - 4\sqrt{x} + C. \end{aligned}$$

□

Example 5.2 (Trigonometric Indefinite Integral). Evaluate

$$\int \left(3\sec^2 \theta - 5e^\theta + \frac{4}{1+\theta^2} \right) d\theta.$$

Solution. Integrate term-by-term using the table:

$$\int \left(3\sec^2 \theta - 5e^\theta + \frac{4}{1+\theta^2} \right) d\theta = 3\tan \theta - 5e^\theta + 4\tan^{-1} \theta + C.$$

□

5.3 The Net Change Theorem

FTC 2 says something useful even outside of area problems. If F' is the rate of change of a quantity F , integrating the rate recovers the total change in F .

Theorem 5.1 (Net Change Theorem)

If F' is continuous on $[a, b]$, its definite integral is the net change of F :

$$\int_a^b F'(x) dx = F(b) - F(a).$$

Summing local changes yields global change

- If $\frac{dn}{dt}$ is a population growth rate: $\int_{t_1}^{t_2} \frac{dn}{dt} dt = n(t_2) - n(t_1)$ is the net change in population.
- If $\frac{dT}{dt}$ is a heating or cooling rate: $\int_{t_1}^{t_2} \frac{dT}{dt} dt = T(t_2) - T(t_1)$ is the net change in temperature.
- If $\frac{dV}{dt}$ is a flow rate: $\int_{t_1}^{t_2} \frac{dV}{dt} dt = V(t_2) - V(t_1)$ is the net volume that has flowed.

Example 5.3 (Rate of Flow). Water leaks from a storage tank at a rate of $r(t) = 1 + t$ liters per hour, where t is the number of hours after midday ($t = 0$). How much water is lost between 1:00 PM ($t = 1$) and 4:00 PM ($t = 4$)?

Solution. The water lost is the integral of the leak rate:

$$\begin{aligned} \text{Water lost} &= \int_1^4 r(t) dt = \int_1^4 (1 + t) dt = \left[t + \frac{t^2}{2} \right]_1^4 \\ &= \left(4 + \frac{4^2}{2} \right) - \left(1 + \frac{1^2}{2} \right) = 12 - 1.5 = 10.5 \text{ liters.} \end{aligned}$$

□

5.4 Kinematic Applications: Displacement vs. Total Distance

For a particle moving along a straight line with position $s(t)$ and velocity $v(t) = s'(t)$, the Net Change Theorem gives the particle's *net* change in position. Motion backwards cancels motion forwards, so to measure the total ground covered we integrate the speed $|v(t)|$ instead.

Displacement vs. total distance

- **Displacement (net change in position):**

$$\text{Displacement} = s(t_2) - s(t_1) = \int_{t_1}^{t_2} v(t) dt.$$

• **Total distance traveled:**

$$\text{Total Distance} = \int_{t_1}^{t_2} |v(t)| dt = \int_{t_1}^{t_2} (\text{speed}) dt.$$

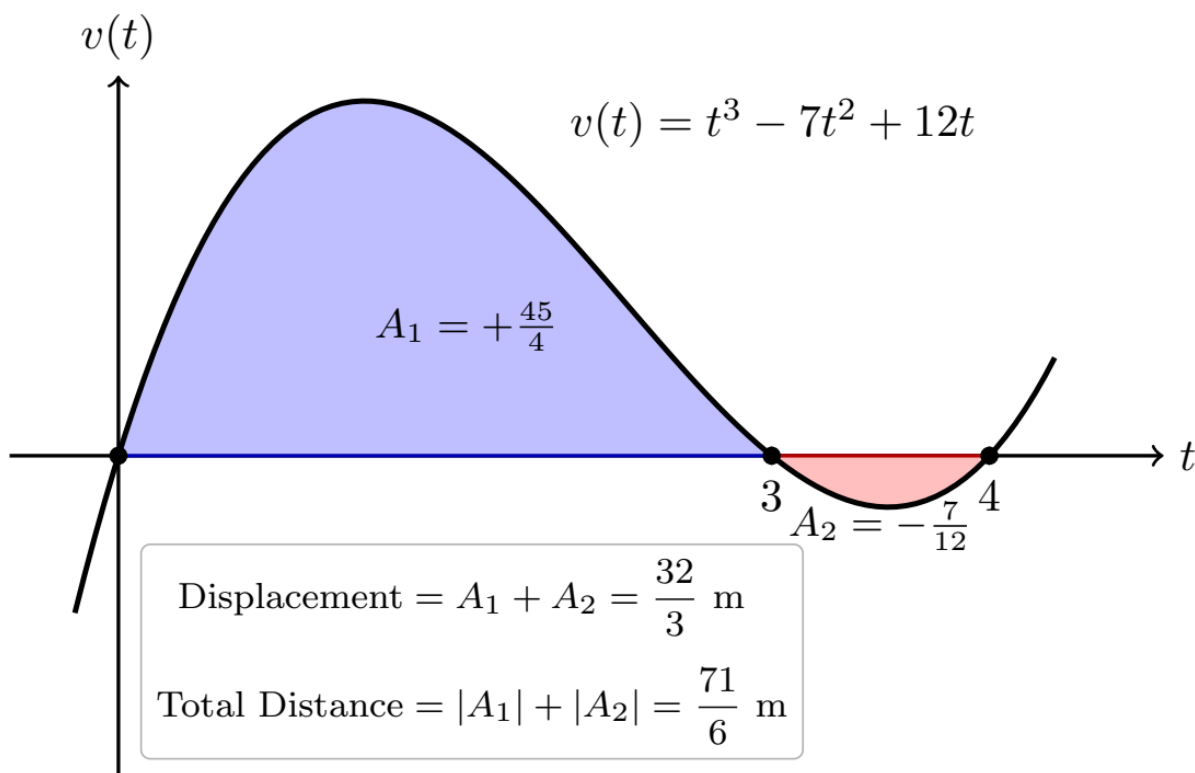


Figure 12: Velocity curve $v(t) = t^3 - 7t^2 + 12t$ on $[0, 4]$. The positive area on $[0, 3]$ represents forward motion, and the negative area on $[3, 4]$ represents backward motion.

Example 5.4 (Displacement vs. Total Distance). A particle moves along a straight line with velocity $v(t) = t^3 - 7t^2 + 12t$ (in m/s).

1. Find the displacement of the particle from $t = 0$ to $t = 4$.
2. Find the total distance traveled by the particle from $t = 0$ to $t = 4$.

Solution. First, find when the particle is at rest by factoring the velocity:

$$v(t) = t(t^2 - 7t + 12) = t(t - 3)(t - 4) = 0 \implies t = 0, t = 3, t = 4.$$

- On $(0, 3)$, $v(t) > 0$ (moving in the positive direction).
- On $(3, 4)$, $v(t) < 0$ (moving in the negative direction).

1. Displacement:

$$\begin{aligned}
 \text{Displacement} &= \int_0^4 (t^3 - 7t^2 + 12t) dt = \left[\frac{t^4}{4} - \frac{7t^3}{3} + 6t^2 \right]_0^4 \\
 &= \left(\frac{256}{4} - \frac{7(64)}{3} + 6(16) \right) - 0 \\
 &= 64 - \frac{448}{3} + 96 = 160 - \frac{448}{3} = \frac{32}{3} \text{ m} \approx 10.67 \text{ m}.
 \end{aligned}$$

2. Total distance traveled: Split the integral at $t = 3$, where the velocity changes sign:

$$\text{Total Distance} = \int_0^4 |v(t)| dt = \int_0^3 v(t) dt - \int_3^4 v(t) dt.$$

Forward motion on $[0, 3]$:

$$\begin{aligned}
 \int_0^3 (t^3 - 7t^2 + 12t) dt &= \left[\frac{t^4}{4} - \frac{7t^3}{3} + 6t^2 \right]_0^3 \\
 &= \frac{81}{4} - 63 + 54 = \frac{45}{4} \text{ m}.
 \end{aligned}$$

Backward motion on $[3, 4]$:

$$\int_3^4 v(t) dt = \int_0^4 v(t) dt - \int_0^3 v(t) dt = \frac{32}{3} - \frac{45}{4} = \frac{128 - 135}{12} = -\frac{7}{12} \text{ m}.$$

Adding the magnitudes:

$$\text{Total Distance} = \frac{45}{4} + \left| -\frac{7}{12} \right| = \frac{135}{12} + \frac{7}{12} = \frac{142}{12} = \frac{71}{6} \text{ m} \approx 11.83 \text{ m}.$$

□

Lesson 6: The Substitution Rule

Goals of this lesson.

- Review the Chain Rule and its relationship to integration.
- Use u -substitution to evaluate indefinite integrals.
- Apply the Substitution Rule to definite integrals by transforming the limits of integration.
- Handle algebraic rearrangements and less obvious substitutions.

6.1 Reversing the Chain Rule

The table in [Lesson 5](#) reverses the basic derivative rules, but it cannot handle compositions like $(x^2 + 3)^4 \cdot 2x$. Those come from the Chain Rule, so we reverse that too.

Recall that for a composite function $F(g(x))$:

$$\frac{d}{dx} [F(g(x))] = F'(g(x)) \cdot g'(x).$$

If $F'(u) = f(u)$, then reading this backwards:

$$\int f(g(x)) \cdot g'(x) dx = F(g(x)) + C.$$

Example 6.1 (Motivating the Substitution Method). Consider $f(x) = \frac{1}{5}(x^2 + 3)^5$. Differentiating with the Chain Rule and $u = x^2 + 3$:

$$f'(x) = \frac{d}{du} \left[\frac{1}{5} u^5 \right] \cdot \frac{du}{dx} = u^4 \cdot (2x) = (x^2 + 3)^4 \cdot 2x.$$

To reverse this process and find $\int (x^2 + 3)^4 (2x) dx$, we change variables:

$$\begin{aligned} \text{Let } u = x^2 + 3 &\implies du = 2x dx. \\ \int (x^2 + 3)^4 (2x) dx &= \int u^4 du = \frac{1}{5} u^5 + C = \frac{1}{5} (x^2 + 3)^5 + C. \end{aligned}$$

Theorem 6.1 (The Substitution Rule for Indefinite Integrals)

If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\int f(g(x)) g'(x) dx = \int f(u) du.$$

How to choose u

- **Look for a derivative match:** pick $u = g(x)$ so that $g'(x)$ also appears in the integrand (up to a constant factor).
- **Look for an “inside” function:** an expression under a radical, inside parentheses

raised to a power, in an exponent, in a denominator, or inside a trigonometric function.

- **Try again if needed:** if the new integral still contains x or is no simpler, try a different u or rearrange algebraically first.

6.2 Indefinite Integration Examples

Example 6.2. Evaluate $\int x^2 \sqrt{x^3 + 4} \, dx$.

Solution. Let $u = x^3 + 4$, so $du = 3x^2 \, dx$ and $x^2 \, dx = \frac{1}{3} du$:

$$\begin{aligned} \int x^2 \sqrt{x^3 + 4} \, dx &= \int \sqrt{u} \cdot \frac{1}{3} du = \frac{1}{3} \int u^{1/2} du \\ &= \frac{1}{3} \left(\frac{u^{3/2}}{3/2} \right) + C = \frac{2}{9} u^{3/2} + C \\ &= \frac{2}{9} (x^3 + 4)^{3/2} + C. \end{aligned}$$

□

Example 6.3. Evaluate $\int e^{1-4x} \, dx$.

Solution. Let $u = 1 - 4x$, so $du = -4 \, dx$ and $dx = -\frac{1}{4} du$:

$$\int e^{1-4x} \, dx = \int e^u \left(-\frac{1}{4} du \right) = -\frac{1}{4} e^u + C = -\frac{1}{4} e^{1-4x} + C.$$

□

Example 6.4. Evaluate $\int \tan x \, dx$.

Solution. Rewrite $\tan x = \frac{\sin x}{\cos x}$ and let $u = \cos x$, so $du = -\sin x \, dx$:

$$\begin{aligned} \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx = \int \frac{-du}{u} = -\ln |u| + C \\ &= -\ln |\cos x| + C = \ln |\cos x|^{-1} + C = \ln |\sec x| + C. \end{aligned}$$

□

Example 6.5. Evaluate $\int \frac{e^{2x}}{e^{2x} + 3} \, dx$.

Solution. Let $u = e^{2x} + 3$, so $du = 2e^{2x} \, dx$ and $e^{2x} \, dx = \frac{1}{2} du$:

$$\int \frac{e^{2x}}{e^{2x} + 3} \, dx = \int \frac{\frac{1}{2} du}{u} = \frac{1}{2} \ln |u| + C = \frac{1}{2} \ln(e^{2x} + 3) + C.$$

(The absolute value can be dropped since $e^{2x} + 3 > 0$ for all x .)

□

Example 6.6 (Algebraic Back-Substitution). Evaluate $\int x\sqrt{x-5} \, dx$.

Solution. Let $u = x - 5$. Then $du = dx$, and the leftover x can be written as $x = u + 5$:

$$\begin{aligned} \int x\sqrt{x-5} \, dx &= \int (u+5)\sqrt{u} \, du = \int (u^{3/2} + 5u^{1/2}) \, du \\ &= \frac{u^{5/2}}{5/2} + 5 \left(\frac{u^{3/2}}{3/2} \right) + C \\ &= \frac{2}{5}(x-5)^{5/2} + \frac{10}{3}(x-5)^{3/2} + C. \end{aligned}$$

□

Example 6.7 (A Less Obvious Substitution). Evaluate $\int (\ln x + 1)x^x \, dx$.

Solution. Recall that $x^x = e^{x \ln x}$. Differentiating $u = x^x$:

$$\frac{du}{dx} = \frac{d}{dx} [e^{x \ln x}] = e^{x \ln x} \cdot \left(\ln x + x \cdot \frac{1}{x} \right) = x^x (\ln x + 1).$$

So $du = (\ln x + 1)x^x \, dx$ is exactly the integrand, and

$$\int (\ln x + 1)x^x \, dx = \int du = u + C = x^x + C.$$

□

6.3 The Substitution Rule for Definite Integrals

For a definite integral, we can change the limits along with the variable. Then there is no need to convert back to x at the end.

Theorem 6.2 (Substitution Rule for Definite Integrals)

If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x)) g'(x) \, dx = \int_{g(a)}^{g(b)} f(u) \, du.$$

Example 6.8. Evaluate $\int_{e^{36}}^{e^{81}} \frac{1}{x\sqrt{\ln x}} \, dx$.

Solution. Let $u = \ln x$, so $du = \frac{1}{x} \, dx$. Transform the limits:

- Lower limit: $x = e^{36} \implies u = 36$.
- Upper limit: $x = e^{81} \implies u = 81$.

Then

$$\int_{e^{36}}^{e^{81}} \frac{1}{x\sqrt{\ln x}} \, dx = \int_{36}^{81} u^{-1/2} \, du = \left[2\sqrt{u} \right]_{36}^{81} = 2(9) - 2(6) = 6.$$

□

Example 6.9. Evaluate $\int_0^{\sqrt{\pi/2}} x \cos(x^2) dx$.

Solution. Let $u = x^2$, so $du = 2x dx$ and $x dx = \frac{1}{2} du$. Transform the limits:

- $x = 0 \implies u = 0$.
- $x = \sqrt{\pi/2} \implies u = \frac{\pi}{2}$.

Then

$$\int_0^{\sqrt{\pi/2}} x \cos(x^2) dx = \frac{1}{2} \int_0^{\pi/2} \cos u du = \frac{1}{2} \left[\sin u \right]_0^{\pi/2} = \frac{1}{2} (1 - 0) = \frac{1}{2}.$$

□

Lesson 7: Areas Between Curves

Goals of this lesson.

- Set up and evaluate integrals for the area between two curves with respect to x (vertical slicing: top – bottom).
- Find intersection points to determine the limits of integration.
- Set up and evaluate integrals with respect to y (horizontal slicing: right – left).
- Compute areas of piecewise or symmetric regions.

7.1 Area Between Curves with Respect to x (Vertical Slicing)

Recall that if $f(x) \geq 0$ on $[a, b]$, the area under $y = f(x)$ is

$$\text{Area} = \int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

Think of each rectangle as a thin strip of width dx and height $f(x)$, with area $d\text{Area} = f(x) dx$. The integral adds up all the strips.

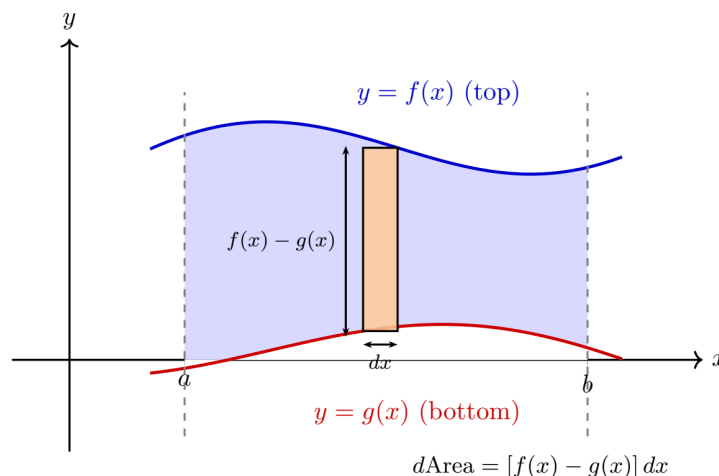


Figure 13: Vertical approximating rectangle between $y = f(x)$ (top) and $y = g(x)$ (bottom).

The same idea works for a region between two curves. If $f(x) \geq g(x)$ on $[a, b]$, a vertical strip of width dx now runs from the bottom curve up to the top curve, so its height is $f(x) - g(x)$ (Figure 13):

$$d\text{Area} = [f(x) - g(x)] dx.$$

Theorem 7.1 (Area Between Curves via Vertical Slicing)

If f and g are continuous and $f(x) \geq g(x)$ for all $x \in [a, b]$, then the area of the region bounded by $y = f(x)$, $y = g(x)$, $x = a$, and $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx = \int_a^b (y_{\text{top}} - y_{\text{bottom}}) dx.$$

Example 7.1. Find the area of the region enclosed between the parabola $y = x^2$ and the line $y = 2x$.

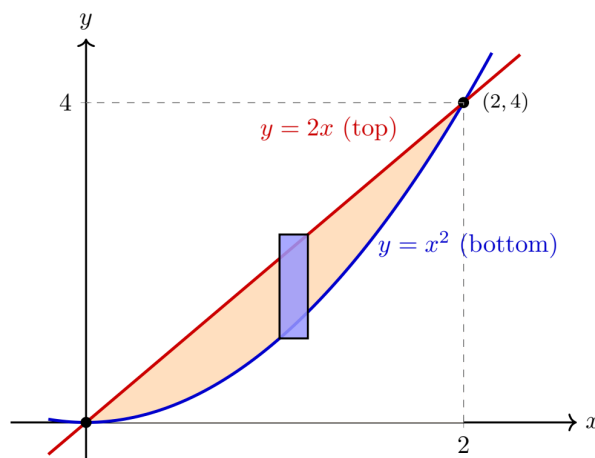


Figure 14: Region bounded between $y = x^2$ and $y = 2x$.

Solution. **Step 1: Find the points of intersection.**

$$x^2 = 2x \implies x(x - 2) = 0 \implies x = 0 \quad \text{and} \quad x = 2.$$

Step 2: Determine the top and bottom curves. On $(0, 2)$, test $x = 1$:

$$y_{\text{line}} = 2, \quad y_{\text{parabola}} = 1 \implies y_{\text{top}} = 2x, \quad y_{\text{bottom}} = x^2.$$

Step 3: Integrate.

$$A = \int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}.$$

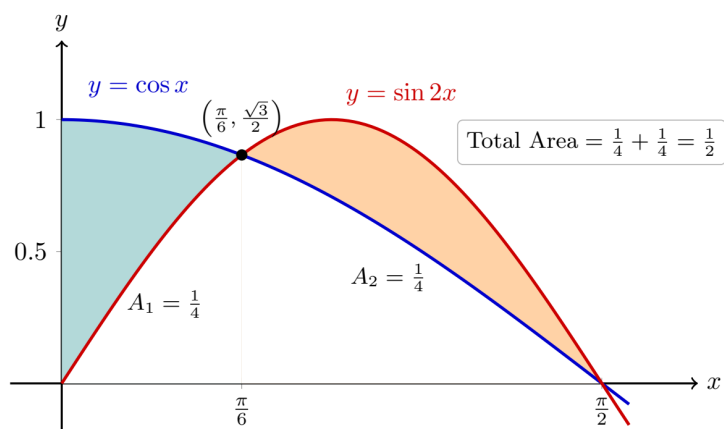
□

Example 7.2. Find the area of the region bounded between $y = \cos x$ and $y = \sin 2x$ on the interval $[0, \frac{\pi}{2}]$.

Solution. **Step 1: Find the points of intersection on $[0, \frac{\pi}{2}]$.** Using the double-angle identity $\sin 2x = 2 \sin x \cos x$:

$$\cos x = \sin 2x \implies \cos x - 2 \sin x \cos x = 0 \implies \cos x(1 - 2 \sin x) = 0.$$

This gives two solutions on $[0, \frac{\pi}{2}]$:

Figure 15: Region between $y = \cos x$ and $y = \sin 2x$ on $[0, \pi/2]$.

- $\cos x = 0 \implies x = \frac{\pi}{2}$
- $1 - 2\sin x = 0 \implies \sin x = \frac{1}{2} \implies x = \frac{\pi}{6}$

Step 2: Compare the curves on each subinterval.

- On $[0, \frac{\pi}{6}]$: $\cos x \geq \sin 2x$ (at $x = 0$, $\cos 0 = 1 > \sin 0 = 0$).
- On $[\frac{\pi}{6}, \frac{\pi}{2}]$: $\sin 2x \geq \cos x$ (at $x = \frac{\pi}{3}$, $\sin \frac{2\pi}{3} = \frac{\sqrt{3}}{2} > \cos \frac{\pi}{3} = \frac{1}{2}$).

The curves swap places, so we split the integral.

Step 3: Integrate over each piece.

$$\begin{aligned}
 A_1 &= \int_0^{\pi/6} (\cos x - \sin 2x) dx = \left[\sin x + \frac{1}{2} \cos 2x \right]_0^{\pi/6} \\
 &= \left(\frac{1}{2} + \frac{1}{4} \right) - \left(0 + \frac{1}{2} \right) = \frac{1}{4}. \\
 A_2 &= \int_{\pi/6}^{\pi/2} (\sin 2x - \cos x) dx = \left[-\frac{1}{2} \cos 2x - \sin x \right]_{\pi/6}^{\pi/2} \\
 &= \left(\frac{1}{2} - 1 \right) - \left(-\frac{1}{4} - \frac{1}{2} \right) = -\frac{1}{2} + \frac{3}{4} = \frac{1}{4}.
 \end{aligned}$$

The total enclosed area is

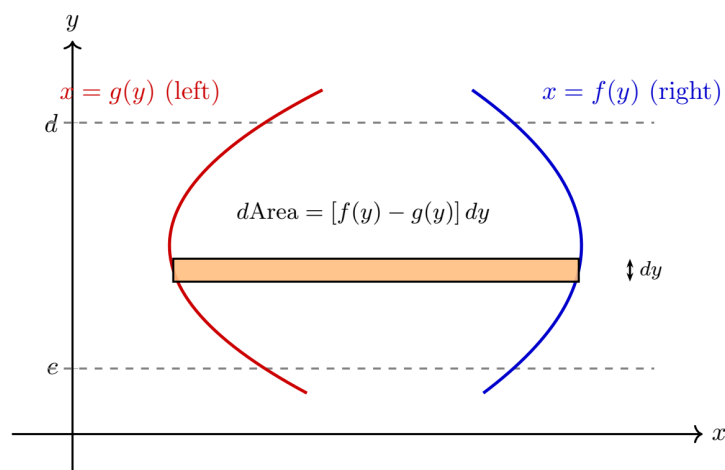
$$A = A_1 + A_2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}.$$

□

7.2 Area Between Curves with Respect to y (Horizontal Slicing)

If a region is bounded by curves written more naturally as $x = f(y)$ and $x = g(y)$, it is easier to use **horizontal strips** of height dy and length $f(y) - g(y)$ (Figure 16):

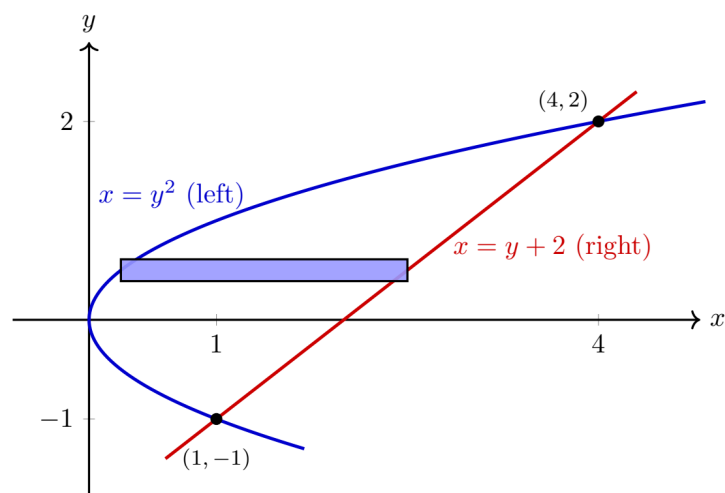
$$d\text{Area} = [f(y) - g(y)] dy = (x_{\text{right}} - x_{\text{left}}) dy.$$

Figure 16: Horizontal approximating strip between $x = f(y)$ (right) and $x = g(y)$ (left).**Theorem 7.2 (Area Between Curves via Horizontal Slicing)**

If f and g are continuous and $f(y) \geq g(y)$ for all $y \in [c, d]$, the enclosed area is

$$A = \int_c^d [f(y) - g(y)] dy = \int_c^d (x_{\text{right}} - x_{\text{left}}) dy.$$

Example 7.3. Find the area enclosed by the parabola $x = y^2$ and the line $y = x - 2$.

Figure 17: Region bounded between $x = y^2$ and $y = x - 2$.

Solution. **Step 1: Express both curves as functions of y .**

$$x = y^2 \quad \text{and} \quad x = y + 2.$$

Step 2: Find the intersection limits in y .

$$y^2 = y + 2 \implies (y - 2)(y + 1) = 0 \implies y = -1 \quad \text{and} \quad y = 2.$$

On $[-1, 2]$, $x_{\text{right}} = y + 2$ and $x_{\text{left}} = y^2$.

Step 3: Integrate with respect to y .

$$\begin{aligned} A &= \int_{-1}^2 ((y + 2) - y^2) dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 \\ &= \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} + \frac{7}{6} = \frac{9}{2}. \end{aligned}$$

(With vertical strips, this region would need two separate integrals, because the bottom boundary changes at $x = 1$.) $\square$

Example 7.4. Find the area enclosed by $x = \sin y$ and $x = 1 - \cos y$ for $0 \leq y \leq \frac{\pi}{2}$.

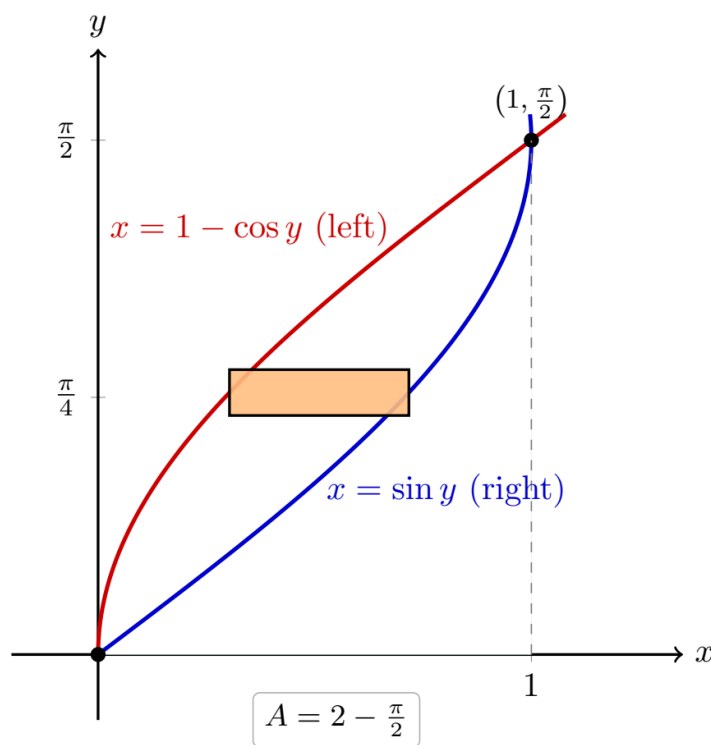


Figure 18: Region bounded between the curves $x = \sin y$ and $x = 1 - \cos y$ for $0 \leq y \leq \pi/2$.

Solution. **Step 1: Compare the boundaries on $[0, \frac{\pi}{2}]$.** Testing at $y = \frac{\pi}{4}$:

$$\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} \approx 0.707, \quad 1 - \cos\left(\frac{\pi}{4}\right) = 1 - \frac{\sqrt{2}}{2} \approx 0.293.$$

Thus $x_{\text{right}} = \sin y$ and $x_{\text{left}} = 1 - \cos y$ (Figure 18).

Step 2: Integrate.

$$\begin{aligned}
 A &= \int_0^{\pi/2} [\sin y - (1 - \cos y)] dy = \int_0^{\pi/2} (\sin y + \cos y - 1) dy \\
 &= \left[-\cos y + \sin y - y \right]_0^{\pi/2} \\
 &= \left(0 + 1 - \frac{\pi}{2} \right) - (-1 + 0 - 0) = 2 - \frac{\pi}{2}.
 \end{aligned}$$

□

7.3 Geometric Evaluations and Circular Symmetry

Example 7.5. Sketch the region whose area is represented by $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} (\sqrt{1-x^2} - |x|) dx$, and evaluate the integral using geometry.

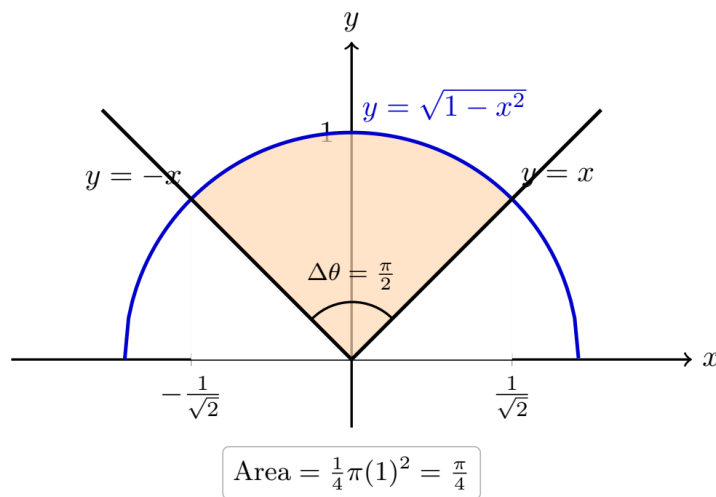


Figure 19: Circular sector between the upper unit semicircle $y = \sqrt{1-x^2}$ and the lines $y = |x|$.

Solution. **Step 1: Identify the boundary curves.**

- Top boundary: $y = \sqrt{1-x^2}$, the upper half of the unit circle $x^2 + y^2 = 1$.
- Bottom boundary: $y = |x|$, made of the rays $y = x$ (for $x \geq 0$) and $y = -x$ (for $x \leq 0$).

Step 2: Interpret geometrically. The ray $y = x$ makes an angle of $\frac{\pi}{4}$ with the positive x -axis, and the ray $y = -x$ makes an angle of $\frac{3\pi}{4}$. The region between them and the circle is a circular sector (Figure 19) with central angle

$$\frac{3\pi}{4} - \frac{\pi}{4} = \frac{\pi}{2}.$$

This is exactly one quarter of the unit disk:

$$\text{Area} = \frac{1}{4}\pi(1)^2 = \frac{\pi}{4}.$$

□

Lesson 8: Volumes by Slicing: Cross-Sections and the Disk Method

Goals of this lesson.

- Extend the “slice and add” idea from areas to volumes.
- Compute volumes of solids with known cross-sections (e.g., semicircles, squares).
- Derive and apply the **Disk Method** for solids of revolution about a coordinate axis.
- Evaluate volumes of unbounded solids of revolution (e.g., Gabriel’s Horn).

8.1 The Slicing Principle

We computed areas by adding up thin strips:

$$\text{Area} = \int_a^b d\text{Area} = \int_a^b f(x) dx.$$

Volumes work the same way, with thin slabs in place of thin strips. Suppose a solid lies between planes perpendicular to the x -axis at $x = a$ and $x = b$, and let $A(x)$ be the area of its cross-section at x . A slab of thickness dx has volume

$$dV = A(x) dx.$$

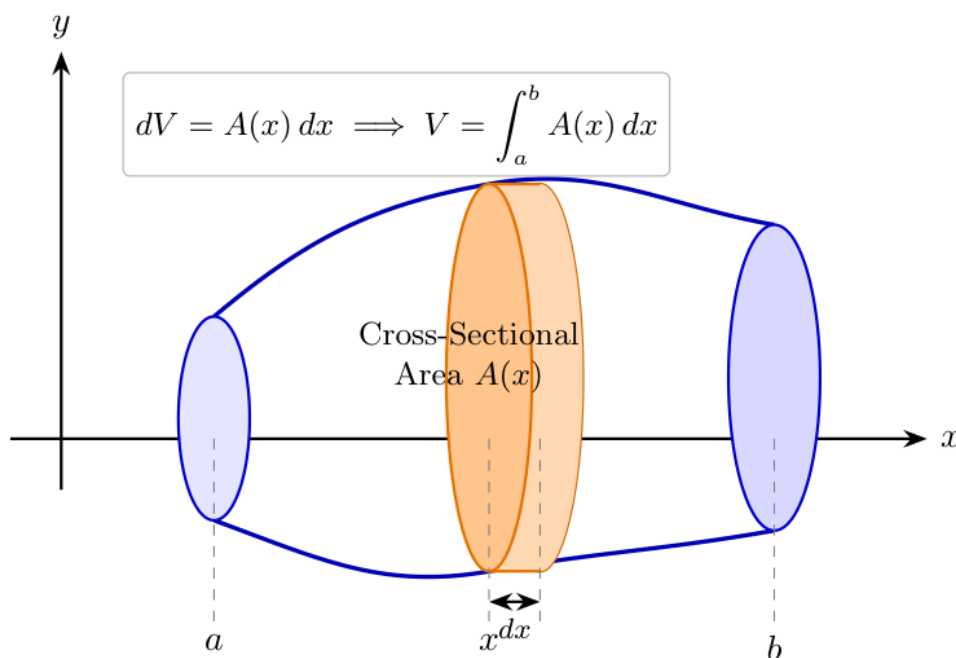


Figure 20: The Slicing Principle: accumulating cross-sectional slices of area $A(x)$ and thickness dx from $x = a$ to $x = b$.

Theorem 8.1 (Volume by Cross-Sectional Slicing)

Let S be a solid that lies between $x = a$ and $x = b$. If the area of the cross-section of S in the plane through x perpendicular to the x -axis is a continuous function $A(x)$, then the volume of S is

$$V = \int_a^b dV = \int_a^b A(x) dx.$$

Main idea

Every volume problem in this lesson and the next comes down to one question: *what is the area $A(x)$ of a typical slice?* Once you have $A(x)$, the rest is integration.

8.2 Solids with Known Cross-Sections

Example 8.1 (Semicircular Cross-Sections). A solid has as its base the triangular region in the xy -plane bounded by the lines $y = 1 - x$, $y = -1 + x$, and the y -axis ($x \in [0, 1]$). Each cross-section perpendicular to the x -axis is a semicircle whose diameter lies in the base. Find the volume of the solid.

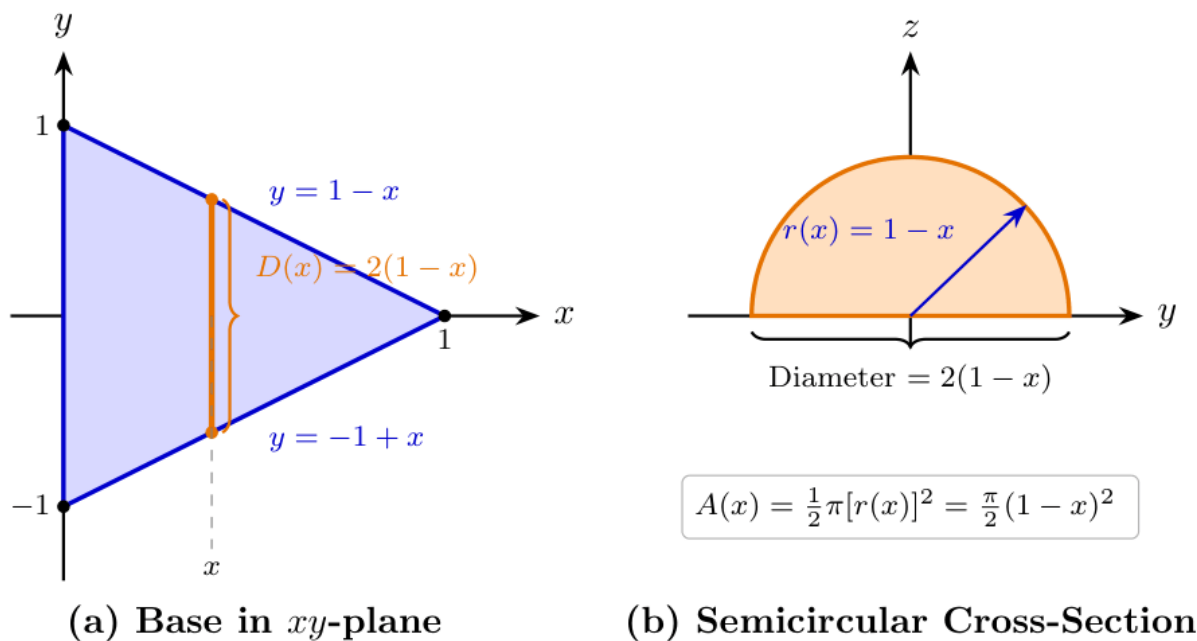


Figure 21: Base of the solid in Example 8.1 with a representative vertical chord of length $2(1 - x)$.

Solution. Step 1: Find the diameter and radius of a cross-section. At each $x \in [0, 1]$, the cross-section runs from $y = -1 + x$ up to $y = 1 - x$, so its diameter is

$$D(x) = (1 - x) - (-1 + x) = 2(1 - x), \quad r(x) = \frac{D(x)}{2} = 1 - x.$$

Step 2: Find the cross-sectional area $A(x)$. Each slice is a semicircle of radius $1 - x$:

$$A(x) = \frac{1}{2}\pi[r(x)]^2 = \frac{\pi}{2}(1 - x)^2.$$

Step 3: Integrate over $[0, 1]$.

$$V = \int_0^1 \frac{\pi}{2}(1 - x)^2 dx = \frac{\pi}{2} \left[-\frac{(1 - x)^3}{3} \right]_0^1 = \frac{\pi}{2} \left(0 + \frac{1}{3} \right) = \frac{\pi}{6}.$$

□

Example 8.2 (Square Pyramid via Similar Triangles). Find the volume of a pyramid of height 6 ft whose base is a square with area 4 ft².

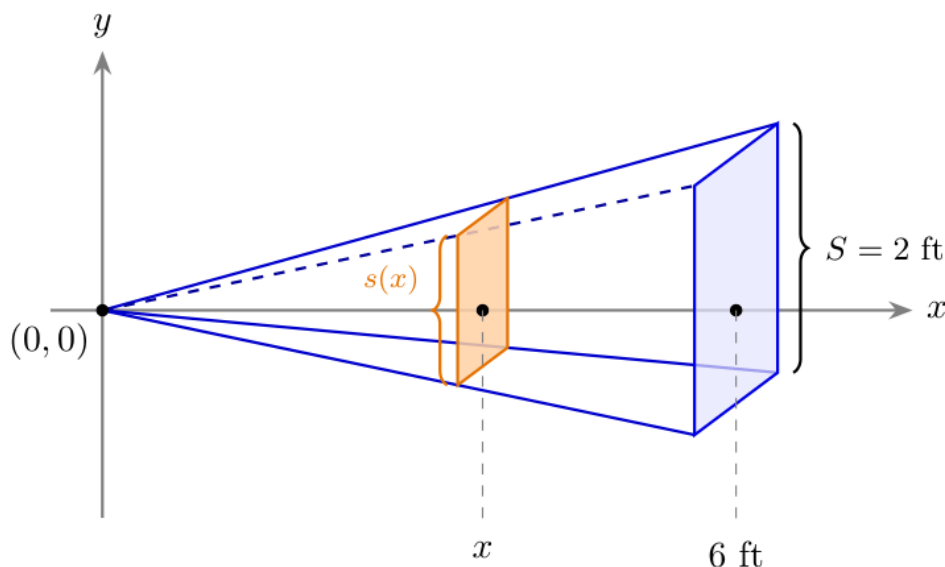


Figure 22: Square pyramid aligned along the x -axis with apex at the origin and base at $x = 6$.

Solution. **Step 1: Set up coordinates.** Lay the pyramid along the x -axis with its apex at the origin and its square base at $x = 6$. The base has area 4 ft², so each side has length 2 ft.

Step 2: Find the side length $s(x)$ using similar triangles. At distance x from the apex, the cross-section is a square of side $s(x)$. By similar triangles (Figure 22):

$$\frac{s(x)}{x} = \frac{2}{6} = \frac{1}{3} \implies s(x) = \frac{x}{3}.$$

Step 3: Find $A(x)$ and integrate.

$$A(x) = [s(x)]^2 = \frac{x^2}{9}, \quad V = \int_0^6 \frac{x^2}{9} dx = \left[\frac{x^3}{27} \right]_0^6 = \frac{216}{27} = 8 \text{ ft}^3.$$

Check: The pyramid formula gives $V = \frac{1}{3}Bh = \frac{1}{3}(4)(6) = 8 \text{ ft}^3$.

□

8.3 Solids of Revolution: The Disk Method

When the region under $y = f(x)$ on $[a, b]$ is revolved about the x -axis, each cross-section perpendicular to the axis is a **circular disk** of radius $R(x) = f(x)$. Its area is $A(x) = \pi[R(x)]^2 = \pi[f(x)]^2$.

Theorem 8.2 (The Disk Method (Rotation about the x -axis))

The volume of the solid obtained by rotating the region under $y = f(x) \geq 0$ on $[a, b]$ about the x -axis is

$$V = \int_a^b \pi[R(x)]^2 dx = \pi \int_a^b [f(x)]^2 dx.$$

Example 8.3 (A Standard Solid of Revolution). Find the volume of the solid obtained by revolving the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ about the x -axis.

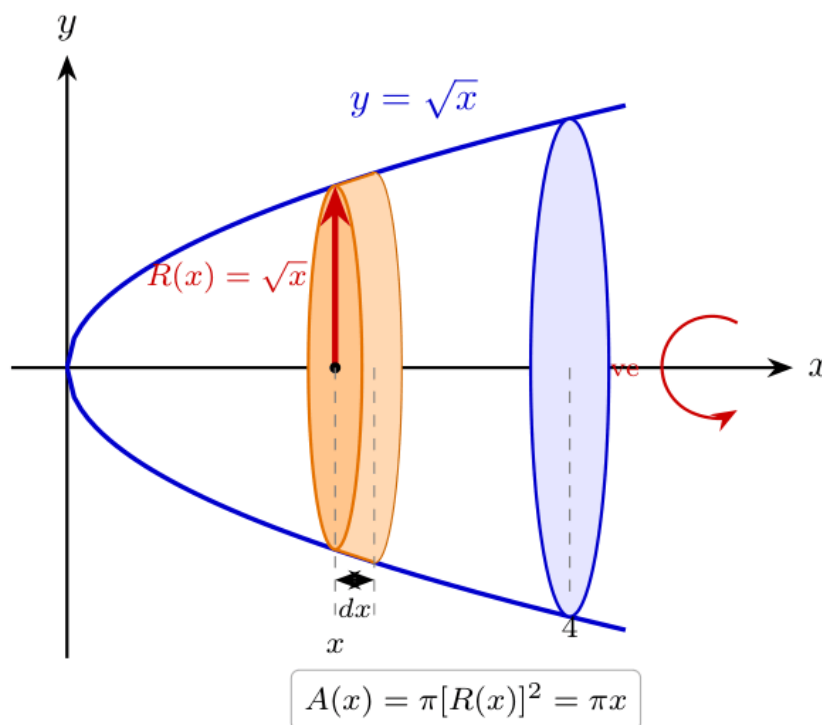


Figure 23: The curve $y = \sqrt{x}$ rotated about the x -axis produces a solid of revolution.

Solution. At each $x \in [0, 4]$, the disk has radius $R(x) = \sqrt{x}$, so

$$A(x) = \pi(\sqrt{x})^2 = \pi x, \quad V = \int_0^4 \pi x dx = \pi \left[\frac{x^2}{2} \right]_0^4 = 8\pi.$$

□

Example 8.4 (Gabriel's Horn: An Unbounded Region). Find the volume of the solid obtained by rotating the unbounded region under $y = \frac{1}{x}$ for $x \in [1, \infty)$ about the x -axis.

Solution. Step 1: Set up the integral. At each $x \geq 1$, the disk has radius $R(x) = \frac{1}{x}$, so $A(x) = \frac{\pi}{x^2}$.

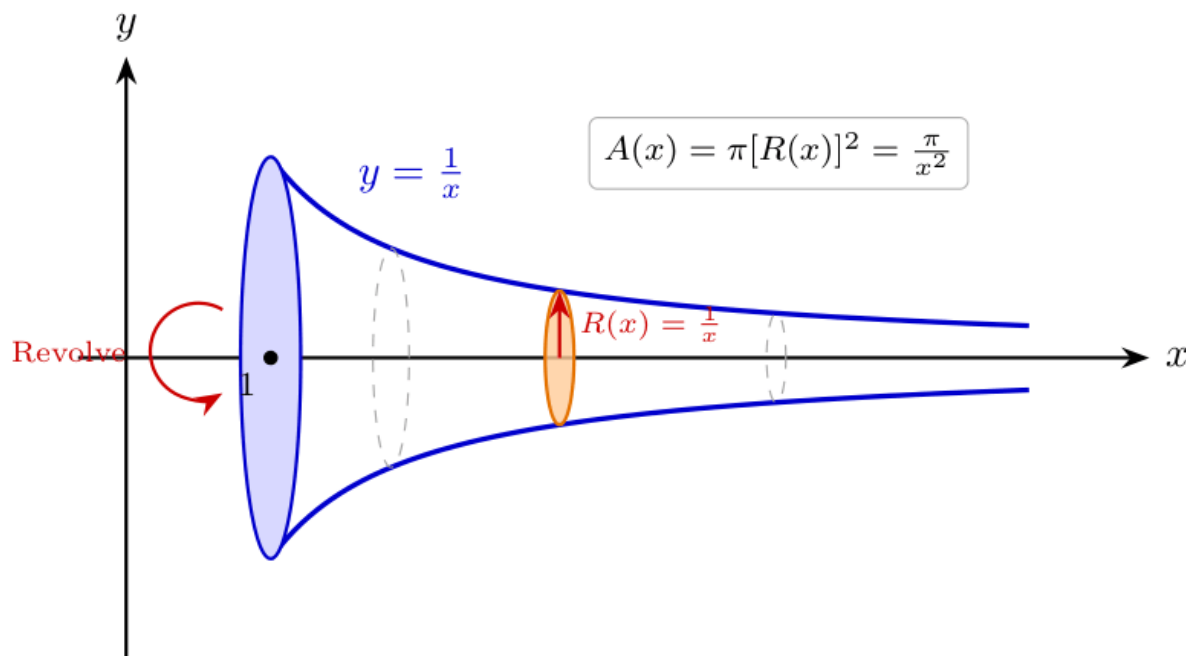


Figure 24: The hyperbola $y = 1/x$ for $1 \leq x < \infty$ rotated about the x -axis. The resulting solid is called Gabriel's Horn.

Step 2: Evaluate using a limit.

$$\begin{aligned} V &= \int_1^{\infty} \frac{\pi}{x^2} dx = \lim_{t \rightarrow \infty} \int_1^t \pi x^{-2} dx = \lim_{t \rightarrow \infty} \left[-\frac{\pi}{x} \right]_1^t \\ &= \lim_{t \rightarrow \infty} \left(-\frac{\pi}{t} + \pi \right) = \pi. \end{aligned}$$

Note

This solid is known as **Gabriel's Horn** (or Torricelli's Trumpet). It is famous for having a **finite volume** ($V = \pi$) but an **infinite surface area**.

□

Lesson 9: Volumes Continued: Washers and Cylindrical Shells

Goals of this lesson.

- Use the **Washer Method** for solids of revolution with hollow interiors.
- Compute volumes of solids revolved around shifted horizontal and vertical axes.
- Introduce and apply the **Method of Cylindrical Shells**.
- Choose between washers and shells based on the region and the axis of revolution.

9.1 The Washer Method (Revolving Around Coordinate Axes)

The slicing principle still applies:

$$V = \int_a^b A(x) dx.$$

What changes is the shape of the slice. When the region between an outer curve $y = f(x)$ and an inner curve $y = g(x)$ (with $f(x) \geq g(x) \geq 0$ on $[a, b]$) is revolved about the x -axis, the region no longer touches the axis. Each cross-section is a disk with a hole in it, called a **washer**:

$$\begin{aligned} A(x) &= (\text{area of outer disk}) - (\text{area of inner disk}) \\ &= \pi[R_{\text{outer}}(x)]^2 - \pi[r_{\text{inner}}(x)]^2 \\ &= \pi([f(x)]^2 - [g(x)]^2). \end{aligned}$$

Theorem 9.1 (The Washer Method (Rotation about the x -axis))

Suppose f and g are continuous and $f(x) \geq g(x) \geq 0$ on $[a, b]$. The volume obtained by revolving the region between them about the x -axis is

$$V = \int_a^b \pi ([R_{\text{outer}}(x)]^2 - [r_{\text{inner}}(x)]^2) dx = \pi \int_a^b ([f(x)]^2 - [g(x)]^2) dx.$$

Note

Square each radius *before* subtracting. In general $[f(x)]^2 - [g(x)]^2 \neq [f(x) - g(x)]^2$.

Example 9.1 (Rotation about the x -axis). Find the volume of the solid obtained by rotating the region enclosed between $y = x$ and $y = x^2$ about the x -axis.

Solution. **Step 1: Find the points of intersection.**

$$x = x^2 \implies x(x - 1) = 0 \implies x = 0 \quad \text{and} \quad x = 1.$$

On $(0, 1)$, $x > x^2$ (Figure 26), so

$$R_{\text{outer}}(x) = x \quad \text{and} \quad r_{\text{inner}}(x) = x^2.$$

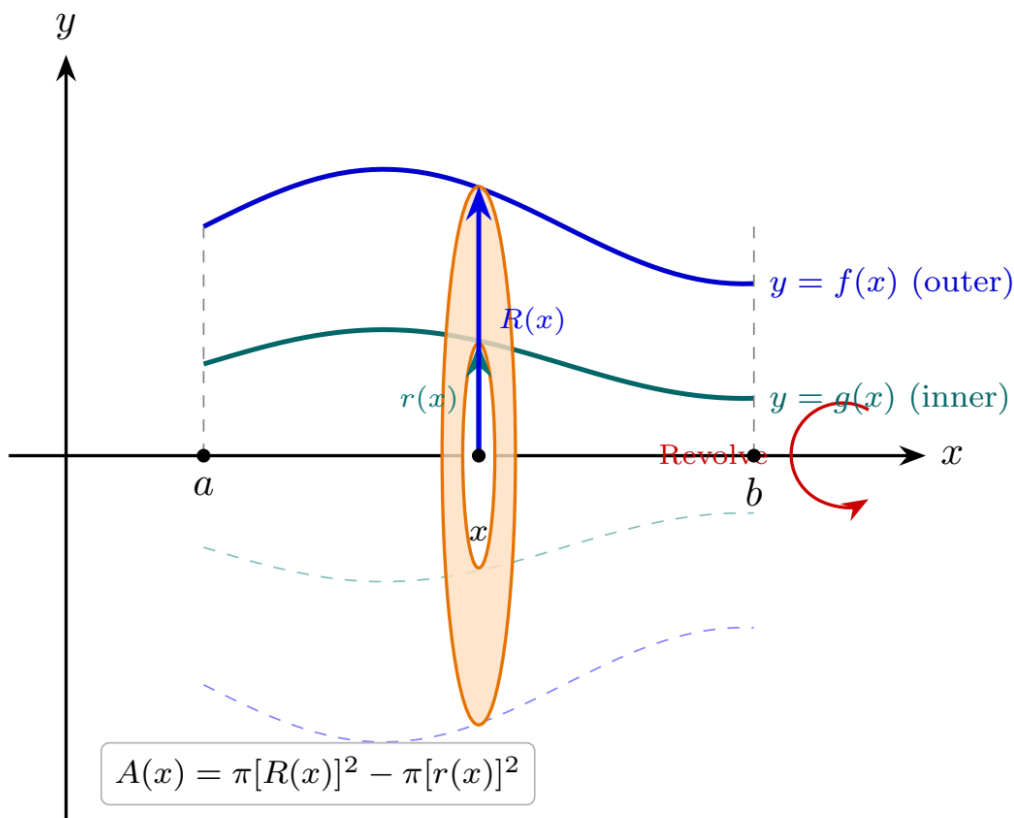


Figure 25: The Washer Method: region between $y = f(x)$ and $y = g(x)$ revolved about the x -axis.

Step 2: Set up and evaluate the integral.

$$\begin{aligned}
 V &= \pi \int_0^1 (x^2 - (x^2)^2) dx = \pi \int_0^1 (x^2 - x^4) dx \\
 &= \pi \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \pi \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2\pi}{15}.
 \end{aligned}$$

□

9.2 Revolving Around Shifted Axes

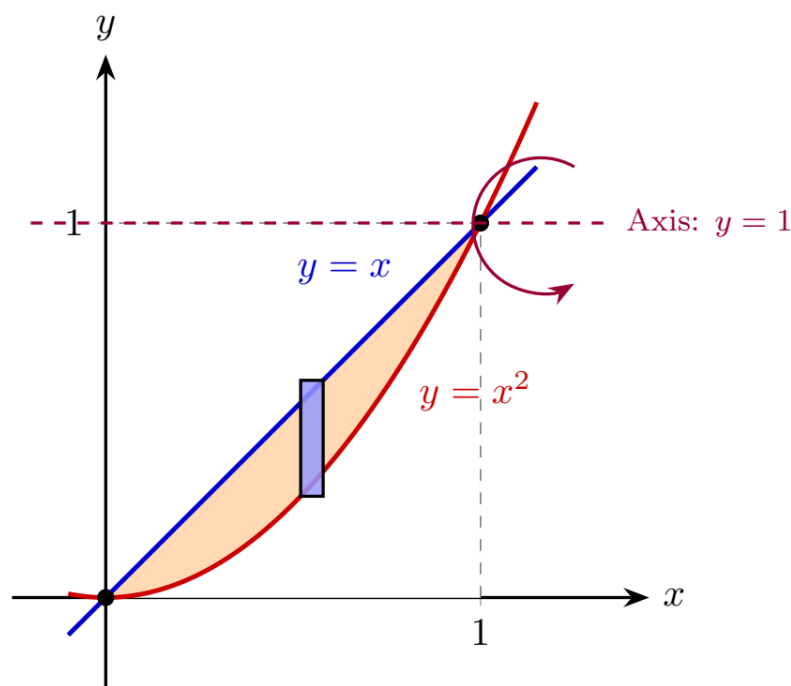
When rotating around a line other than a coordinate axis, a radius is still just a distance from the curve to the axis.

Finding a radius

$$\text{Radius} = |\text{curve} - \text{axis}|.$$

For a horizontal axis $y = k$ this means $|f(x) - k|$; for a vertical axis $x = h$ it means $|g(y) - h|$.

Example 9.2 (Rotation about the Horizontal Line $y = 1$). Find the volume of the solid obtained by revolving the region between $y = x$ and $y = x^2$ around the line $y = 1$.



About x-axis: $R(x) = x$, $r(x) = x^2 \implies V = \frac{2\pi}{15}$ About $y = 1$: $R(x) = 1 - x^2$, $r(x) = 1 - x \implies V = \frac{\pi}{5}$
--

Figure 26: The region between $y = x$ and $y = x^2$ on $[0, 1]$, showing the rotations in [Examples 9.1](#) and [9.2](#).

Solution. As seen in [Figure 26](#), the region lies entirely below the axis $y = 1$ for $x \in [0, 1]$.

- The curve *farther* from $y = 1$ is the parabola $y = x^2$, so it gives the outer radius:

$$R_{\text{outer}}(x) = 1 - x^2.$$

- The curve *closer* to $y = 1$ is the line $y = x$, so it gives the inner radius:

$$r_{\text{inner}}(x) = 1 - x.$$

Notice that the roles of the two curves have switched compared with [Example 9.1](#). Integrating over

$[0, 1]$:

$$\begin{aligned}
 V &= \pi \int_0^1 ([1 - x^2]^2 - [1 - x]^2) dx \\
 &= \pi \int_0^1 ((1 - 2x^2 + x^4) - (1 - 2x + x^2)) dx \\
 &= \pi \int_0^1 (x^4 - 3x^2 + 2x) dx \\
 &= \pi \left[\frac{x^5}{5} - x^3 + x^2 \right]_0^1 = \pi \left(\frac{1}{5} - 1 + 1 \right) = \frac{\pi}{5}.
 \end{aligned}$$

□

Example 9.3 (Rotation about the y -axis via Horizontal Disks). Find the volume of the solid obtained by rotating the region bounded by $y = x^3 + 1$, $y = 9$, and $x = 0$ about the y -axis.

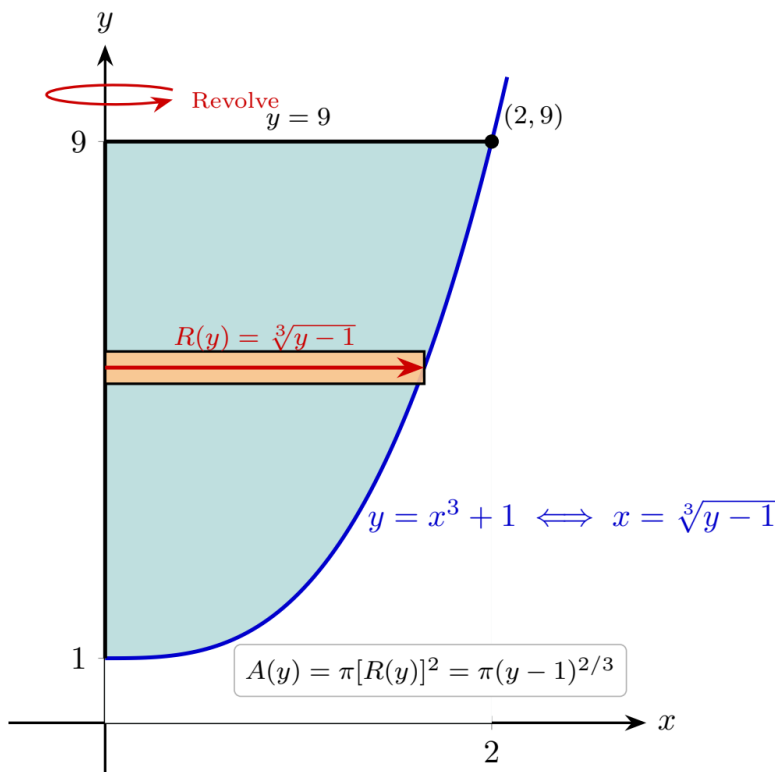


Figure 27: Horizontal disk at height y with radius $R(y) = \sqrt[3]{y-1}$, rotated about the y -axis.

Solution. The axis of revolution is vertical, so slices perpendicular to it are horizontal with thickness dy (Figure 27):

- Solve for x in terms of y : $y = x^3 + 1 \implies x = \sqrt[3]{y-1}$.
- Limits: when $x = 0$, $y = 1$, so y runs from 1 to 9.
- At height y , the disk has radius $R(y) = (y-1)^{1/3}$.

The cross-sectional area is $A(y) = \pi(y-1)^{2/3}$, so

$$V = \int_1^9 \pi(y-1)^{2/3} dy = \pi \left[\frac{3}{5}(y-1)^{5/3} \right]_1^9 = \frac{3\pi}{5} (8^{5/3}) = \frac{3\pi}{5} (32) = \frac{96\pi}{5}.$$

□

9.3 The Method of Cylindrical Shells

Sometimes slicing perpendicular to the axis is awkward, for example when it would require solving $y = f(x)$ for x . Shells offer another way: slice *parallel* to the axis instead.

When a vertical strip of thickness dx is revolved around a vertical axis (such as the y -axis), it sweeps out a thin **cylindrical shell**. Cutting the shell open and unrolling it gives an almost-flat slab with

$$\begin{aligned} \text{length} &= \text{circumference} = 2\pi r, \\ \text{height} &= h, \\ \text{thickness} &= dx. \end{aligned}$$

Therefore

$$dV = (\text{circumference}) \cdot (\text{height}) \cdot (\text{thickness}) = 2\pi r \cdot h \cdot dx.$$

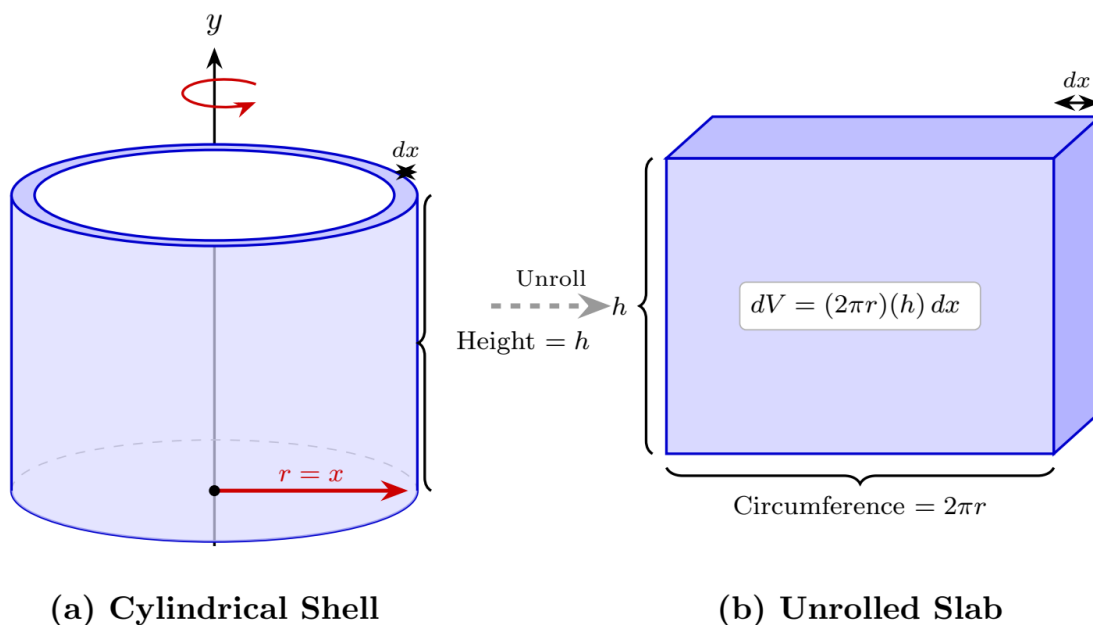


Figure 28: The Method of Cylindrical Shells: (a) a shell of radius r , height h , and thickness dx ; (b) the unrolled slab with volume $dV = (2\pi r)(h) dx$.

Theorem 9.2 (Method of Cylindrical Shells (Rotation about the y -axis))

The volume of the solid obtained by revolving the region under $y = f(x) \geq 0$ on $[a, b]$ ($0 \leq a < b$) about the y -axis is

$$V = \int_a^b 2\pi(\text{radius})(\text{height}) dx = 2\pi \int_a^b x f(x) dx.$$

Example 9.4 (Cylindrical Shells). Use cylindrical shells to find the volume of the solid obtained by rotating the region bounded by $y = 2 - x$, $y = 0$, and $x = 0$ about the y -axis.

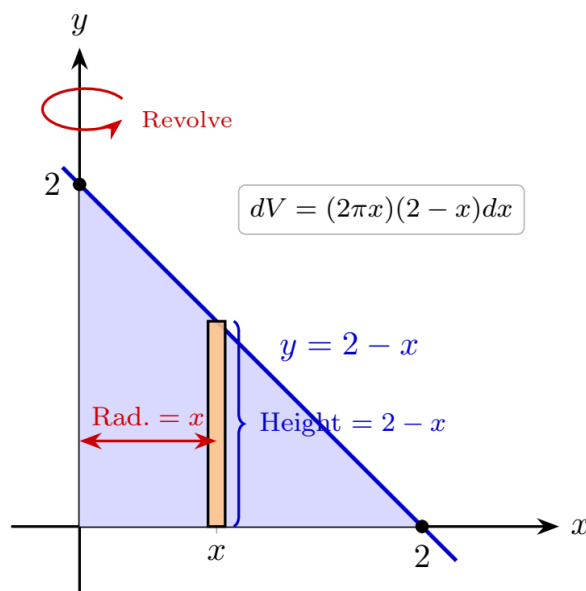


Figure 29: Shells for [Example 9.4](#): a vertical strip at radius x has height $2 - x$ and thickness dx .

Solution. The region lies in the first quadrant over $x \in [0, 2]$ ([Figure 29](#)). A vertical strip at distance x from the y -axis has

- radius = x ,
- height = $2 - x$,
- thickness = dx .

So

$$\begin{aligned} V &= 2\pi \int_0^2 x(2 - x) dx = 2\pi \int_0^2 (2x - x^2) dx \\ &= 2\pi \left[x^2 - \frac{x^3}{3} \right]_0^2 = 2\pi \left(4 - \frac{8}{3} \right) = \frac{8\pi}{3}. \end{aligned}$$

Check: The solid is a cone with base radius 2 and height 2, whose volume is $\frac{1}{3}\pi(2)^2(2) = \frac{8\pi}{3}$. $\square$

Washers or shells?

- **Washers/disks:** slices are *perpendicular* to the axis. Integrate along the axis.
- **Shells:** strips are *parallel* to the axis. Integrate across the axis.

Pick whichever method gives the simpler integral, or avoids solving for x in terms of y (or vice versa).

Lesson 10: Work

Goals of this lesson.

- Understand the units of mass, force, and work in both metric and US customary systems.
- Compute the work done by a constant force and by a variable force.
- Set up and evaluate integrals for work done against gravity, such as lifting a leaking bucket.

10.1 Work Done by a Constant Force

Work is the energy transferred by a force acting on an object through a distance.

By Newton's Second Law, force equals mass times acceleration:

$$F = ma = m \frac{d^2 s}{dt^2}.$$

When an object is lifted vertically, the force needed to oppose gravity is the object's weight:

$$F = mg,$$

where $g \approx 9.8 \text{ m/s}^2$ is the acceleration due to gravity on Earth.

Definition 10.1 (Work Done by a Constant Force)

If an object moves a distance d along a straight line under a constant force F in the direction of motion, the work done is

$$W = F \cdot d.$$

Quantity	SI Metric Units	US Customary Units
Distance (d)	meter (m)	foot (ft)
Mass (m)	kilogram (kg)	slug
Force (F)	Newton ($\text{N} = \text{kg} \cdot \text{m/s}^2$)	pound (lb)
Work / Energy (W)	Joule ($\text{J} = \text{N} \cdot \text{m}$)	foot-pound (ft-lb)

Note

In the US customary system, the pound (lb) is already a unit of force (weight), so you do *not* multiply by g . In the metric system, you must multiply mass (kg) by $g = 9.8 \text{ m/s}^2$ to get force (N).

Example 10.1 (Constant Force).

1. How much work is done lifting a 2 kg book from the floor to a desk 1.2 m high?
2. How much work is done lifting a 20 lb weight 6 ft off the ground?

Solution.

1. In metric units, the force is the weight $F = mg = (2)(9.8) = 19.6$ N, so

$$W = F \cdot d = (19.6 \text{ N})(1.2 \text{ m}) = 23.52 \text{ J}.$$

2. In US units, the weight is already a force ($F = 20$ lb), so

$$W = F \cdot d = (20 \text{ lb})(6 \text{ ft}) = 120 \text{ ft-lb}.$$

□

10.2 Work Done by a Variable Force

The formula $W = F \cdot d$ only works when the force is constant. If the force $F(x)$ changes as the object moves from $x = a$ to $x = b$, we use the same trick as always: break the path into small pieces of length dx . On each piece the force is nearly constant, so the work done on that piece is

$$dW = F(x) dx.$$

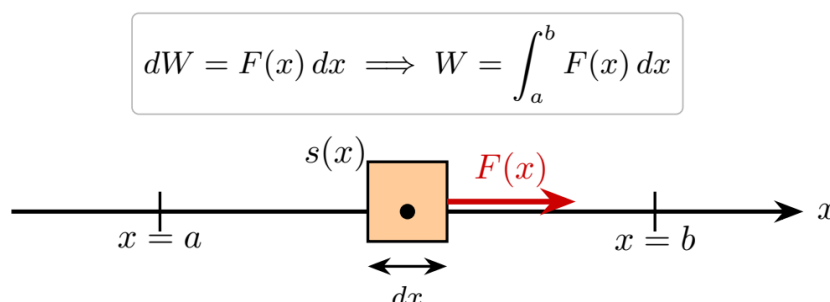


Figure 30: A variable force $F(x)$ acting on an object at position $s(x)$ over the interval $[a, b]$, giving differential work $dW = F(x) dx$.

Work done by a variable force

Adding up the work on all the pieces (Figure 30):

$$W = \int dW = \int_a^b F(x) dx.$$

Example 10.2. A force $F(x) = 3x^{-2}$ pounds is applied to an object moving along a line, where x is the distance in feet from the origin. Find the work done moving the object from $x = 1$ ft to $x = 10$ ft.

Solution.

$$W = \int_1^{10} 3x^{-2} dx = \left[-\frac{3}{x} \right]_1^{10} = -\frac{3}{10} + 3 = \frac{27}{10} = 2.7 \text{ ft-lb}.$$

□

Example 10.3 (Hooke's Law). A spring requires a force of 18 N to stretch it 0.06 m beyond its natural length. Find the work needed to stretch it from 0.02 m to 0.10 m beyond that length.

Solution. Hooke's law says the required force is $F(x) = kx$, where x is the extension. Thus $k = 18/0.06 = 300$ N/m, and

$$W = \int_{0.02}^{0.10} 300x \, dx = 150(0.10^2 - 0.02^2) = \boxed{1.44 \text{ J}}.$$

□

Example 10.4 (Lifting a Leaking Bucket). A bucket weighing 3 lb is attached to a rope of negligible weight and used to draw water from a well 80 ft deep. The bucket starts with 36 lb of water and is pulled up at a constant rate of 2 ft/s. As it rises, water leaks out at a constant rate of 0.2 lb/s. Find the work required to pull the bucket to the top of the well.

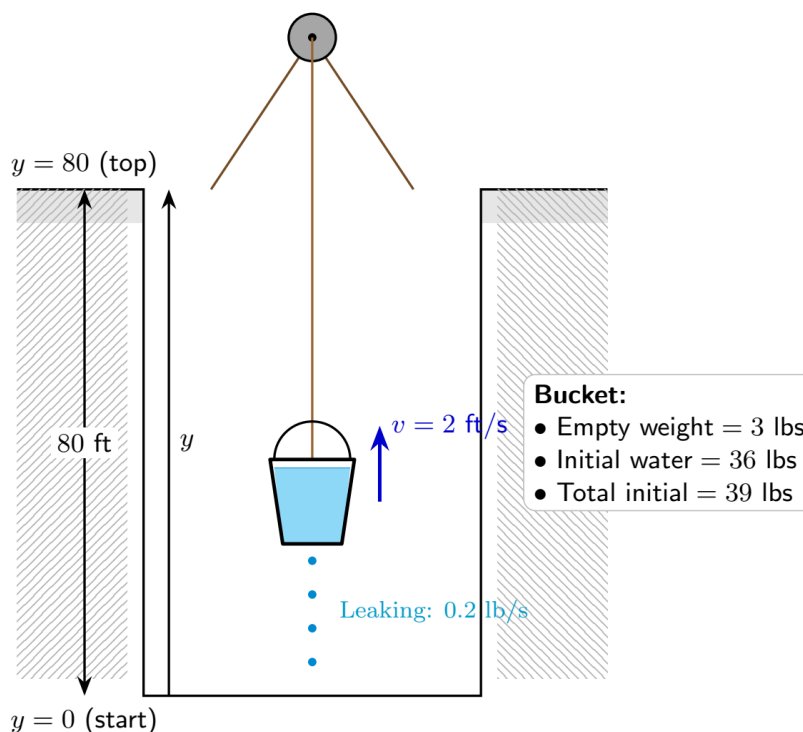


Figure 31: A water well and pulley system showing a leaking bucket pulled upward along an 80-foot vertical axis.

Solution. The force changes because the bucket loses water, so we first express the force in terms of the height. Let x be the height of the bucket above the bottom of the well (Figure 31), so x runs from 0 to 80 ft.

The bucket rises at 2 ft/s, so reaching height x takes

$$t = \frac{x}{2} \text{ seconds.}$$

Water leaks at 0.2 lb/s, so by the time the bucket reaches height x it has lost

$$(0.2 \text{ lb/s}) \cdot \left(\frac{x}{2} \text{ s}\right) = 0.1x \text{ lb}$$

of water. The water remaining weighs $36 - 0.1x$ lb. Adding the 3 lb bucket gives the total force at height x :

$$F(x) = (36 - 0.1x) + 3 = 39 - 0.1x \text{ lb.}$$

Now integrate from $x = 0$ to $x = 80$:

$$\begin{aligned} W &= \int_0^{80} (39 - 0.1x) dx = [39x - 0.05x^2]_0^{80} \\ &= 39(80) - 0.05(80^2) = 3120 - 320 = 2800 \text{ ft-lb.} \end{aligned}$$

□

Lesson 11: Work: Pumping Liquids and Applications

Goals of this lesson.

- Review units and the general work integral.
- Calculate the work required to empty tanks of various shapes (hemispherical and conical) by pumping out liquid.
- Set up work integrals using thin horizontal layers of liquid.

11.1 Review: Units and Variable Force

Recall from [Lesson 10](#): if a variable force $F(x)$ acts along a straight line from $x = a$ to $x = b$, the total work is ([Figure 30](#))

$$W = \int_a^b F(x) dx.$$

Quantity	SI Metric Units	US Customary Units
Length	meter (m)	foot (ft)
Mass	kilogram (kg)	slug
Force / Weight	Newton ($\text{N} = \text{kg} \cdot \text{m}/\text{s}^2$)	pound (lb)
Work / Energy	Joule ($\text{J} = \text{N} \cdot \text{m}$)	foot-pound (ft-lb)

Note

In US units, pounds are already a unit of force, so you do not multiply by g . In metric units, multiply mass (kg) by $g = 9.8 \text{ m/s}^2$ to get force in Newtons.

11.2 A New Situation: Different Pieces Move Different Distances

In [Lesson 10](#), the whole object moved together, and the *force* was what changed. When we build a column or empty a tank, the opposite happens: different pieces of material travel *different distances*. So we slice the material into thin horizontal layers, find the work to move each layer, and add:

$$dW = (\text{weight of the layer}) \times (\text{distance that layer moves}).$$

Example 11.1 (Building a Cylindrical Column). Find the work required against gravity to build a solid cylindrical column of height 4 m and radius 0.8 m out of material with density 600 kg/m^3 . Assume the material is lifted from ground level, and use $g = 9.8 \text{ m/s}^2$.

Solution. Place the ground at $y = 0$ and the top of the column at $y = 4$. For a horizontal layer of thickness dy at height y :

$$dV = \pi(0.8)^2 dy = 0.64\pi dy \text{ m}^3,$$

$$dm = (\text{density}) \cdot dV = 600(0.64\pi) dy = 384\pi dy \text{ kg},$$

$$dF = g dm = 384\pi(9.8) dy = 3763.2\pi dy \text{ N}.$$

This layer must be lifted from the ground to height y , a distance of y . So

$$W = \int_0^4 3763.2\pi y \, dy = 3763.2\pi \left[\frac{y^2}{2} \right]_0^4 = 3763.2\pi(8) = 30105.6\pi \approx 94,580 \text{ J.}$$

□

11.3 Pumping Liquids from Tanks

Strategy for pumping problems

1. Choose a coordinate axis and slice the liquid horizontally into thin layers of thickness dy .
2. Find the cross-sectional area $A(y)$ of a layer, so its volume is $dV = A(y) \, dy$.
3. Multiply the volume by the weight density to get the layer's weight dF .
4. Multiply dF by the vertical distance the layer must travel to leave the tank.
5. Integrate over the depth of the liquid.

Example 11.2 (Pumping from a Hemispherical Tank). A hemispherical tank of radius 10 ft is full of water. A spout extends 2 ft above the top of the tank. Find the work required to pump all the water out through the spout. (Use the weight density of water, 62.4 lb/ft^3 .)

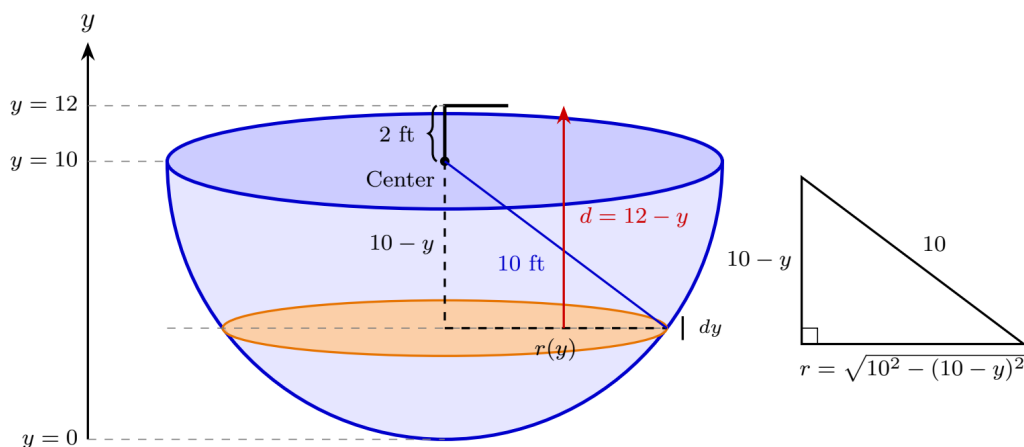


Figure 32: Hemispherical tank of radius 10 ft with a representative water layer of thickness dy at height y , lifted through a 2 ft spout.

Solution. Step 1: Set up coordinates. Place $y = 0$ at the bottom of the bowl (Figure 32). The rim is at $y = 10$ and the spout is at $y = 12$. The water occupies $y \in [0, 10]$.

Step 2: Find the radius of a layer. At height y , the layer is $10 - y$ below the center of the sphere. By the Pythagorean theorem, its radius $r(y)$ satisfies

$$[r(y)]^2 + (10 - y)^2 = 10^2 \implies [r(y)]^2 = 100 - (10 - y)^2 = 20y - y^2.$$

Step 3: Find the weight and the distance.

$$\begin{aligned} dV &= \pi[r(y)]^2 dy = \pi(20y - y^2) dy, \\ dF &= 62.4 dV = 62.4\pi(20y - y^2) dy \text{ lb.} \end{aligned}$$

The layer at height y must be lifted to the spout at $y = 12$, a distance of $12 - y$.

Step 4: Integrate.

$$\begin{aligned} W &= \int_0^{10} 62.4\pi(20y - y^2)(12 - y) dy \\ &= 62.4\pi \int_0^{10} (240y - 32y^2 + y^3) dy \\ &= 62.4\pi \left[120y^2 - \frac{32}{3}y^3 + \frac{y^4}{4} \right]_0^{10} \\ &= 62.4\pi \left(12000 - \frac{32000}{3} + 2500 \right) \\ &= 62.4\pi \left(\frac{11500}{3} \right) = 239200\pi \approx 7.51 \times 10^5 \text{ ft-lb.} \end{aligned}$$

□

Example 11.3 (Pumping from a Conical Tank (Vertex Up)). An upright conical tank has height 8 m and base radius 1 m, with its vertex at the top. It is full of water. Find the work required to pump all the water out through the vertex. (Use water density 1000 kg/m^3 and $g = 9.8 \text{ m/s}^2$.)

Solution. **Step 1: Set up coordinates.** Set $y = 0$ at the flat base and $y = 8$ at the vertex. The water occupies $y \in [0, 8]$.

Step 2: Find the radius of a layer using similar triangles. At height y , the layer is $8 - y$ below the vertex. By similar triangles (Figure 33):

$$\frac{r(y)}{8 - y} = \frac{1}{8} \implies r(y) = \frac{8 - y}{8}.$$

Step 3: Find the weight and the distance.

$$\begin{aligned} dV &= \pi[r(y)]^2 dy = \frac{\pi}{64}(8 - y)^2 dy, \\ dm &= 1000 dV, \\ dF &= g dm = \frac{9800\pi}{64}(8 - y)^2 dy \text{ N.} \end{aligned}$$

Each layer is lifted to the vertex at $y = 8$, a distance of $8 - y$.

Step 4: Integrate.

$$W = \int_0^8 \frac{9800\pi}{64}(8 - y)^2(8 - y) dy = \frac{9800\pi}{64} \int_0^8 (8 - y)^3 dy.$$

Let $u = 8 - y$, so $du = -dy$. When $y = 0$, $u = 8$; when $y = 8$, $u = 0$:

$$\begin{aligned} W &= \frac{9800\pi}{64} \int_8^0 u^3 (-du) = \frac{9800\pi}{64} \int_0^8 u^3 du = \frac{9800\pi}{64} \left[\frac{u^4}{4} \right]_0^8 \\ &= \frac{9800\pi}{64} \cdot 1024 = 156800\pi \approx 4.93 \times 10^5 \text{ J.} \end{aligned}$$

□

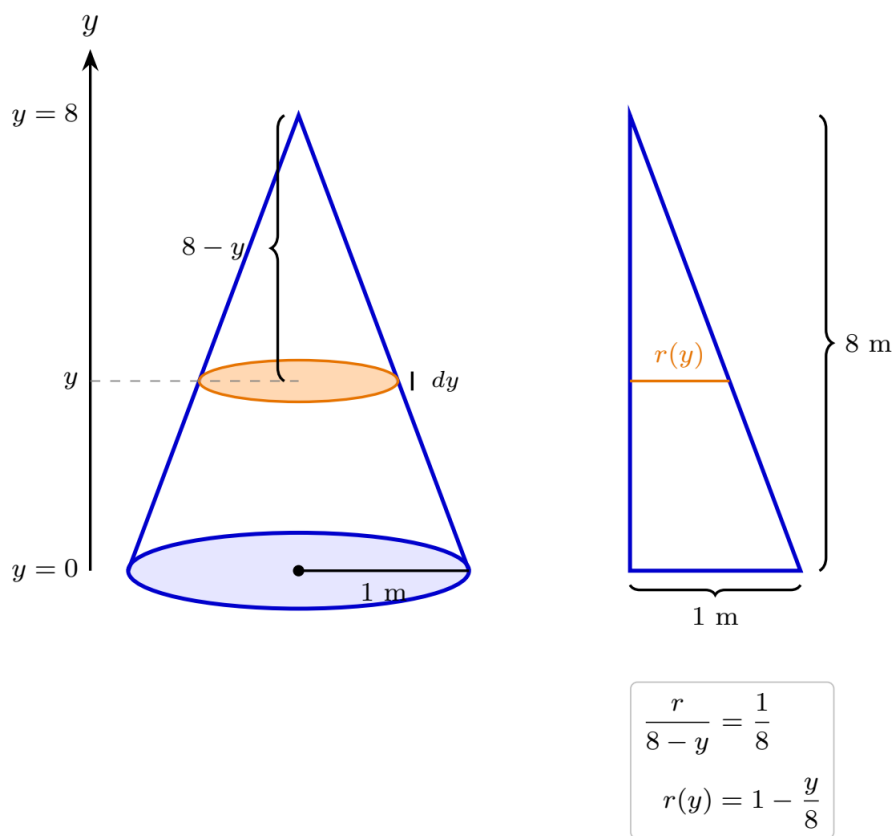


Figure 33: Upright conical tank with height 8 m and base radius 1 m. Similar triangles give the layer radius $r(y) = 1 - \frac{y}{8}$.

Lesson 12: Average Value of a Function

Goals of this lesson.

- Define and compute the average value of a continuous function on an interval.
- State, understand, and apply the Mean Value Theorem for Integrals.
- Interpret the Mean Value Theorem geometrically and solve problems that combine several intervals.

12.1 Definition of Average Value

The average of n numbers $y_1, y_2, \dots, y_n$ is

$$y_{\text{avg}} = \frac{y_1 + y_2 + \cdots + y_n}{n}.$$

A function has infinitely many values, so we cannot add them all up directly. Instead, we sample f at n points x_i^* , one in each of n equal subintervals of width $\Delta x = \frac{b-a}{n}$. Since $\frac{1}{n} = \frac{\Delta x}{b-a}$,

$$\frac{1}{n} \sum_{i=1}^n f(x_i^*) = \frac{1}{b-a} \sum_{i=1}^n f(x_i^*) \Delta x.$$

The right-hand side is a Riemann sum divided by $b-a$. Letting $n \rightarrow \infty$ averages over more and more sample points and gives the definition below.

Definition 12.1 (Average Value of a Function)

Let f be continuous on $[a, b]$. The **average value** of f on $[a, b]$ is

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx.$$

Example 12.1. Find the average value of $f(x) = \frac{1}{x^2}$ on the interval $[1, 5]$.

Solution. Here $b-a = 4$:

$$f_{\text{avg}} = \frac{1}{4} \int_1^5 x^{-2} dx = \frac{1}{4} \left[-\frac{1}{x} \right]_1^5 = \frac{1}{4} \left(-\frac{1}{5} + 1 \right) = \frac{1}{5}.$$

□

Example 12.2. Find the average value of $f(x) = x^n$ on the interval $[0, 1]$, where $n \geq 0$.

Solution.

$$f_{\text{avg}} = \frac{1}{1-0} \int_0^1 x^n dx = \left[\frac{x^{n+1}}{n+1} \right]_0^1 = \frac{1}{n+1}.$$

□

12.2 The Mean Value Theorem for Integrals

Theorem 12.1 (Mean Value Theorem for Integrals)

If $a < b$ and f is continuous on $[a, b]$, then there exists a number $c \in [a, b]$ such that

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx,$$

or equivalently,

$$\int_a^b f(x) dx = f(c)(b-a).$$

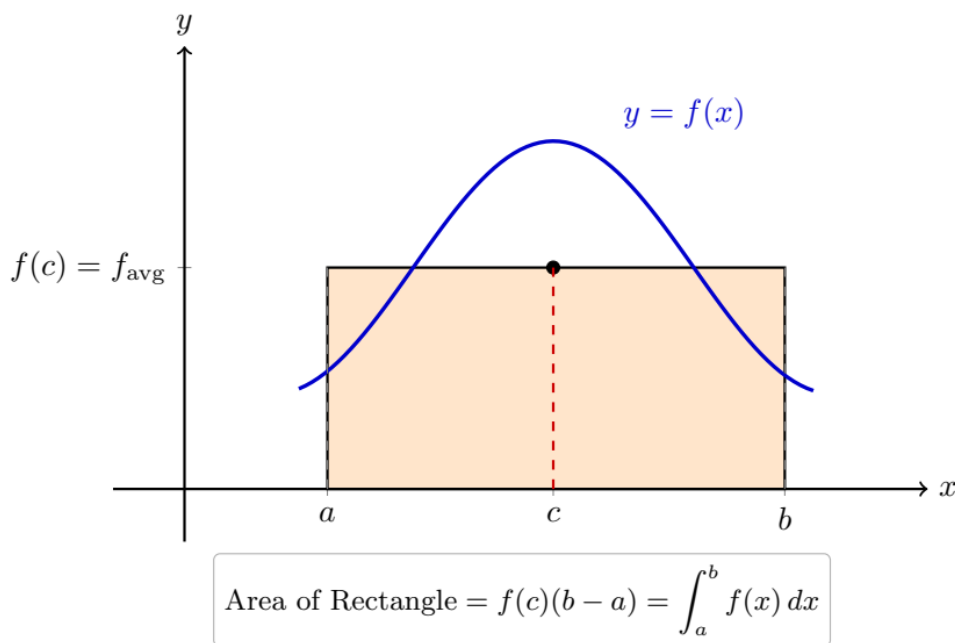


Figure 34: Geometric meaning of the Mean Value Theorem for Integrals: the area under the curve equals the area of a rectangle with base $b - a$ and height $f(c) = f_{\text{avg}}$.

Intuition

Imagine the region under the curve as water in a tank, with the curve as the frozen surface of a wave. When the wave settles, the water becomes flat at height f_{avg} (Figure 34). That level lies between the highest crest and the lowest trough, so the flat line $y = f_{\text{avg}}$ must cross the continuous curve at least once, at some $x = c$.

Example 12.3. Consider $f(x) = \tan x$ on the interval $[0, \frac{\pi}{4}]$.

1. Find the average value f_{avg} .
2. Find the value $c \in [0, \frac{\pi}{4}]$ guaranteed by the Mean Value Theorem for Integrals.

Solution. **1. Average value:**

$$f_{\text{avg}} = \frac{1}{\frac{\pi}{4} - 0} \int_0^{\pi/4} \tan x \, dx = \frac{4}{\pi} \left[\ln |\sec x| \right]_0^{\pi/4}.$$

Since $\sec(\pi/4) = \sqrt{2}$ and $\sec(0) = 1$:

$$f_{\text{avg}} = \frac{4}{\pi} (\ln \sqrt{2} - \ln 1) = \frac{4}{\pi} \cdot \frac{\ln 2}{2} = \frac{2 \ln 2}{\pi}.$$

2. Finding c : Set $f(c) = f_{\text{avg}}$:

$$\tan c = \frac{2 \ln 2}{\pi} \implies c = \tan^{-1} \left(\frac{2 \ln 2}{\pi} \right).$$

Since $\frac{2 \ln 2}{\pi} \approx 0.4413$, we get $c \approx 0.4158$, which is indeed in $(0, \frac{\pi}{4})$. $\square$

Example 12.4. Suppose the average value of a continuous function f over $[1, 2]$ is 7, and its average value over $[2, 4]$ is 6. Find $\int_1^4 f(x) \, dx$.

Solution. Multiply each average by the length of its interval to recover the integral:

$$\begin{aligned} \frac{1}{2-1} \int_1^2 f(x) \, dx &= 7 \implies \int_1^2 f(x) \, dx = 7, \\ \frac{1}{4-2} \int_2^4 f(x) \, dx &= 6 \implies \int_2^4 f(x) \, dx = 12. \end{aligned}$$

By interval additivity:

$$\int_1^4 f(x) \, dx = 7 + 12 = 19.$$

(Note that the answer is *not* $3 \cdot \frac{7+6}{2}$: the two intervals have different lengths.) $\square$

Example 12.5 (Challenge Problem). Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(x) \, dx = 0$. Prove that there exist distinct points $x_1, x_2 \in [0, 1]$ such that

$$f(x_1) + f(x_2) = 0.$$

Solution. By the Mean Value Theorem for Integrals, there exists $c \in [0, 1]$ such that

$$f(c) = \frac{1}{1-0} \int_0^1 f(x) \, dx = 0.$$

- If $f(x) = 0$ for all $x \in [0, 1]$, choose any two distinct points, e.g. $x_1 = 0$ and $x_2 = 1$. Then $f(0) + f(1) = 0$.
- Otherwise, since $\int_0^1 f(x) \, dx = 0$, the function must take both strictly positive and strictly negative values. Let $M = \max_{[0,1]} f > 0$ and $m = \min_{[0,1]} f < 0$.
- If $|m| \leq M$, choose x_2 with $f(x_2) = m$. Then $-f(x_2) = |m| \in (0, M]$. By the Intermediate Value Theorem, f takes every value between $0 = f(c)$ and M , so there is some x_1 with $f(x_1) = -f(x_2)$, i.e. $f(x_1) + f(x_2) = 0$. Since $f(x_1) > 0 > f(x_2)$, $x_1 \neq x_2$.
- If $|m| > M$, choose x_1 with $f(x_1) = M$. Since $-M \in (m, 0)$, the Intermediate Value Theorem gives some x_2 with $f(x_2) = -M = -f(x_1)$, and again $x_1 \neq x_2$. $\square$

Lesson 13: Integration by Parts

Goals of this lesson.

- Derive the Integration by Parts formula as the integral version of the Product Rule.
- Apply Integration by Parts to indefinite and definite integrals.
- Learn how to choose u and dv effectively.
- Handle repeated integration by parts and integrals that “loop back” on themselves.

13.1 Motivation and Derivation

Consider the integral

$$\int \ln x \, dx.$$

None of our tools apply. It is not in the table, and there is no inner function whose derivative appears, so substitution does not help. A common first guess is $\frac{1}{x} + C$, but that mixes up differentiation with integration:

$$\int \frac{1}{x} \, dx = \ln |x| + C, \quad \text{while} \quad \frac{d}{dx}(\ln x) = \frac{1}{x}.$$

A computer algebra system reports

$$\int \ln x \, dx = x \ln x - x + C,$$

and we can confirm this with the Product Rule:

$$\frac{d}{dx}(x \ln x - x) = \ln x + x \left(\frac{1}{x} \right) - 1 = \ln x.$$

The appearance of the Product Rule is the clue. Substitution reversed the Chain Rule; to find antiderivatives like this one, we reverse the Product Rule.

If $u(x)$ and $v(x)$ are differentiable,

$$\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}, \quad \text{or in differential form} \quad d(uv) = v \, du + u \, dv.$$

Integrating both sides:

$$uv = \int v \, du + \int u \, dv.$$

Rearranging gives the Integration by Parts formula.

Theorem 13.1 (Integration by Parts)

For continuously differentiable functions u and v on the interval of integration:

$$\int u \, dv = uv - \int v \, du.$$

For definite integrals on $[a, b]$:

$$\int_a^b u \, dv = \left[uv \right]_a^b - \int_a^b v \, du.$$

How to choose u and dv

The formula trades $\int u \, dv$ for $\int v \, du$, so the goal is to make the new integral *easier*:

- choose u to be something that gets simpler when differentiated (like $\ln x$ or a polynomial);
- choose dv to be something you can integrate (like $e^x dx$, $\cos x dx$, or $x^n dx$).

13.2 Examples

Example 13.1. Evaluate $I = \int \ln x \, dx$.

Solution. We cannot integrate $\ln x$, but we can differentiate it, so let $u = \ln x$ and take $dv = dx$:

$$\begin{array}{ll} u = \ln x & dv = dx \\ du = \frac{1}{x} dx & v = x \end{array}$$

Applying the formula:

$$I = x \ln x - \int x \left(\frac{1}{x} \right) dx = x \ln x - \int 1 \, dx = x \ln x - x + C.$$

□

Example 13.2. Evaluate $I = \int x \ln x \, dx$.

Solution. Choose:

$$\begin{array}{ll} u = \ln x & dv = x \, dx \\ du = \frac{1}{x} dx & v = \frac{x^2}{2} \end{array}$$

Applying the formula:

$$\begin{aligned} I &= \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \left(\frac{1}{x} \right) dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx \\ &= \frac{x^2}{2} \ln x - \frac{1}{4} x^2 + C. \end{aligned}$$

□

Example 13.3. Evaluate $I = \int_0^{\pi/2} x \cos x \, dx$.

Solution. Differentiating x makes it simpler, so choose:

$$\begin{array}{ll} u = x & dv = \cos x \, dx \\ du = dx & v = \sin x \end{array}$$

Applying the formula:

$$\begin{aligned} I &= \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} \sin x \, dx \\ &= \frac{\pi}{2} - \left[-\cos x \right]_0^{\pi/2} \\ &= \frac{\pi}{2} + \left(\cos \frac{\pi}{2} - \cos 0 \right) = \frac{\pi}{2} - 1. \end{aligned}$$

□

Example 13.4 (An Integral That Loops Back). Evaluate $J = \int e^x \sin x \, dx$.

Solution. Choose:

$$\begin{array}{ll} u = \sin x & dv = e^x \, dx \\ du = \cos x \, dx & v = e^x \end{array}$$

This gives

$$J = e^x \sin x - \int e^x \cos x \, dx.$$

The new integral is no simpler, so apply integration by parts again, making the same kind of choice:

$$\begin{array}{ll} u = \cos x & dv = e^x \, dx \\ du = -\sin x \, dx & v = e^x \end{array}$$

so

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + J.$$

The original integral J has reappeared. Substituting back:

$$J = e^x \sin x - (e^x \cos x + J) = e^x \sin x - e^x \cos x - J.$$

Now solve for J algebraically:

$$2J = e^x(\sin x - \cos x) \implies J = \frac{e^x}{2}(\sin x - \cos x) + C.$$

□

Lesson 14: Trigonometric Integrals

Goals of this lesson.

- Evaluate integrals of products of powers of sine and cosine: $\int \sin^m x \cos^n x dx$.
- Evaluate integrals of products of powers of tangent and secant: $\int \tan^m x \sec^n x dx$.
- Use Pythagorean and half-angle identities to set up substitutions.
- Compute the classic antiderivatives $\int \sec x dx$ and $\int \sec^3 x dx$.

14.1 Integrals Involving Sine and Cosine

To evaluate integrals of the form

$$\int \sin^m x \cos^n x dx,$$

the idea is to make a u -substitution work. Since $\frac{d}{dx} \sin x = \cos x$, a single leftover factor of $\cos x$ can serve as du when $u = \sin x$ (and similarly with the roles swapped). Everything else must then be written in terms of u , which we do with the identities below.

Trigonometric identities

$$\begin{array}{ll} \sin^2 x + \cos^2 x = 1 & \text{(Pythagorean)} \\ \sin^2 x = \frac{1}{2}(1 - \cos 2x), \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x) & \text{(half-angle)} \\ \sin 2x = 2 \sin x \cos x, \quad \cos 2x = \cos^2 x - \sin^2 x & \text{(double-angle)} \end{array}$$

Strategy for $\int \sin^m x \cos^n x dx$

- **If the power of cosine is odd:** save one $\cos x$ for du , convert the remaining (even) power of cosine to sine using $\cos^2 x = 1 - \sin^2 x$, and let $u = \sin x$.
- **If the power of sine is odd:** save one $\sin x$ for du , convert the rest to cosine using $\sin^2 x = 1 - \cos^2 x$, and let $u = \cos x$.
- **If both powers are even:** substitution does not work. Lower the powers using the half-angle identities.

Example 14.1 (An Odd Power of Cosine). Evaluate $I = \int \sin^2 x \cos^5 x dx$.

Solution. The power of cosine is odd ($n = 5$), so save one factor of $\cos x$ for $du = \cos x dx$ and rewrite the remaining $\cos^4 x$ in terms of sine:

$$\cos^4 x = (\cos^2 x)^2 = (1 - \sin^2 x)^2.$$

Then, with $u = \sin x$ and $du = \cos x \, dx$:

$$\begin{aligned} I &= \int \sin^2 x (1 - \sin^2 x)^2 \cos x \, dx = \int u^2 (1 - u^2)^2 \, du \\ &= \int (u^2 - 2u^4 + u^6) \, du = \frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} + C \\ &= \frac{1}{3} \sin^3 x - \frac{2}{5} \sin^5 x + \frac{1}{7} \sin^7 x + C. \end{aligned}$$

□

Example 14.2 (Both Powers Even). Evaluate $I = \int \sin^2 x \cos^2 x \, dx$.

Solution. Both powers are even, so apply the half-angle identities:

$$I = \int \left(\frac{1 - \cos 2x}{2} \right) \left(\frac{1 + \cos 2x}{2} \right) dx = \frac{1}{4} \int (1 - \cos^2 2x) \, dx = \frac{1}{4} \int \sin^2 2x \, dx.$$

Apply the half-angle identity once more (the angle doubles from $2x$ to $4x$):

$$\sin^2 2x = \frac{1 - \cos 4x}{2}.$$

Therefore

$$I = \frac{1}{8} \int (1 - \cos 4x) \, dx = \frac{1}{8} \left(x - \frac{1}{4} \sin 4x \right) + C = \frac{x}{8} - \frac{1}{32} \sin 4x + C.$$

□

14.2 Integrals Involving Tangent and Secant

For integrals of the form

$$\int \tan^m x \sec^n x \, dx,$$

the same idea applies, now using the derivatives

$$\frac{d}{dx}(\tan x) = \sec^2 x \quad \text{and} \quad \frac{d}{dx}(\sec x) = \sec x \tan x.$$

Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$ gives the identity that links them:

$$\tan^2 x + 1 = \sec^2 x.$$

Strategy for $\int \tan^m x \sec^n x \, dx$

- **If the power of secant is even:** save $\sec^2 x$ for du , convert the rest to tangent using $\sec^2 x = 1 + \tan^2 x$, and let $u = \tan x$.
- **If the power of tangent is odd:** save $\sec x \tan x$ for du , convert the rest to secant using $\tan^2 x = \sec^2 x - 1$, and let $u = \sec x$.

Example 14.3 (Antiderivative of Tangent). Evaluate $\int \tan x \, dx$.

Solution. Write $\tan x = \frac{\sin x}{\cos x}$ and let $u = \cos x$, so $du = -\sin x \, dx$:

$$\int \tan x \, dx = \int \frac{-du}{u} = -\ln |\cos x| + C = \ln |\sec x| + C.$$

□

Example 14.4 (Odd Power of Tangent). Evaluate $\int \tan^3 x \, dx$.

Solution. Split off $\tan^2 x$ and replace it with $\sec^2 x - 1$:

$$\int \tan^3 x \, dx = \int \tan x (\sec^2 x - 1) \, dx = \int \tan x \sec^2 x \, dx - \int \tan x \, dx.$$

For the first integral, let $u = \tan x$, so $du = \sec^2 x \, dx$:

$$\int \tan x \sec^2 x \, dx = \int u \, du = \frac{1}{2} \tan^2 x.$$

Combining with the previous example:

$$\int \tan^3 x \, dx = \frac{1}{2} \tan^2 x - \ln |\sec x| + C.$$

□

Example 14.5 (Even Power of Secant). Evaluate $\int \tan^2 x \sec^4 x \, dx$.

Solution. The power of secant is even ($n = 4$), so save $\sec^2 x$ for du and convert the remaining $\sec^2 x$ to tangent:

$$\int \tan^2 x (1 + \tan^2 x) \sec^2 x \, dx.$$

With $u = \tan x$ and $du = \sec^2 x \, dx$:

$$\int u^2 (1 + u^2) \, du = \frac{u^3}{3} + \frac{u^5}{5} + C = \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C.$$

□

Example 14.6 (Antiderivative of Secant). Evaluate $I = \int \sec x \, dx$.

Solution. This one needs a trick: multiply and divide by $\sec x + \tan x$.

$$I = \int \sec x \cdot \frac{\sec x + \tan x}{\sec x + \tan x} \, dx = \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx.$$

The numerator is exactly the derivative of the denominator. Let $u = \sec x + \tan x$, so $du = (\sec x \tan x + \sec^2 x) \, dx$:

$$I = \int \frac{du}{u} = \ln |u| + C = \ln |\sec x + \tan x| + C.$$

□

Example 14.7 (Secant Cubed via Integration by Parts). Evaluate $I = \int \sec^3 x \, dx$.

Solution. Write $\sec^3 x = \sec x \cdot \sec^2 x$ and integrate by parts (Lesson 13):

$$\begin{array}{ll} u = \sec x & dv = \sec^2 x \, dx \\ du = \sec x \tan x \, dx & v = \tan x \end{array}$$

Then

$$\begin{aligned} I &= \sec x \tan x - \int \sec x \tan^2 x \, dx \\ &= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\ &= \sec x \tan x - I + \ln |\sec x + \tan x|. \end{aligned}$$

As in the $\int e^x \sin x \, dx$ example, the original integral has looped back. Add I to both sides:

$$2I = \sec x \tan x + \ln |\sec x + \tan x| \implies I = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C.$$

□

Lesson 15: Trigonometric Substitution

Goals of this lesson.

- Recognize quadratic radicals as coming from circles or right triangles.
- Use trigonometric substitutions to eliminate square roots from integrands.
- Use the three standard trigonometric substitutions:
 - $\sqrt{a^2 - x^2} \implies x = a \sin \theta$ (or $x = a \cos \theta$)
 - $\sqrt{a^2 + x^2} \implies x = a \tan \theta$
 - $\sqrt{x^2 - a^2} \implies x = a \sec \theta$
- Convert answers back to the original variable using reference right triangles.

15.1 Geometric Motivation and the Three Standard Forms

An ordinary u -substitution handles $\int x\sqrt{a^2 - x^2} dx$, because the factor x is (up to a constant) the derivative of $a^2 - x^2$. Without that extra factor of x , as in $\int \sqrt{a^2 - x^2} dx$, the direct substitution $u = a^2 - x^2$ does not simplify the integral.

The fix comes from geometry. The curve $y = \sqrt{a^2 - x^2}$ is the upper half of the circle

$$x^2 + y^2 = a^2,$$

and points on a circle are naturally described by an angle. Setting $x = a \sin \theta$ turns the *difference* under the radical into a single square, and the square root disappears:

$$\sqrt{a^2 - x^2} = \sqrt{a^2(1 - \sin^2 \theta)} = \sqrt{a^2 \cos^2 \theta} = a \cos \theta.$$

The other two forms work the same way, each using a different Pythagorean identity.

The three standard substitutions

Expression	Substitution	Differential	Identity used
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$dx = a \cos \theta d\theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$dx = a \sec^2 \theta d\theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$dx = a \sec \theta \tan \theta d\theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

Angle Ranges and Signs

Assume $a > 0$. For $x = a \sin \theta$, choose $-\pi/2 \leq \theta \leq \pi/2$, so $\cos \theta \geq 0$. For $x = a \tan \theta$, choose $-\pi/2 < \theta < \pi/2$, so $\sec \theta > 0$. For $x = a \sec \theta$ with $x > a$, choose $0 < \theta < \pi/2$, so $\tan \theta > 0$. On the branch $x < -a$, use $x = -a \sec \theta$ with the same angle range. In the cosine substitutions below, take $0 < t < \pi$, so $\sin t > 0$. Square roots simplify only after choosing these ranges.

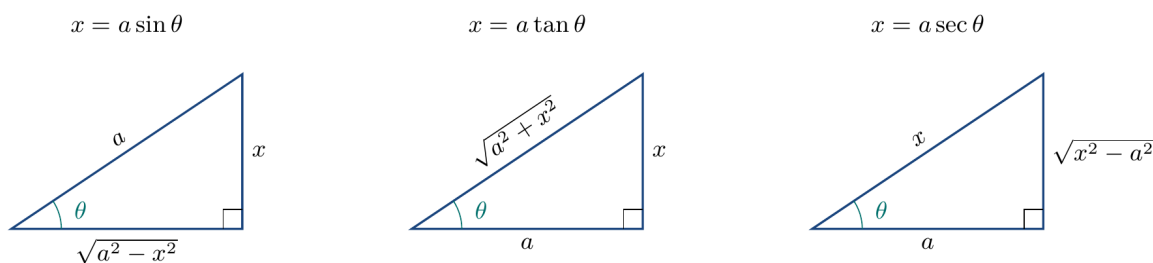


Figure 35: Reference triangles for positive x , with the relevant ratio defining θ . For negative x , use the chosen angle range and signed trigonometric ratios.

15.2 Integrals Involving $\sqrt{a^2 - x^2}$

Example 15.1. Evaluate $I = \int \frac{1}{\sqrt{1 - x^2}} dx$.

Solution. Let $x = \cos t$, so $dx = -\sin t dt$ and

$$\sqrt{1 - x^2} = \sqrt{1 - \cos^2 t} = \sqrt{\sin^2 t} = \sin t.$$

Then

$$I = \int \frac{1}{\sin t} (-\sin t dt) = \int -1 dt = -t + C = -\arccos(x) + C.$$

Note: The standard substitution $x = \sin \theta$ gives $\arcsin(x) + C$ instead. Both answers are correct: they differ only by a constant, since $\arcsin x + \arccos x = \frac{\pi}{2}$. $\square$

Example 15.2. Evaluate $I = \int \frac{x^2}{(1 - x^2)^{3/2}} dx$.

Solution. Let $x = \cos t$, so $dx = -\sin t dt$ and $(1 - x^2)^{3/2} = (\sin^2 t)^{3/2} = \sin^3 t$:

$$I = \int \frac{\cos^2 t}{\sin^3 t} (-\sin t dt) = - \int \frac{\cos^2 t}{\sin^2 t} dt = - \int \cot^2 t dt.$$

Using the identity $\cot^2 t = \csc^2 t - 1$:

$$I = \int (1 - \csc^2 t) dt = t + \cot t + C.$$

To return to x , use $\cos t = x$ and $\sin t = \sqrt{1 - x^2}$ on $0 < t < \pi$. For $x > 0$, these ratios can also be read from a right triangle. Then

$$t = \arccos x, \quad \cot t = \frac{\text{adjacent}}{\text{opposite}} = \frac{x}{\sqrt{1 - x^2}}.$$

Therefore

$$I = \arccos(x) + \frac{x}{\sqrt{1 - x^2}} + C.$$

$\square$

Example 15.3 (A Scaled Variable). Evaluate $I = \int_0^{1/2} \frac{x^2}{\sqrt{1-2x^2}} dx$.

Solution. Rewrite $1 - 2x^2 = 1 - (\sqrt{2}x)^2$, which has the form $1 - (\text{something})^2$. Set

$$\sqrt{2}x = \sin \theta \implies x = \frac{1}{\sqrt{2}} \sin \theta, \quad dx = \frac{1}{\sqrt{2}} \cos \theta d\theta.$$

Transform the limits:

- When $x = 0$: $\sin \theta = 0 \implies \theta = 0$.
- When $x = \frac{1}{2}$: $\sin \theta = \frac{\sqrt{2}}{2} \implies \theta = \frac{\pi}{4}$.

Under this substitution $\sqrt{1 - 2x^2} = \cos \theta$, so

$$\begin{aligned} I &= \int_0^{\pi/4} \frac{\frac{1}{2} \sin^2 \theta}{\cos \theta} \cdot \frac{1}{\sqrt{2}} \cos \theta d\theta = \frac{1}{2\sqrt{2}} \int_0^{\pi/4} \sin^2 \theta d\theta \\ &= \frac{1}{2\sqrt{2}} \int_0^{\pi/4} \frac{1 - \cos 2\theta}{2} d\theta = \frac{1}{4\sqrt{2}} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/4} \\ &= \frac{1}{4\sqrt{2}} \left(\frac{\pi}{4} - \frac{1}{2} \right) = \frac{\pi - 2}{16\sqrt{2}}. \end{aligned}$$

□

15.3 Integrals Involving $a^2 + x^2$

Example 15.4 (A Familiar Integral). Evaluate $I = \int \frac{1}{1+x^2} dx$.

Solution. Let $x = \tan \theta$, so $dx = \sec^2 \theta d\theta$ and $1 + x^2 = \sec^2 \theta$:

$$I = \int \frac{\sec^2 \theta}{\sec^2 \theta} d\theta = \int 1 d\theta = \theta + C = \arctan(x) + C.$$

This recovers the formula from our table of integrals, and explains where it comes from.

□

Example 15.5. Evaluate $I = \int \frac{1}{x^2 \sqrt{4x^2 + 36}} dx$.

Solution. Work first on the branch $x > 0$ with $0 < \theta < \pi/2$. Factor out the constants:

$$\sqrt{4x^2 + 36} = \sqrt{4(x^2 + 9)} = 2\sqrt{x^2 + 9}.$$

Set $x = 3 \tan \theta$, so $dx = 3 \sec^2 \theta d\theta$ and $\sqrt{x^2 + 9} = 3 \sec \theta$:

$$I = \int \frac{3 \sec^2 \theta}{(9 \tan^2 \theta)(2 \cdot 3 \sec \theta)} d\theta = \frac{1}{18} \int \frac{\sec \theta}{\tan^2 \theta} d\theta = \frac{1}{18} \int \frac{\cos \theta}{\sin^2 \theta} d\theta.$$

Let $u = \sin \theta$, so $du = \cos \theta d\theta$:

$$I = \frac{1}{18} \int u^{-2} du = -\frac{1}{18u} + C = -\frac{1}{18} \csc \theta + C.$$

From $\tan \theta = \frac{x}{3}$, draw a right triangle with opposite side x , adjacent side 3, and hypotenuse $\sqrt{x^2 + 9}$:

$$\csc \theta = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{\sqrt{x^2 + 9}}{x}.$$

Therefore

$$I = -\frac{\sqrt{x^2 + 9}}{18x} + C.$$

□

15.4 Integrals Involving $\sqrt{x^2 - a^2}$

Example 15.6. Evaluate $J = \int \frac{1}{(x^2 - 4)^{3/2}} dx$.

Solution. Work first with $x > 2$ and $0 < \theta < \pi/2$. Here $a = 2$. Set $x = 2 \sec \theta$, so $dx = 2 \sec \theta \tan \theta d\theta$ and

$$(x^2 - 4)^{3/2} = (4 \sec^2 \theta - 4)^{3/2} = (4 \tan^2 \theta)^{3/2} = 8 \tan^3 \theta.$$

Then

$$\begin{aligned} J &= \int \frac{2 \sec \theta \tan \theta}{8 \tan^3 \theta} d\theta = \frac{1}{4} \int \frac{\sec \theta}{\tan^2 \theta} d\theta \\ &= \frac{1}{4} \int \frac{\cos \theta}{\sin^2 \theta} d\theta = -\frac{1}{4 \sin \theta} + C = -\frac{1}{4} \csc \theta + C. \end{aligned}$$

Since $\sec \theta = \frac{x}{2}$, draw a right triangle with hypotenuse x , adjacent side 2, and opposite side $\sqrt{x^2 - 4}$:

$$\csc \theta = \frac{x}{\sqrt{x^2 - 4}}.$$

Thus

$$J = -\frac{x}{4\sqrt{x^2 - 4}} + C.$$

Differentiating verifies the same formula on $x < -2$, with an independent constant on that interval.

□

15.5 Geometric Application: Area of a Circle

Example 15.7. Find the area enclosed by a circle of radius r .

Solution. The circle has equation $x^2 + y^2 = r^2$. By symmetry, the total area is four times the area in the first quadrant:

$$A = 4 \int_0^r \sqrt{r^2 - x^2} dx.$$

Let $x = r \sin \theta$, so $dx = r \cos \theta d\theta$ and $\sqrt{r^2 - x^2} = r \cos \theta$. The limits become:

- $x = 0 \implies \theta = 0$
- $x = r \implies \theta = \frac{\pi}{2}$

Then

$$\begin{aligned} A &= 4 \int_0^{\pi/2} (r \cos \theta)(r \cos \theta) d\theta = 4r^2 \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta \\ &= 2r^2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\pi/2} = 2r^2 \cdot \frac{\pi}{2} = \pi r^2. \end{aligned}$$

Calculus confirms the familiar formula.

□

Lesson 16: Integration by Partial Fractions (Part 1: Linear Factors)

16.1 Goals

- Understand Partial Fraction Decomposition (PFD) as fraction addition run in reverse.
- Identify the prerequisite steps for PFD: ensuring proper degree and complete denominator factorization.
- Formulate and solve decompositions for Case I (distinct linear factors).
- Formulate and solve decompositions for Case II (repeated linear factors).
- Integrate resulting rational functions into logarithmic and reciprocal power forms.

16.2 Motivation: Addition in Reverse

Consider the elementary addition of two rational expressions:

$$\frac{2}{x-2} - \frac{3}{x+2} = \frac{2(x+2) - 3(x-2)}{(x-2)(x+2)} = \frac{2x+4-3x+6}{x^2-4} = \frac{10-x}{x^2-4}.$$

Now look at these expressions from the perspective of integration:

$$\int \frac{10-x}{x^2-4} dx \quad \longleftrightarrow \quad \int \left(\frac{2}{x-2} - \frac{3}{x+2} \right) dx.$$

Integrating the single combined fraction is not obvious. On the other hand, integrating the separated sum of simple fractions is immediate:

$$\int \left(\frac{2}{x-2} - \frac{3}{x+2} \right) dx = 2 \ln |x-2| - 3 \ln |x+2| + C.$$

The technique of **Partial Fraction Decomposition** takes a complicated rational function and decomposes it back into simple constituent fractions that we know how to integrate.

16.3 Prerequisites Before Decomposing

Before setting up any partial fraction decomposition for a rational function $\frac{P(x)}{Q(x)}$, verify two criteria:

1. **Proper Rational Function:** The degree of the numerator must be strictly less than the degree of the denominator:

$$\deg P(x) < \deg Q(x).$$

If $\deg P(x) \geq \deg Q(x)$, perform polynomial long division first:

$$\frac{P(x)}{Q(x)} = (\text{Polynomial}) + \frac{R(x)}{Q(x)}, \quad \text{where } \deg R(x) < \deg Q(x).$$

Apply partial fractions only to the remainder term $\frac{R(x)}{Q(x)}$.

2. **Complete Denominator Factorization:** Factor the denominator $Q(x)$ completely into linear factors and irreducible quadratic factors (e.g., $x^3 + x = x(x^2 + 1)$).

16.4 Case I: Distinct Linear Factors

Suppose the denominator factors completely into non-repeated linear factors:

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_kx + b_k).$$

Then the rational function can be written in the form:

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \cdots + \frac{A_k}{a_kx + b_k},$$

where $A_1, A_2, \dots, A_k$ are constants to be determined.

Example 16.1. Evaluate:

$$\int \frac{10 - x}{x^2 - 4} dx.$$

Solution. **Step 1: Check degree and factor denominator.**

The fraction is proper since $\deg(10 - x) = 1 < 2 = \deg(x^2 - 4)$. Factoring gives $x^2 - 4 = (x - 2)(x + 2)$.

Step 2: Set up the decomposition form.

$$\frac{10 - x}{(x - 2)(x + 2)} = \frac{A}{x - 2} + \frac{B}{x + 2}.$$

Step 3: Clear denominators.

Multiply both sides by $(x - 2)(x + 2)$:

$$10 - x = A(x + 2) + B(x - 2).$$

Step 4: Solve for coefficients.

We can solve for A and B by evaluating at the roots of the linear factors:

- Let $x = 2$:

$$10 - 2 = A(2 + 2) + B(0) \implies 8 = 4A \implies A = 2.$$

- Let $x = -2$:

$$10 - (-2) = A(0) + B(-2 - 2) \implies 12 = -4B \implies B = -3.$$

Step 5: Integrate.

$$\begin{aligned} \int \frac{10 - x}{x^2 - 4} dx &= \int \left(\frac{2}{x - 2} - \frac{3}{x + 2} \right) dx \\ &= 2 \ln |x - 2| - 3 \ln |x + 2| + C. \end{aligned}$$

□

16.5 Case II: Repeated Linear Factors

If a linear factor $(ax + b)$ is repeated, as in $\frac{x}{(x-1)^2}$, attempting a naive form like $\frac{A}{x-1} + \frac{B}{x-1}$ fails because combining terms gives $\frac{A+B}{x-1}$, which cannot produce a denominator with power 2.

Instead, every repeated factor requires a term for each successive power up to its multiplicity:

$$\frac{x}{(x - 1)^2} = \frac{1}{x - 1} + \frac{1}{(x - 1)^2}.$$

Theorem 16.1 (Rule for Repeated Linear Factors)

If a linear factor $(ax + b)$ appears with multiplicity r in the denominator $Q(x)$, it contributes a sum of r partial fractions:

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_r}{(ax + b)^r}.$$

Example 16.2 (Setting Up the PFD Form). Determine the partial fraction decomposition form for:

$$\frac{x^3 - x + 1}{x^2(x - 1)^3}.$$

Solution. The degree of the numerator is 3, and the degree of the denominator is $2 + 3 = 5$. Since $3 < 5$, the fraction is proper.

- The factor x is repeated twice ($r = 2$), contributing:

$$\frac{A_1}{x} + \frac{A_2}{x^2}.$$

- The factor $(x - 1)$ is repeated three times ($r = 3$), contributing:

$$\frac{B_1}{x - 1} + \frac{B_2}{(x - 1)^2} + \frac{B_3}{(x - 1)^3}.$$

Combining these gives the complete decomposition form:

$$\frac{x^3 - x + 1}{x^2(x - 1)^3} = \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{B_1}{x - 1} + \frac{B_2}{(x - 1)^2} + \frac{B_3}{(x - 1)^3}.$$

□

Example 16.3 (Evaluating an Integral with Repeated and Distinct Linear Factors).

Evaluate:

$$I = \int \frac{4x^2 - 21x}{(x - 3)^2(2x + 3)} dx.$$

Solution. **Step 1: Check degree and set up the decomposition.**

The numerator degree is 2, and the denominator degree is $2 + 1 = 3$. The fraction is proper.

Set up the decomposition:

$$\frac{4x^2 - 21x}{(x - 3)^2(2x + 3)} = \frac{A}{x - 3} + \frac{B}{(x - 3)^2} + \frac{C}{2x + 3}.$$

Step 2: Clear denominators.

Multiply both sides by $(x - 3)^2(2x + 3)$:

$$4x^2 - 21x = A(x - 3)(2x + 3) + B(2x + 3) + C(x - 3)^2.$$

Step 3: Solve for coefficients.

Substitute strategic values of x :

- Let $x = 3$:

$$\begin{aligned} 4(3)^2 - 21(3) &= A(0) + B(2(3) + 3) + C(0)^2 \\ 36 - 63 &= B(9) \\ -27 &= 9B \implies B = -3. \end{aligned}$$

- Let $x = -\frac{3}{2}$:

$$\begin{aligned} 4\left(-\frac{3}{2}\right)^2 - 21\left(-\frac{3}{2}\right) &= A(0) + B(0) + C\left(-\frac{3}{2} - 3\right)^2 \\ 4\left(\frac{9}{4}\right) + \frac{63}{2} &= C\left(-\frac{9}{2}\right)^2 \\ 9 + \frac{63}{2} &= C\left(\frac{81}{4}\right) \\ \frac{81}{2} &= \frac{81}{4}C \implies C = 2. \end{aligned}$$

- To find A , compare coefficients of x^2 :

$$\begin{aligned} 4x^2 &= A(2x^2) + C(x^2) \\ 4 &= 2A + C. \end{aligned}$$

Substitute $C = 2$:

$$4 = 2A + 2 \implies 2A = 2 \implies A = 1.$$

Step 4: Integrate the decomposed terms.

Substituting $A = 1$, $B = -3$, and $C = 2$:

$$\begin{aligned} I &= \int \left(\frac{1}{x-3} - \frac{3}{(x-3)^2} + \frac{2}{2x+3} \right) dx \\ &= \int \frac{1}{x-3} dx - 3 \int (x-3)^{-2} dx + \int \frac{2}{2x+3} dx \\ &= \ln|x-3| - 3 \left(-\frac{1}{x-3} \right) + \ln|2x+3| + C \\ &= \ln|x-3| + \frac{3}{x-3} + \ln|2x+3| + C. \end{aligned}$$

Using logarithm rules, this can also be written as:

$$I = \ln|(x-3)(2x+3)| + \frac{3}{x-3} + C.$$

□

Lesson 17: Integration by Partial Fractions (Part 2: Irreducible Quadratic Factors)

17.1 Goals

- Decide whether a quadratic factor is irreducible using the discriminant.
- Formulate and solve decompositions for Case III (distinct irreducible quadratic factors).
- Formulate and solve decompositions for Case IV (repeated irreducible quadratic factors).
- Integrate terms of the form $\frac{Ax + B}{ax^2 + bx + c}$ using u -substitution, completing the square, and the arctangent formula.
- Combine all four cases in a single decomposition.

17.2 Motivation: Factors That Do Not Split

In Part 1, every denominator factored into linear pieces. But consider

$$\int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx.$$

The factor $x^2 + 4$ has no real roots, so it cannot be factored further over the real numbers. A term like $\frac{A}{x^2+4}$ is not flexible enough to capture all possible numerators, so we must allow a *linear* numerator over each quadratic factor.

17.3 Recall: Irreducible Quadratics

A quadratic $ax^2 + bx + c$ is **irreducible** (over the real numbers) exactly when its discriminant is negative:

$$b^2 - 4ac < 0.$$

For example, $x^2 + 4$ ($0 - 16 < 0$) and $x^2 + x + 1$ ($1 - 4 < 0$) are irreducible, while $x^2 - 4$ ($0 + 16 > 0$) is not. **Always factor completely first**; only then treat what remains as irreducible.

17.4 Case III: Distinct Irreducible Quadratic Factors

Theorem 17.1 (Rule for Irreducible Quadratic Factors)

Each non-repeated irreducible quadratic factor $ax^2 + bx + c$ in the denominator $Q(x)$ contributes one partial fraction with a *linear* numerator:

$$\frac{Ax + B}{ax^2 + bx + c}.$$

Example 17.1 (Linear and Quadratic Factors). Evaluate:

$$\int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx.$$

Solution. **Step 1: Check degree and factorization.**

The numerator has degree $2 < 3$, so the fraction is proper. The factor $x^2 + 4$ has discriminant $-16 < 0$, so it is irreducible.

Step 2: Set up the decomposition.

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}.$$

Step 3: Clear denominators.

$$2x^2 - x + 4 = A(x^2 + 4) + (Bx + C)x.$$

Step 4: Solve for coefficients.

- Let $x = 0$: $4 = 4A \implies A = 1$.
- Expand the right side and collect powers of x :

$$2x^2 - x + 4 = (A + B)x^2 + Cx + 4A.$$

- Compare x^2 coefficients: $2 = A + B \implies B = 1$.
- Compare x coefficients: $-1 = C$.

Thus

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{1}{x} + \frac{x - 1}{x^2 + 4}.$$

Step 5: Integrate. Split the quadratic term into a piece handled by u -substitution and a piece handled by the arctangent formula:

$$\int \frac{x - 1}{x^2 + 4} dx = \int \frac{x}{x^2 + 4} dx - \int \frac{1}{x^2 + 4} dx.$$

For the first integral, let $u = x^2 + 4$, so $du = 2x dx$. Therefore

$$\begin{aligned} \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx &= \int \frac{1}{x} dx + \int \frac{x}{x^2 + 4} dx - \int \frac{1}{x^2 + 4} dx \\ &= \ln|x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) + C. \end{aligned}$$

□

17.5 Integration Toolkit for Quadratic Terms

Every integral arising from Cases III and IV reduces to combinations of the following.

- **Log part** (numerator is the derivative of the denominator):

$$\int \frac{2ax + b}{ax^2 + bx + c} dx = \ln|ax^2 + bx + c| + C.$$

- **Arctangent part:**

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C.$$

- **Completing the square:** if the quadratic is not of the form $x^2 + a^2$, rewrite it as $(x+h)^2 + k^2$ before using the arctangent formula.

Strategy for $\int \frac{Ax + B}{ax^2 + bx + c} dx$:

1. Rewrite the numerator as (constant) $\cdot$ (derivative of the denominator) + (constant).
2. The first piece gives a logarithm.
3. Complete the square in the denominator of the second piece to obtain an arctangent.

Example 17.2 (Completing the Square). Evaluate:

$$\int \frac{3x^2 + 3x + 2}{(x-1)(x^2 + 2x + 5)} dx.$$

Solution. **Step 1: Check degree and factorization.**

The numerator has degree $2 < 3$. The discriminant of $x^2 + 2x + 5$ is $4 - 20 = -16 < 0$, so it is irreducible.

Step 2: Set up and clear denominators.

$$\frac{3x^2 + 3x + 2}{(x-1)(x^2 + 2x + 5)} = \frac{A}{x-1} + \frac{Bx + C}{x^2 + 2x + 5},$$

$$3x^2 + 3x + 2 = A(x^2 + 2x + 5) + (Bx + C)(x-1).$$

Step 3: Solve for coefficients.

- Let $x = 1$: $3 + 3 + 2 = A(8) \implies A = 1$.
- Compare x^2 coefficients: $3 = A + B \implies B = 2$.
- Compare constant terms: $2 = 5A - C \implies C = 3$.

(Check with the x coefficients: $3 = 2A + C - B = 2 + 3 - 2 = 3$. $\checkmark$)

Step 4: Integrate.

$$\int \left(\frac{1}{x-1} + \frac{2x+3}{x^2+2x+5} \right) dx.$$

The derivative of $x^2 + 2x + 5$ is $2x + 2$, so write $2x + 3 = (2x + 2) + 1$:

$$\int \frac{2x+3}{x^2+2x+5} dx = \int \frac{2x+2}{x^2+2x+5} dx + \int \frac{1}{x^2+2x+5} dx.$$

Complete the square: $x^2 + 2x + 5 = (x+1)^2 + 4$. Then with $u = x+1$,

$$\int \frac{1}{(x+1)^2 + 4} dx = \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C.$$

Combining,

$$\int \frac{3x^2 + 3x + 2}{(x-1)(x^2 + 2x + 5)} dx = \ln|x-1| + \ln(x^2 + 2x + 5) + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C.$$

□

17.6 Case IV: Repeated Irreducible Quadratic Factors

Theorem 17.2 (Rule for Repeated Irreducible Quadratic Factors)

If an irreducible quadratic $ax^2 + bx + c$ appears with multiplicity r in the denominator $Q(x)$, it contributes a sum of r partial fractions, each with a linear numerator:

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}.$$

Example 17.3 (Setting Up the PFD Form with All Four Cases). Determine the partial fraction decomposition form for:

$$\frac{x^2 + 3}{x(x - 2)(x^2 + x + 1)^2}.$$

Solution. The numerator has degree 2 and the denominator has degree $1 + 1 + 4 = 6$, so the fraction is proper. The discriminant of $x^2 + x + 1$ is $1 - 4 = -3 < 0$, so it is irreducible.

- The linear factor x (multiplicity 1) contributes $\frac{A}{x}$.
- The linear factor $x - 2$ (multiplicity 1) contributes $\frac{B}{x - 2}$.
- The irreducible quadratic $x^2 + x + 1$ (multiplicity 2) contributes

$$\frac{C_1x + D_1}{x^2 + x + 1} + \frac{C_2x + D_2}{(x^2 + x + 1)^2}.$$

The complete form is

$$\frac{x^2 + 3}{x(x - 2)(x^2 + x + 1)^2} = \frac{A}{x} + \frac{B}{x - 2} + \frac{C_1x + D_1}{x^2 + x + 1} + \frac{C_2x + D_2}{(x^2 + x + 1)^2}.$$

This decomposition has 6 unknowns, matching the degree of the denominator. (In general, the number of unknowns always equals $\deg Q(x)$.) $\square$

17.7 An Extra Integral: Powers of $x^2 + a^2$

Repeated quadratic factors produce integrals of the form $\int \frac{1}{(x^2 + a^2)^2} dx$. These are handled with the trigonometric substitution $x = a \tan \theta$ from Section 7.3. For $a = 1$: with $x = \tan \theta$ and $dx = \sec^2 \theta d\theta$,

$$\int \frac{1}{(x^2 + 1)^2} dx = \int \frac{\sec^2 \theta}{\sec^4 \theta} d\theta = \int \cos^2 \theta d\theta = \frac{\theta}{2} + \frac{\sin \theta \cos \theta}{2} + C.$$

Converting back ($\theta = \arctan x$, $\sin \theta \cos \theta = \frac{x}{x^2 + 1}$):

$$\boxed{\int \frac{1}{(x^2 + 1)^2} dx = \frac{x}{2(x^2 + 1)} + \frac{1}{2} \arctan x + C.}$$

Also note that a term of the form $\frac{x}{(x^2 + 1)^2}$ is handled by the substitution $u = x^2 + 1$:

$$\int \frac{x}{(x^2 + 1)^2} dx = -\frac{1}{2(x^2 + 1)} + C.$$

Example 17.4 (Repeated Quadratic Factor). Evaluate:

$$I = \int \frac{x^3 + 2x^2 + 4x + 6}{(x^2 + 1)^2} dx.$$

Solution. **Step 1: Check degree and set up the decomposition.**

The numerator has degree $3 < 4$, and $x^2 + 1$ is irreducible. Set

$$\frac{x^3 + 2x^2 + 4x + 6}{(x^2 + 1)^2} = \frac{Ax + B}{x^2 + 1} + \frac{Cx + D}{(x^2 + 1)^2}.$$

Step 2: Clear denominators.

$$x^3 + 2x^2 + 4x + 6 = (Ax + B)(x^2 + 1) + Cx + D.$$

Expanding the right side:

$$x^3 + 2x^2 + 4x + 6 = Ax^3 + Bx^2 + (A + C)x + (B + D).$$

Step 3: Solve for coefficients.

Since $x^2 + 1$ has no real roots, we cannot use strategic values to isolate the unknowns; we compare coefficients instead:

- x^3 : $A = 1$.
- x^2 : $B = 2$.
- x^1 : $A + C = 4 \implies C = 3$.
- x^0 : $B + D = 6 \implies D = 4$.

Thus

$$\frac{x^3 + 2x^2 + 4x + 6}{(x^2 + 1)^2} = \frac{x + 2}{x^2 + 1} + \frac{3x + 4}{(x^2 + 1)^2}.$$

Step 4: Integrate each piece.

$$\begin{aligned} \int \frac{x + 2}{x^2 + 1} dx &= \frac{1}{2} \ln(x^2 + 1) + 2 \arctan x, \\ \int \frac{3x}{(x^2 + 1)^2} dx &= -\frac{3}{2(x^2 + 1)}, \\ \int \frac{4}{(x^2 + 1)^2} dx &= \frac{2x}{x^2 + 1} + 2 \arctan x. \end{aligned}$$

Step 5: Combine.

$$\begin{aligned} I &= \frac{1}{2} \ln(x^2 + 1) + 4 \arctan x + \frac{2x}{x^2 + 1} - \frac{3}{2(x^2 + 1)} + C \\ &= \frac{1}{2} \ln(x^2 + 1) + 4 \arctan x + \frac{4x - 3}{2(x^2 + 1)} + C. \end{aligned}$$

□

17.8 Summary of the Four Cases

Factor in $Q(x)$	Contribution to the decomposition
$(ax + b)$	$\frac{A}{ax + b}$
$(ax + b)^r$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_r}{(ax + b)^r}$
$(ax^2 + bx + c)$, irreducible	$\frac{Ax + B}{ax^2 + bx + c}$
$(ax^2 + bx + c)^r$, irreducible	$\frac{A_1x + B_1}{ax^2 + bx + c} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$

17.9 Common Mistakes

- Using a constant numerator $\frac{A}{x^2+4}$ instead of $\frac{Ax+B}{x^2+4}$.
- Failing to check the discriminant: x^2-4 is *not* irreducible and must be factored as $(x-2)(x+2)$.
- Forgetting the lower powers of a repeated quadratic (writing only $\frac{Ax+B}{(x^2+1)^2}$).
- Forgetting to perform long division when $\deg P \geq \deg Q$.
- Writing $\int \frac{1}{x^2+4} dx = \frac{1}{2} \ln |x^2 + 4|$ instead of using the arctangent formula.

Lesson 18: Strategies for Integration

18.1 Goals

- Systematize the five core techniques of integration: Substitution, Integration by Parts, Trigonometric Integrals, Trigonometric Substitution, and Partial Fractions.
- Master the technique of completing the square to transform irreducible quadratic denominators into standard arctangent and logarithmic forms.
- Synthesize multiple techniques in multistep integrals (e.g., polynomial long division followed by partial fractions, or integration by parts combined with algebraic tricks).

18.2 The Toolkit: Five Foundational Techniques

When confronted with an arbitrary integral, work systematically through the following catalog of strategies:

1. **Direct Substitution (u -substitution):** Look for an internal expression $u = g(x)$ whose derivative $g'(x)$ appears as a factor in the integrand:

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

2. **Integration by Parts:** Useful for products of different classes of functions (polynomial, logarithmic, exponential, trigonometric) or single inverse/logarithmic functions:

$$\int u dv = uv - \int v du.$$

3. **Trigonometric Integrals:**

- For $\int \sin^m x \cos^n x dx$:
 - If the power of cosine is odd ($n = 2k + 1$), save $\cos x dx$ for du and convert the rest via $\cos^2 x = 1 - \sin^2 x$.
 - If the power of sine is odd ($m = 2k + 1$), save $\sin x dx$ for du and convert the rest via $\sin^2 x = 1 - \cos^2 x$.
 - If both powers are even, reduce powers using the half-angle formulas:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}.$$

- For $\int \tan^m x \sec^n x dx$, utilize $\tan^2 x + 1 = \sec^2 x$ paired with $d(\tan x) = \sec^2 x dx$ or $d(\sec x) = \sec x \tan x dx$.

4. **Trigonometric Substitution:** Match quadratic radicals with Pythagorean identities:

$$\begin{aligned} \sqrt{a^2 - x^2} &\implies x = a \sin \theta & (1 - \sin^2 \theta = \cos^2 \theta) \\ \sqrt{a^2 + x^2} &\implies x = a \tan \theta & (1 + \tan^2 \theta = \sec^2 \theta) \\ \sqrt{x^2 - a^2} &\implies x = a \sec \theta & (\sec^2 \theta - 1 = \tan^2 \theta) \end{aligned}$$

5. **Partial Fraction Decomposition (PFD):** For rational functions $\frac{P(x)}{Q(x)}$:

- Ensure the rational expression is proper ($\deg P < \deg Q$); if not, perform polynomial long division first.
- Factor $Q(x)$ completely.
- A linear factor $(ax + b)^r$ contributes:

$$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \cdots + \frac{A_r}{(ax + b)^r}.$$

- An irreducible quadratic factor $(ax^2 + bx + c)^r$ ($b^2 - 4ac < 0$) contributes:

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}.$$

18.3 Completing the Square

When an irreducible quadratic expression $ax^2 + bx + c$ appears under a radical or in a denominator, completing the square expresses it in the form $a(x - h)^2 + d$, with $d = (4ac - b^2)/(4a)$. For a positive irreducible quadratic ($a > 0$), $d > 0$ and can be written as k^2 . This typically enables an immediate arctangent or inverse trigonometric substitution.

Theorem 18.1 (Completing the Square Algorithm)

For $x^2 + bx + c$ (where $a = 1$):

1. Group the variable terms: $(x^2 + bx) + c$.
2. Halve the linear coefficient b and square it: $(\frac{b}{2})^2$.
3. Add and subtract this term:

$$x^2 + bx + c = \left(x^2 + bx + \frac{b^2}{4}\right) + \left(c - \frac{b^2}{4}\right) = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right).$$

When $a \neq 1$, factor out a from the first two terms before completing the square:

$$ax^2 + bx + c = a \left(x + \frac{b}{2a}\right)^2 + \left(\frac{4ac - b^2}{4a}\right).$$

Example 18.1 (Introductory: Standard Arctangent Form). Evaluate:

$$I = \int \frac{1}{x^2 + 2x + 5} dx.$$

Solution. Complete the square on the denominator:

$$x^2 + 2x + 5 = (x^2 + 2x + 1) + 5 - 1 = (x + 1)^2 + 4 = (x + 1)^2 + 2^2.$$

Recall the standard antiderivative $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$. With $u = x + 1$, $du = dx$, and $a = 2$:

$$I = \int \frac{dx}{(x + 1)^2 + 2^2} = \frac{1}{2} \arctan\left(\frac{x + 1}{2}\right) + C.$$

□

18.4 Worked Examples

Example 18.2 (Partial Fractions with an Irreducible Quadratic). Evaluate:

$$I = \int \frac{3x^2 + 3x + 2}{(x-1)(x^2 + 2x + 5)} dx.$$

Solution. **Step 1: Set up the partial fraction decomposition.**

The fraction is proper ($\deg = 2 < 3$). The quadratic $x^2 + 2x + 5$ has discriminant $2^2 - 4(1)(5) = -16 < 0$, making it irreducible over the real numbers.

$$\frac{3x^2 + 3x + 2}{(x-1)(x^2 + 2x + 5)} = \frac{A}{x-1} + \frac{Bx + C}{x^2 + 2x + 5}.$$

Step 2: Clear denominators and solve for constants.

$$3x^2 + 3x + 2 = A(x^2 + 2x + 5) + (Bx + C)(x-1).$$

- Let $x = 1$:

$$3(1)^2 + 3(1) + 2 = A(1 + 2 + 5) + 0 \implies 8 = 8A \implies A = 1.$$

- Let $x = 0$:

$$2 = 1(5) + C(-1) \implies 2 = 5 - C \implies C = 3.$$

- Compare coefficients of x^2 :

$$3 = A + B \implies 3 = 1 + B \implies B = 2.$$

Step 3: Integrate each component.

$$I = \int \frac{1}{x-1} dx + \int \frac{2x+3}{x^2+2x+5} dx.$$

The first integral is $\ln|x-1|$. For the second integral, the derivative of the denominator is $2x+2$. Split the numerator as $(2x+2) + 1$:

$$\begin{aligned} \int \frac{2x+3}{x^2+2x+5} dx &= \int \frac{2x+2}{x^2+2x+5} dx + \int \frac{1}{(x+1)^2+4} dx \\ &= \ln(x^2+2x+5) + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right). \end{aligned}$$

Combining all parts:

$$I = \ln|x-1| + \ln(x^2+2x+5) + \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + C.$$

□

Example 18.3 (Integration by Parts and an Inverse Trigonometric Function). Evaluate

$$\int \frac{x \arcsin x}{\sqrt{1-x^2}} dx \text{ for } -1 < x < 1.$$

Solution. Choose $u = \arcsin x$ and $dv = x/\sqrt{1-x^2} dx$. Then

$$du = \frac{dx}{\sqrt{1-x^2}}, \quad v = -\sqrt{1-x^2}.$$

Integration by parts gives

$$\int \frac{x \arcsin x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} \arcsin x + \int 1 dx = \boxed{x - \sqrt{1-x^2} \arcsin x + C}.$$

The derivative of the final answer recovers the integrand. □

Example 18.4 (Integration by Parts with Power/Logarithm). Evaluate:

$$I = \int x^2 \ln(x^2 + 1) dx.$$

Solution. **Step 1: Integration by Parts.**

Choose:

$$\begin{aligned} u = \ln(x^2 + 1) &\implies du = \frac{2x}{x^2 + 1} dx, \\ dv = x^2 dx &\implies v = \frac{x^3}{3}. \end{aligned}$$

Then:

$$I = \frac{x^3}{3} \ln(x^2 + 1) - \frac{2}{3} \int \frac{x^4}{x^2 + 1} dx.$$

Step 2: Algebraic division on the remaining integral.

Divide x^4 by $x^2 + 1$ using polynomial division or by adding and subtracting terms:

$$\frac{x^4}{x^2 + 1} = \frac{(x^4 - 1) + 1}{x^2 + 1} = \frac{(x^2 - 1)(x^2 + 1) + 1}{x^2 + 1} = x^2 - 1 + \frac{1}{x^2 + 1}.$$

Integrating term-by-term:

$$\int \left(x^2 - 1 + \frac{1}{x^2 + 1} \right) dx = \frac{x^3}{3} - x + \arctan x.$$

Step 3: Combine all terms.

$$I = \frac{x^3}{3} \ln(x^2 + 1) - \frac{2}{3} \left(\frac{x^3}{3} - x + \arctan x \right) + C = \frac{x^3}{3} \ln(x^2 + 1) - \frac{2x^3}{9} + \frac{2x}{3} - \frac{2}{3} \arctan x + C.$$

□

Example 18.5 (Improper Degree and Repeated Quadratic Factors). Evaluate:

$$I = \int \frac{x^4}{(x^2 + 4)^2} dx.$$

Solution. **Step 1: Simplify via algebraic manipulation / long division.**

Notice that the degree of the numerator is 4, which equals the degree of the denominator $(x^2 + 4)^2 = x^4 + 8x^2 + 16$. Rewrite the numerator in terms of $x^2 + 4$:

$$x^4 = (x^2 + 4 - 4)^2 = (x^2 + 4)^2 - 8(x^2 + 4) + 16.$$

Dividing by $(x^2 + 4)^2$:

$$\frac{x^4}{(x^2 + 4)^2} = 1 - \frac{8}{x^2 + 4} + \frac{16}{(x^2 + 4)^2}.$$

Step 2: Trigonometric substitution.

We evaluate the three integrals individually:

$$\begin{aligned} \int 1 \, dx &= x, \\ \int \frac{8}{x^2 + 4} \, dx &= 8 \left(\frac{1}{2} \arctan \left(\frac{x}{2} \right) \right) = 4 \arctan \left(\frac{x}{2} \right). \end{aligned}$$

For $\int \frac{16}{(x^2 + 4)^2} \, dx$, let $x = 2 \tan \theta$, so $dx = 2 \sec^2 \theta \, d\theta$ and $x^2 + 4 = 4 \sec^2 \theta$:

$$\begin{aligned} \int \frac{16}{16 \sec^4 \theta} (2 \sec^2 \theta \, d\theta) &= 2 \int \frac{1}{\sec^2 \theta} \, d\theta = 2 \int \cos^2 \theta \, d\theta \\ &= 2 \int \frac{1 + \cos 2\theta}{2} \, d\theta = \theta + \frac{1}{2} \sin 2\theta \\ &= \theta + \sin \theta \cos \theta. \end{aligned}$$

Using a reference triangle with $\tan \theta = \frac{x}{2}$: $\sin \theta = \frac{x}{\sqrt{x^2 + 4}}$ and $\cos \theta = \frac{2}{\sqrt{x^2 + 4}}$, so:

$$\theta + \sin \theta \cos \theta = \arctan \left(\frac{x}{2} \right) + \frac{2x}{x^2 + 4}.$$

Step 3: Combine all terms.

$$\begin{aligned} I &= x - 4 \arctan \left(\frac{x}{2} \right) + \left[\arctan \left(\frac{x}{2} \right) + \frac{2x}{x^2 + 4} \right] + C \\ &= x - 3 \arctan \left(\frac{x}{2} \right) + \frac{2x}{x^2 + 4} + C. \end{aligned}$$

□

Lesson 19: Improper Integrals

Goals of this lesson.

- Recognize infinite intervals and unbounded integrands.
- Define improper integrals using limits and decide convergence.
- Split an integral at every singularity and every infinite end.

19.1 Why the Usual Definition Needs a Limit

Our definite integral was defined for a continuous function on a closed, bounded interval. What changes if the interval extends forever, or the graph becomes unbounded near an endpoint? First integrate over an ordinary interval, then let its endpoint approach the troublesome place.

Definition 19.1 (Infinite Intervals)

If f is continuous on $[a, \infty)$, define

$$\int_a^\infty f(x) dx = \lim_{b \rightarrow \infty} \int_a^b f(x) dx.$$

Similarly, $\int_{-\infty}^b f(x) dx = \lim_{a \rightarrow -\infty} \int_a^b f(x) dx$. The integral **converges** when the defining limit is a finite real number. Otherwise it **diverges**.

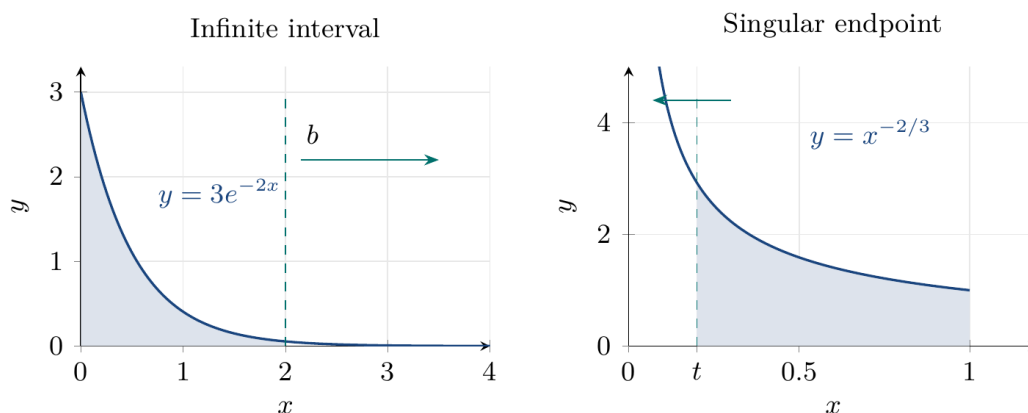


Figure 36: An infinite interval and an unbounded integrand. In each case, begin with a truncated region and take a limit.

Example 19.1 (A Finite Area on an Infinite Interval). Evaluate $\int_0^\infty 3e^{-2x} dx$.

Solution.

$$\lim_{b \rightarrow \infty} \left[-\frac{3}{2} e^{-2x} \right]_0^b = \lim_{b \rightarrow \infty} \frac{3}{2} (1 - e^{-2b}) = \boxed{\frac{3}{2}}.$$

The region is unbounded, but its area is finite. □

Example 19.2 (Both Ends Are Infinite). Evaluate $\int_{-\infty}^{\infty} \frac{dx}{x^2 + 9}$.

Solution. Split at 0. The two limits must converge separately:

$$\begin{aligned}\int_{-\infty}^0 \frac{dx}{x^2 + 9} &= \lim_{a \rightarrow -\infty} \left[\frac{1}{3} \arctan \frac{x}{3} \right]_a^0 = \frac{\pi}{6}, \\ \int_0^{\infty} \frac{dx}{x^2 + 9} &= \lim_{b \rightarrow \infty} \left[\frac{1}{3} \arctan \frac{x}{3} \right]_0^b = \frac{\pi}{6}.\end{aligned}$$

Thus the integral is $\boxed{\pi/3}$. □

19.2 Unbounded Integrands on Finite Intervals

Definition 19.2 (Singular Endpoints)

If f is continuous on $(a, b]$ and unbounded near a , define

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx.$$

If the singular endpoint is b , use $\lim_{t \rightarrow b^-} \int_a^t f(x) dx$. If there is a singularity at $c \in (a, b)$, split into $\int_a^c f(x) dx + \int_c^b f(x) dx$. Both one-sided integrals must converge.

Example 19.3 (An Integrable Singularity). Evaluate $\int_0^8 x^{-2/3} dx$.

Solution. The integrand is unbounded at 0, so

$$\int_0^8 x^{-2/3} dx = \lim_{t \rightarrow 0^+} [3x^{1/3}]_t^8 = \lim_{t \rightarrow 0^+} (6 - 3t^{1/3}) = \boxed{6}.$$

□

Example 19.4 (Cancellation Does Not Repair Divergence). Does $\int_{-2}^3 \frac{dx}{x}$ converge?

Solution. We must split at 0. On the left,

$$\lim_{t \rightarrow 0^-} \int_{-2}^t \frac{dx}{x} = \lim_{t \rightarrow 0^-} (\ln |t| - \ln 2) = -\infty.$$

On the right, $\lim_{t \rightarrow 0^+} \int_t^3 \frac{dx}{x} = +\infty$. The integral $\boxed{\text{diverges}}$. We cannot add $-\infty$ and $+\infty$ as if they were real numbers. □

Check Before Integrating

Mark all singularities, write the required limits, evaluate on the truncated intervals, and then take the limits. A symmetric limit such as $\lim_{b \rightarrow \infty} \int_{-b}^b f$ is a different notion, called a principal value. It does not establish convergence of the improper integral.

Problem 19.5 (Quick Check). Decide whether $\int_1^4 (x-1)^{-3/2} dx$ converges.

Solution. It diverges because $\lim_{t \rightarrow 1^+} [-2(x-1)^{-1/2}]_t^4 = +\infty$. □

Lesson 20: Improper Integrals: Powers and Comparison

Goals of this lesson.

- Use the two power-integral criteria correctly.
- Prove convergence or divergence without finding an antiderivative.
- Explain why the direction of a comparison matters.

20.1 How Fast Must the Graph Decay?

The functions x^{-p} give useful benchmarks. At infinity we need sufficiently fast decay. Near 0 we need sufficiently slow growth.

Theorem 20.1 (Power Integrals)

For a real number p :

Integral	Converges exactly when	Value when convergent
$\int_1^\infty x^{-p} dx$	$p > 1$	$1/(p-1)$
$\int_0^1 x^{-p} dx$	$p < 1$	$1/(1-p)$

Each integral diverges for the remaining values of p .

Proof. For $p \neq 1$, use the antiderivative $x^{1-p}/(1-p)$ and take the appropriate endpoint limit. For $p = 1$, use $\ln x$. The decisive difference is whether x^{1-p} tends to 0 at the endpoint in question. $\square$

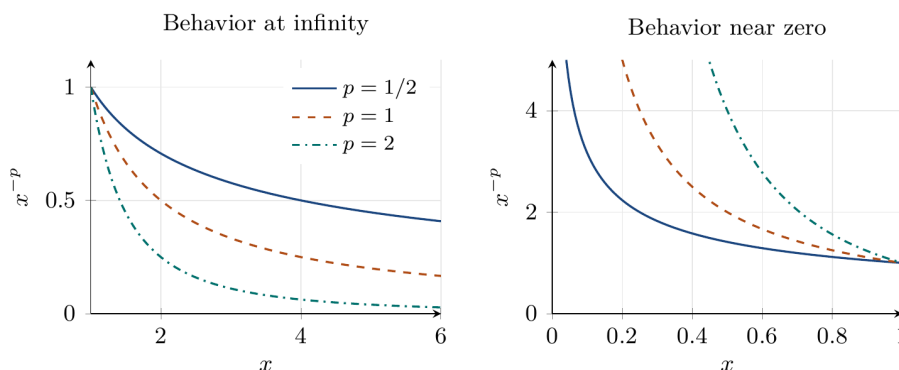


Figure 37: At infinity, larger powers give faster decay. Near zero, larger powers give a stronger singularity. The threshold is $p = 1$.

Example 20.1. For which p does $\int_0^\infty x^{-p} dx$ converge?

Solution. Split at 1. Convergence near 0 requires $p < 1$, while convergence near infinity requires $p > 1$. No value satisfies both conditions, so it diverges for every p . $\square$

20.2 Comparison

Theorem 20.2 (Comparison Test)

Suppose f and g are continuous on $[a, \infty)$ and $0 \leq f(x) \leq g(x)$ there.

- If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
- If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

The same reasoning applies near a singular endpoint.

Proof. For nonnegative functions, the truncated integrals increase as the region grows. A finite upper bound therefore gives a finite limit. If the smaller integrals grow without bound, so do the larger ones. $\square$

Example 20.2 (A Convergent Upper Bound). Test $\int_1^\infty \frac{2 + \sin x}{x^{5/2}} dx$.

Solution. For $x \geq 1$,

$$0 \leq \frac{2 + \sin x}{x^{5/2}} \leq \frac{3}{x^{5/2}}.$$

The integral of the upper bound converges since $5/2 > 1$. By comparison, the given integral converges. $\square$

Example 20.3 (A Divergent Lower Bound). Test $\int_2^\infty \frac{dx}{\sqrt{x^2 + 5}}$.

Solution. For $x \geq 2$, $x^2 + 5 \leq (9/4)x^2$, so

$$\frac{1}{\sqrt{x^2 + 5}} \geq \frac{2}{3x} > 0.$$

The integral of $2/(3x)$ diverges. The given integral therefore diverges by comparison. $\square$

Example 20.4 (Comparison at a Finite Endpoint). Test $\int_0^1 \frac{e^{-x}}{x^{3/4}} dx$.

Solution. For $0 < x \leq 1$, $0 \leq e^{-x}x^{-3/4} \leq x^{-3/4}$. Since $3/4 < 1$, the upper-bound integral near 0 converges. Thus the given integral converges. $\square$

A Complete Comparison Argument

State the inequality, give the benchmark integral's behavior, and apply the Comparison Test. A divergent upper bound or a convergent lower bound is inconclusive.

Quick check. The integral $\int_1^\infty (x^2 - x + 1)^{-1} dx$ converges by comparison with $2/x^2$ for $x \geq 2$. Its integrand is continuous on the remaining interval $[1, 2]$.

Lesson 21: Arc Length

Goals of this lesson.

- Derive arc length by approximating a curve with line segments.
- Choose between integration with respect to x and y .
- Recognize when an arc-length integral is improper.

21.1 Adding Small Lengths

Area used rectangles. Length uses short chords. For a curve $y = f(x)$, the chord over a subinterval of width Δx has length

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}.$$

The Mean Value Theorem gives $\Delta y = f'(c_i)\Delta x$ for some c_i in that subinterval. Thus the chord length is $\sqrt{1 + [f'(c_i)]^2}\Delta x$.

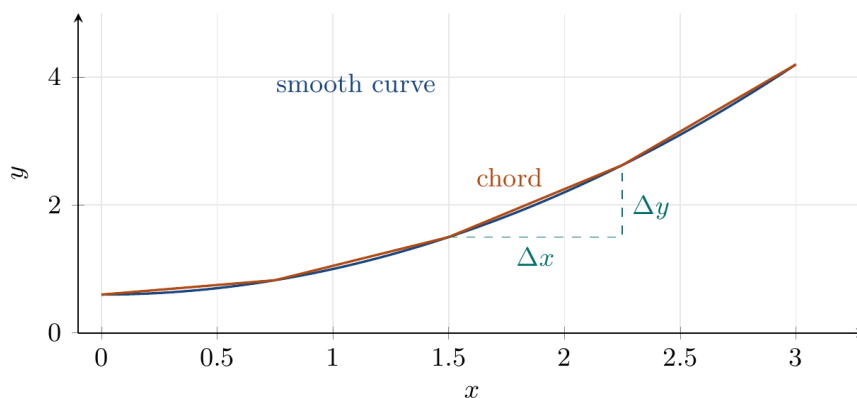


Figure 38: A polygonal approximation to a curve. Each chord is the hypotenuse of a triangle with legs Δx and Δy .

Theorem 21.1 (Arc Length of a Graph)

If f' is continuous on $[a, b]$, the length of $y = f(x)$ is

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

If $x = g(y)$ with g' continuous on $[c, d]$, then

$$L = \int_c^d \sqrt{1 + [g'(y)]^2} dy.$$

Note

The length is the supremum of the lengths of inscribed polygons. For a continuously differentiable graph, the chord sums converge to the displayed integral as the mesh of the partition tends to zero. A longer parameter interval need not mean a longer horizontal interval, so keep track of the variable used.

Example 21.1 (A Perfect Square). Find the length of $y = \frac{1}{4}x^2 - \frac{1}{2}\ln x$ for $1 \leq x \leq 2$.

Solution. Here $f'(x) = x/2 - 1/(2x)$, so

$$1 + [f'(x)]^2 = \left(\frac{x}{2} + \frac{1}{2x}\right)^2.$$

Since $x > 0$, the square root is $x/2 + 1/(2x)$. Therefore

$$L = \int_1^2 \left(\frac{x}{2} + \frac{1}{2x}\right) dx = \left[\frac{x^2}{4} + \frac{1}{2}\ln x\right]_1^2 = \boxed{\frac{3}{4} + \frac{1}{2}\ln 2}.$$

□

Example 21.2 (Integrating with Respect to y). Find the length of $x = \frac{2}{3}y^{3/2}$ for $0 \leq y \leq 3$.

Solution. Since $dx/dy = \sqrt{y}$, we obtain

$$L = \int_0^3 \sqrt{1+y} dy = \left[\frac{2}{3}(1+y)^{3/2}\right]_0^3 = \boxed{\frac{14}{3}}.$$

Solving for y first would make the calculation harder.

□

Example 21.3 (An Improper Arc-Length Integral). Set up the length of $y = x^{2/3}$ for $0 \leq x \leq 1$ and decide whether it is finite.

Solution. The derivative is unbounded at 0, so

$$L = \lim_{t \rightarrow 0^+} \int_t^1 \sqrt{1 + \frac{4}{9x^{2/3}}} dx.$$

For $x > 0$, $\sqrt{1+u^2} \leq 1+u$ when $u \geq 0$. The integrand is at most $1 + \frac{2}{3}x^{-1/3}$, whose integral near 0 converges. The length is finite.

□

Arc-Length Checklist

Sketch the curve, choose the easier variable, differentiate, simplify $1 + (\text{derivative})^2$, and integrate. Always use $\sqrt{u^2} = |u|$. An exact answer may be a definite integral if no elementary antiderivative is available.

Problem 21.4 (Quick Check). Find the length of $y = 3x - 2$ for $-1 \leq x \leq 2$.

Solution. $L = \int_{-1}^2 \sqrt{10} dx = 3\sqrt{10}$, agreeing with the distance between the endpoints.

□

Lesson 22: Arc Length of Parametric Curves

Goals of this lesson.

- Interpret a parametric curve and determine the interval being traced.
- Compute length by integrating speed.
- Distinguish the length of a path from the length of its geometric image.

22.1 The Parameter Describes Motion

A parametric curve is given by $x = x(t)$ and $y = y(t)$. Each value of t specifies a point $(x(t), y(t))$. The curve need not be the graph of a single function of x . Its direction is the direction of increasing t .

For a small change Δt , the displacement is approximately $(x'(t)\Delta t, y'(t)\Delta t)$, with length $\sqrt{[x'(t)]^2 + [y'(t)]^2}\Delta t$.

Theorem 22.1 (Parametric Arc Length)

If x' and y' are continuous on $[\alpha, \beta]$, the length of the path is

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt.$$

The integrand is the **speed**. Repeated traversals are counted repeatedly. If the curve is traced once, this is its geometric length.

Example 22.1 (A Logarithmic Spiral). Find the length of $x = e^t \cos t$, $y = e^t \sin t$ for $0 \leq t \leq \pi$.

Solution.

$$\begin{aligned} x' &= e^t(\cos t - \sin t), & y' &= e^t(\sin t + \cos t), \\ (x')^2 + (y')^2 &= 2e^{2t}. \end{aligned}$$

Thus $L = \sqrt{2} \int_0^{\pi} e^t dt = \boxed{\sqrt{2}(e^{\pi} - 1)}$.

□

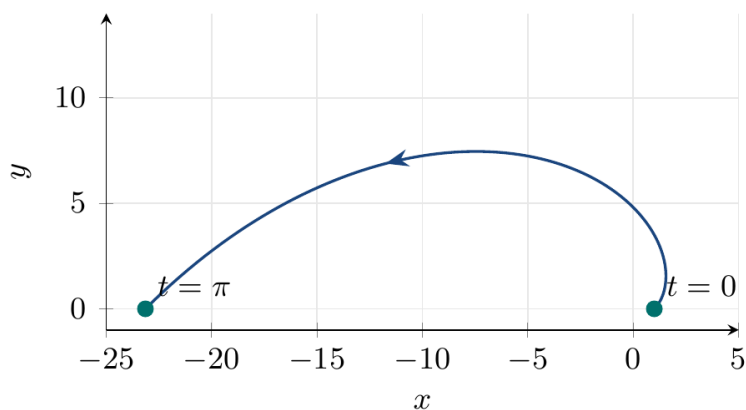


Figure 39: The spiral $r = e^t$ for $0 \leq t \leq \pi$. The radius grows as the point moves counterclockwise.

Example 22.2 (One Arch of a Cycloid). A point on the rim of a rolling circle of radius 2 traces

$$x = 2(t - \sin t), \quad y = 2(1 - \cos t).$$

Find the length of one arch, $0 \leq t \leq 2\pi$.

Solution. The speed is

$$\begin{aligned} \sqrt{4(1 - \cos t)^2 + 4 \sin^2 t} &= \sqrt{8(1 - \cos t)} \\ &= 4|\sin(t/2)| = 4 \sin(t/2) \end{aligned}$$

on this interval. Hence

$$L = \int_0^{2\pi} 4 \sin(t/2) dt = [-8 \cos(t/2)]_0^{2\pi} = \boxed{16}.$$

The speed vanishes at the endpoints, which causes no difficulty. $\square$

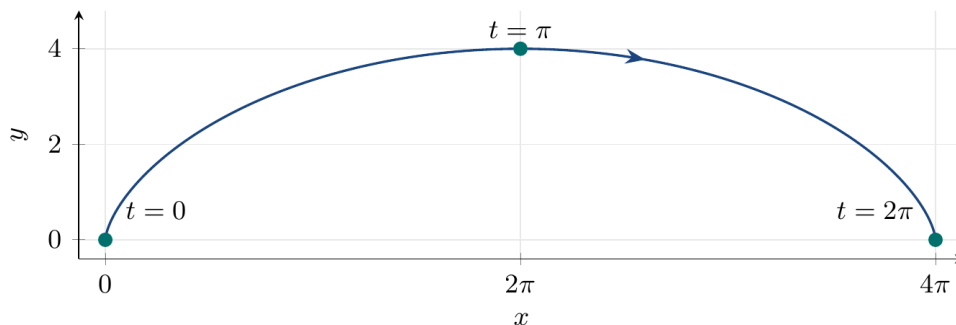


Figure 40: One cycloid arch, with endpoints at $(0, 0)$ and $(4\pi, 0)$ and highest point $(2\pi, 4)$.

Example 22.3 (A Curve Traced Twice). Find the path length of $x = 3 \cos t$, $y = 3 \sin t$ for $0 \leq t \leq 4\pi$.

Solution. The speed is 3, so $L = 3(4\pi) = \boxed{12\pi}$. The geometric image is a circle of circumference 6π , traversed twice. $\square$

Recovering the Graph Formula

For $x(t) = t$, $y(t) = f(t)$, the parametric formula becomes $\int \sqrt{1 + [f'(t)]^2} dt$. Thus the graph formula is a special case of integrating speed.

Problem 22.4 (Quick Check). Find the path length of $x = t^2$, $y = t^2 + 1$ for $-1 \leq t \leq 1$.

Solution. The speed is $2\sqrt{2}|t|$, so the length is $2\sqrt{2}$. The line segment between $(0, 1)$ and $(1, 2)$ is traversed twice. $\square$

Lesson 23: Area of a Surface of Revolution

Goals of this lesson.

- Explain surface area as circumference times a small arc length.
- Identify the radius as distance from the axis of rotation.
- Use the appropriate arc-length factor and check endpoint behavior.

23.1 Rotating a Short Segment

Rotating a short chord creates the lateral surface of a conical frustum. Its area is

$$\pi(r_1 + r_2)\ell = 2\pi \left(\frac{r_1 + r_2}{2} \right) \ell,$$

where ℓ is the chord length. As the pieces become small, the average radius becomes the radius of the curve and ℓ becomes an arc-length element.

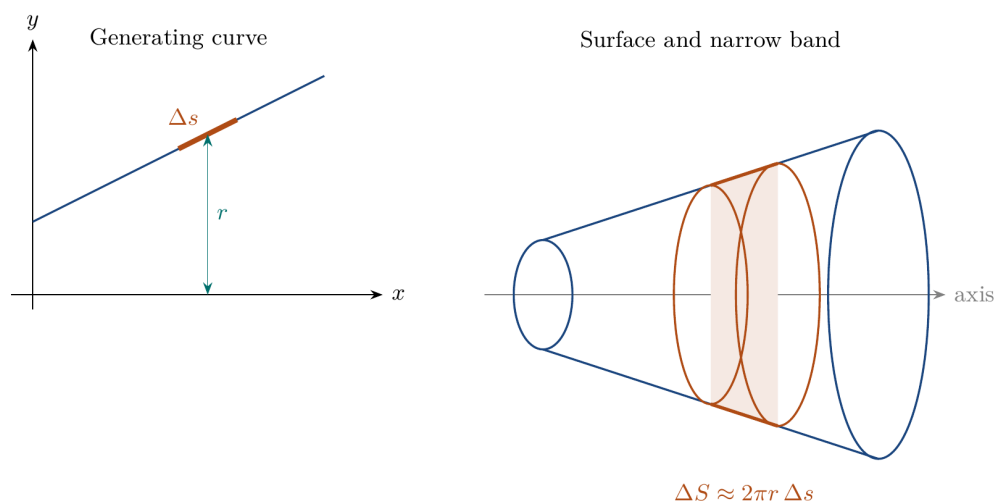


Figure 41: A short segment rotates into a narrow band. Its area is approximately $2\pi r \Delta s$, using slant length rather than horizontal width.

Theorem 23.1 (Surface Area)

For a continuously differentiable curve rotated once to form a surface without repeated coverage,

$$S = \int 2\pi r \, ds,$$

where r is the distance to the rotation axis. In particular, for $y = f(x)$, $a \leq x \leq b$,

$$\text{about } y = k : \quad S = 2\pi \int_a^b |f(x) - k| \sqrt{1 + [f'(x)]^2} dx,$$

$$\text{about } x = h : \quad S = 2\pi \int_a^b |x - h| \sqrt{1 + [f'(x)]^2} dx.$$

For $x = g(y)$, replace ds by $\sqrt{1 + [g'(y)]^2} dy$.

Note

These are lateral surface areas. Add end disks only if the problem asks for a closed surface. The radius is always a nonnegative distance. If different parts generate the same patch, split the curve to avoid counting that patch twice.

Example 23.1 (A Frustum). Rotate $y = 1 + x/2$, $0 \leq x \leq 4$, about the x -axis. Find the surface area.

Solution. The radius is $1 + x/2$, and $ds = (\sqrt{5}/2) dx$. Therefore

$$S = \pi \sqrt{5} \int_0^4 (1 + x/2) dx = \pi \sqrt{5} [x + x^2/4]_0^4 = \boxed{8\pi\sqrt{5}}.$$

The frustum has radii 1 and 3 and slant height $2\sqrt{5}$, giving the same result from $\pi(r_1 + r_2)\ell$. $\square$

Example 23.2 (A Shifted Vertical Axis). Set up the area obtained by rotating $y = x^2$, $0 \leq x \leq 1$, about $x = -1$.

Solution. The radius is $x + 1$, while $ds = \sqrt{1 + 4x^2} dx$. Thus

$$S = 2\pi \int_0^1 (x + 1) \sqrt{1 + 4x^2} dx.$$

The radius comes from the rotation axis. The arc-length factor comes from the curve. $\square$

Example 23.3 (A Spherical Zone). Rotate $y = \sqrt{R^2 - x^2}$ about the x -axis for $u \leq x \leq v$, where $-R < u < v < R$. Find the area.

Solution. Since $f'(x) = -x/\sqrt{R^2 - x^2}$,

$$f(x) \sqrt{1 + [f'(x)]^2} = R.$$

Hence $\boxed{S = 2\pi R(v - u)}$. The area depends only on the height of the zone along the axis. Letting $u \rightarrow -R^+$ and $v \rightarrow R^-$ gives the area $4\pi R^2$ of the whole sphere. Those endpoint limits are needed because the graph has vertical tangents. $\square$

Example 23.4 (Returning to Gabriel's Horn). The horn in Lesson 8 has volume π . Show that its surface area is infinite.

Solution. For $y = 1/x$, $x \geq 1$,

$$S = 2\pi \int_1^\infty \frac{1}{x} \sqrt{1 + \frac{1}{x^4}} dx \geq 2\pi \int_1^\infty \frac{dx}{x}.$$

The lower-bound integral diverges, so the horn has infinite surface area. $\square$

Lesson 24: Hydrostatic Pressure and Force

Goals of this lesson.

- Distinguish pressure, force, mass density, and weight density.
- Use horizontal strips to account for varying pressure.
- Find strip width from geometry and express depth in the chosen variable.

24.1 Pressure Increases with Depth

Water presses harder on a surface farther below the water level. For an incompressible fluid of constant mass density ρ , the *gauge pressure* at depth d is

$$P = \rho g d = \gamma d,$$

where $\gamma = \rho g$ is the weight density. In SI units, water has $\rho \approx 1000 \text{ kg/m}^3$ and $\gamma \approx 9800 \text{ N/m}^3$. Pressure has units N/m^2 , called pascals.

Which Pressure?

Gauge pressure measures the excess over atmospheric pressure. We use it when atmospheric pressure on the two sides cancels, or when the problem asks for the force due to the fluid. Absolute pressure includes the additional atmospheric pressure.

On a vertical plate, pressure varies with height. Slice horizontally so that every point of a thin strip is nearly at the same depth. If the strip has width $w(y)$ and thickness dy , its area is $w(y) dy$.

Hydrostatic Force on One Side

$$F = \gamma \int_a^b d(y)w(y) dy.$$

Depth is the distance *below the water surface*. The bounds cover the submerged portion of the plate. Pressure acts perpendicular to the plate.

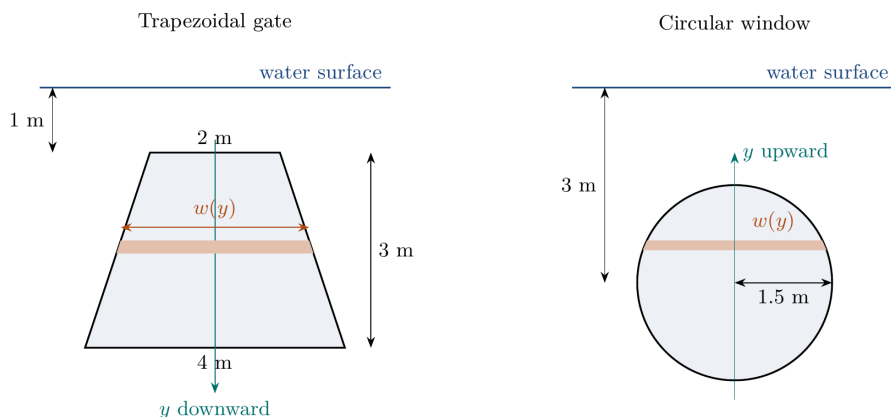


Figure 42: Use horizontal strips. For the trapezoid, depth is $1 + y$. For the circular window, depth is $3 - y$, with y measured upward from its center.

Example 24.1 (A Trapezoidal Gate). A vertical gate is 3 m tall, 2 m wide at its top, and 4 m wide at its bottom. Its top is 1 m below the water surface. Find the force on one side.

Solution. Let y measure downward from the top, so $0 \leq y \leq 3$. The depth is $d(y) = 1 + y$. Linear interpolation gives $w(y) = 2 + \frac{2}{3}y$. Therefore

$$\begin{aligned} F &= 9800 \int_0^3 (1 + y) \left(2 + \frac{2}{3}y \right) dy \\ &= 9800 \int_0^3 \left(2 + \frac{8}{3}y + \frac{2}{3}y^2 \right) dy \\ &= 9800(6 + 12 + 6) = \boxed{235200 \text{ N}}. \end{aligned}$$

□

Example 24.2 (A Circular Window). A circular window of radius 1.5 m lies vertically with its center 3 m below the surface. Set up and evaluate the force on one side.

Solution. Set the center at $(0, 0)$ and take y upward. The circle has equation $x^2 + y^2 = 9/4$. Thus

$$w(y) = 2\sqrt{9/4 - y^2}, \quad d(y) = 3 - y.$$

The force is

$$\begin{aligned} F &= 9800 \int_{-3/2}^{3/2} (3 - y) 2\sqrt{9/4 - y^2} dy \\ &= 9800 \cdot 3 \cdot \pi(3/2)^2 = \boxed{66150\pi \text{ N}}. \end{aligned}$$

The term involving y integrates to 0 by odd symmetry. The remaining integral is the area of the circular window. □

Problem 24.3 (Quick Check). A rectangular plate 2 ft wide and 3 ft tall has its top at the water surface. Use $\gamma = 62.4 \text{ lb/ft}^3$ to find the force on one side.

Solution. Using depth y directly, $F = 62.4 \int_0^3 2y \, dy = \boxed{561.6 \text{ lb}}$. □

Lesson 25: Centers of Mass and Centroids

Goals of this lesson.

- Interpret center of mass as a weighted average.
- Derive moment integrals for a thin plate of uniform density.
- Use symmetry and choose a convenient slicing direction.

25.1 From Point Masses to a Plate

For masses $m_i > 0$ at points (x_i, y_i) , total mass and moments are

$$m = \sum_i m_i, \quad M_y = \sum_i m_i x_i, \quad M_x = \sum_i m_i y_i.$$

The center of mass is

$$\bar{x} = \frac{M_y}{m}, \quad \bar{y} = \frac{M_x}{m}.$$

The subscripts refer to the axis about which the moment is taken. A moment about the y -axis uses the x -coordinate.

Example 25.1 (Point Masses). Masses 2, 1, and 3 kg lie at $(0, 0)$, $(3, 0)$, and $(0, 2)$ m. Find their center of mass.

Solution. $m = 6$, $M_y = 3$, and $M_x = 6$. Therefore $(\bar{x}, \bar{y}) = (1/2, 1)$ m. □

25.2 Vertical Strips

Let a plate of constant surface mass density $\rho > 0$ lie between $y = f(x)$ and $y = g(x)$ on $[a, b]$, with $f \geq g$. A strip has mass approximately $\rho(f(x) - g(x))\Delta x$. Its center is approximately $(x, (f(x) + g(x))/2)$.

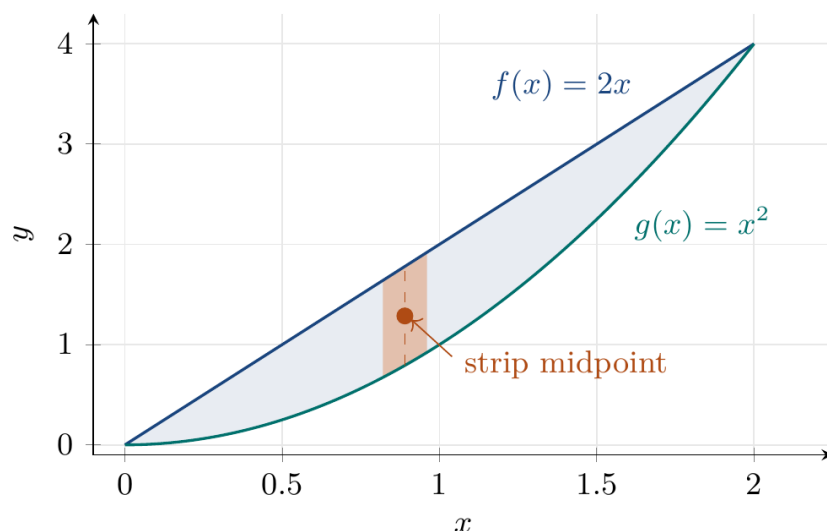


Figure 43: A vertical strip between two curves. Its midpoint height is $(f + g)/2$, and its mass is proportional to $f - g$.

Theorem 25.1 (Mass and Moments of a Uniform Plate)

For continuous $f \geq g$ on $[a, b]$ and a region of positive area,

$$m = \rho \int_a^b (f - g) dx,$$

$$M_y = \rho \int_a^b x(f - g) dx,$$

$$M_x = \frac{\rho}{2} \int_a^b (f^2 - g^2) dx.$$

The center of mass is $(M_y/m, M_x/m)$. For constant density this point is also the **centroid**, and ρ cancels.

The last formula comes from $\frac{f+g}{2}\rho(f - g) = \frac{\rho}{2}(f^2 - g^2)$. It uses the strip's midpoint, which explains why we do *not* square $f - g$.

Example 25.2 (A Region Between Two Curves). Find the centroid of the region between $y = 2x$ and $y = x^2$.

Solution. The intersections are at $x = 0, 2$, and $2x \geq x^2$ there. Take $\rho = 1$:

$$m = \int_0^2 (2x - x^2) dx = \frac{4}{3},$$

$$M_y = \int_0^2 (2x^2 - x^3) dx = \frac{4}{3},$$

$$M_x = \frac{1}{2} \int_0^2 (4x^2 - x^4) dx = \frac{32}{15}.$$

Thus $(\bar{x}, \bar{y}) = (1, 8/5)$. □

25.3 Horizontal Strips and Symmetry

For $h_1(y) \leq x \leq h_2(y)$, $c \leq y \leq d$,

$$\begin{aligned} m &= \rho \int_c^d (h_2 - h_1) dy, \\ M_x &= \rho \int_c^d y(h_2 - h_1) dy, \\ M_y &= \frac{\rho}{2} \int_c^d (h_2^2 - h_1^2) dy. \end{aligned}$$

If both the region and its density are symmetric about a line, the center of mass lies on that line.

Example 25.3 (Horizontal Slicing). Find the centroid of $0 \leq y \leq 1$, $-y^2 \leq x \leq y^2$.

Solution. Symmetry gives $\bar{x} = 0$. With $\rho = 1$, $m = \int_0^1 2y^2 dy = 2/3$ and $M_x = \int_0^1 2y^3 dy = 1/2$. Hence $(\bar{x}, \bar{y}) = (0, 3/4)$. □

Note

A centroid can lie outside the plate. For example, the centroid of a uniform annulus is its center, which lies in the hole. If density varies, it must remain inside the integrals and the uniform-density formulas need adjustment.

Problem 25.4 (Quick Check). A region has area 5 and, with density 1, moments $M_y = -10$, $M_x = 15$. Find its centroid.

Solution. $(-2, 3)$. Negative coordinates are allowed. □

Lesson 26: Taylor Polynomials: Why Study Infinite Series?

Goals of this lesson.

- Approximate a function by matching its derivatives at a center.
- Compute Taylor polynomials and explain the factorials.
- Identify the questions that lead to sequences and series.

26.1 From a Tangent Line to a Polynomial

A tangent line matches a function's value and first derivative at $x = a$:

$$T_1(x) = f(a) + f'(a)(x - a).$$

To match the second derivative as well, add $c_2(x - a)^2$. Its second derivative at a is $2c_2$, so $c_2 = f''(a)/2$. Continuing gives the following formula.

Definition 26.1 (Taylor Polynomial)

If f has derivatives through order n at a , its **Taylor polynomial of order n** centered at a is

$$T_n(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$$

Here $f^{(0)} = f$ and $0! = 1$. It is the unique polynomial of degree *at most* n whose derivatives through order n agree with those of f at a .

The coefficient contains $k!$ because differentiating $(x - a)^k$ exactly k times produces $k!$.

Example 26.1 (A Nonzero Center). Find $T_3(x)$ for $f(x) = \ln x$ centered at $a = 2$.

Solution.

k	$f^{(k)}(x)$	$f^{(k)}(2)$	$f^{(k)}(2)/k!$
0	$\ln x$	$\ln 2$	$\ln 2$
1	$1/x$	$1/2$	$1/2$
2	$-1/x^2$	$-1/4$	$-1/8$
3	$2/x^3$	$1/4$	$1/24$

Therefore

$$T_3(x) = \ln 2 + \frac{x - 2}{2} - \frac{(x - 2)^2}{8} + \frac{(x - 2)^3}{24}.$$

□

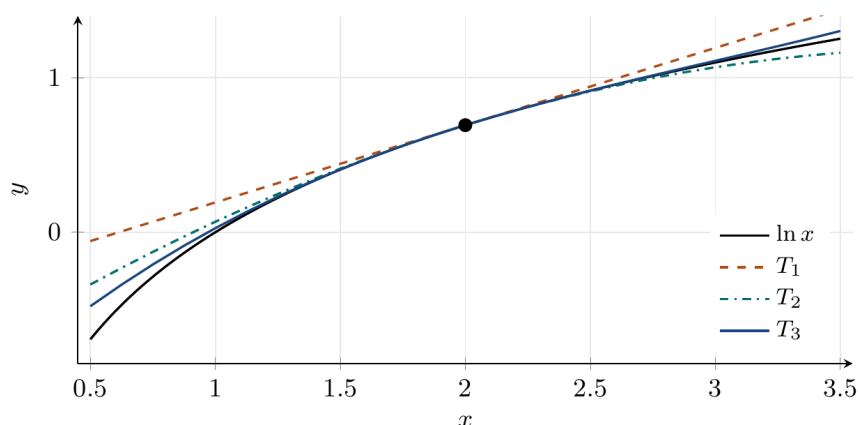


Figure 44: Taylor polynomials for $\ln x$ centered at 2. Matching more derivatives improves the local fit, but accuracy away from the center requires further justification.

Example 26.2 (A Pattern of Coefficients). Find the order-4 Taylor polynomial of $f(x) = e^{2x}$ at 0.

Solution. Since $f^{(k)}(0) = 2^k$,

$$T_4(x) = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4.$$

For example, $T_4(0.1) = 1.2214$ approximates $e^{0.2}$. □

26.2 An Integral We Cannot Evaluate Elementarily

Example 26.3. Use a Taylor polynomial to approximate $\int_0^{1/2} e^{-x^2} dx$.

Solution. Substituting $-x^2$ into the initial terms of the Taylor polynomial for e^u gives the order-4 approximation

$$e^{-x^2} \approx 1 - x^2 + \frac{x^4}{2}.$$

Integrating the polynomial gives

$$\int_0^{1/2} e^{-x^2} dx \approx \left[x - \frac{x^3}{3} + \frac{x^5}{10} \right]_0^{1/2} = \frac{443}{960} \approx 0.46146.$$

This is an approximation. We will justify it and bound its error in [Lesson 37](#). □

The Questions for This Unit

- A **sequence** is an ordered list of numbers. Do its terms approach a limit?
- A **series** adds the terms of a sequence. Do its partial sums approach a finite number?
- A **power series** has terms depending on x . For which x does it converge?
- A **Taylor series** uses the derivatives of f . When does its sum actually equal f ?

Lesson 27: Sequences and Their Limits

Goals of this lesson.

- Describe sequences by formulas and recursive rules.
- Compute limits and distinguish convergence from boundedness.
- Use monotonicity and bounds to prove convergence.

Definition 27.1 (Sequence and Convergence)

A real **sequence** is a function defined on the integers $n \geq n_0$, written $\{a_n\}$. It **converges to** L if, for every $\varepsilon > 0$, there is an integer N such that

$$n \geq N \implies |a_n - L| < \varepsilon.$$

We write $\lim_{n \rightarrow \infty} a_n = L$. If there is no finite limit, the sequence **diverges**.

The definition says that all sufficiently late terms lie in any prescribed band around L .

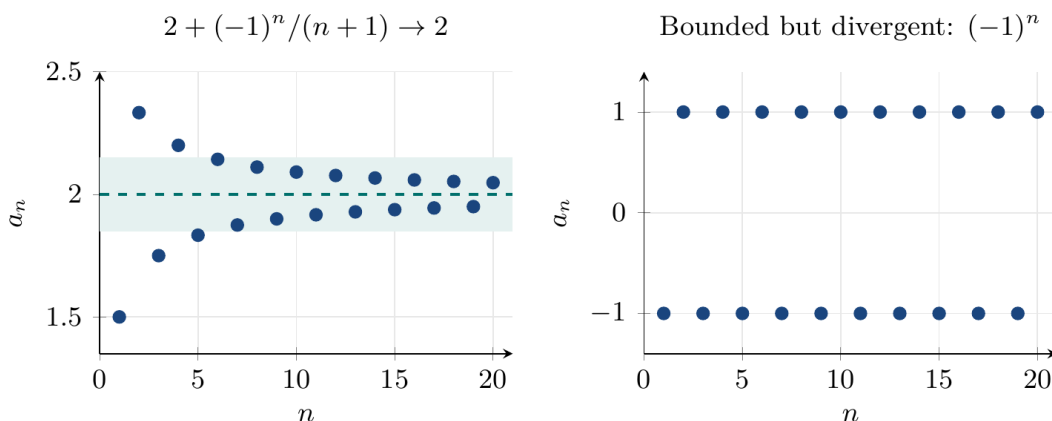


Figure 45: A convergent sequence may oscillate. A bounded sequence may diverge. Sequence graphs consist of points at integer indices.

Example 27.1 (A Limit from the Definition). Show that $a_n = 2 + (-1)^n / (n+1)$ converges to 2.

Solution. Given $\varepsilon > 0$, choose an integer $N > 1/\varepsilon$. For $n \geq N$, $|a_n - 2| = 1/(n+1) < \varepsilon$. Thus $\boxed{a_n \rightarrow 2}$. $\square$

27.1 Computing Limits

The sum, product, and quotient rules for limits apply to sequences. For a quotient, the denominator's limit must be nonzero. The Squeeze Theorem also applies.

Using a Continuous Variable

If $a_n = f(n)$ and $\lim_{x \rightarrow \infty} f(x) = L$, then $a_n \rightarrow L$. The converse can fail: $\sin(2\pi n) = 0$ for every integer n , although $\sin(2\pi x)$ has no limit as $x \rightarrow \infty$. Apply L'Hospital's Rule to a function of a real variable, not directly to $n!$ or to a discrete sequence.

Example 27.2. Find the limits of $a_n = \frac{5n^2 - 3}{2n^2 + 7}$ and $b_n = \frac{\sin(3n)}{\sqrt{n+2}}$.

Solution. Dividing numerator and denominator by n^2 gives $a_n \rightarrow \boxed{5/2}$. Also $|b_n| \leq 1/\sqrt{n+2} \rightarrow 0$, so the Squeeze Theorem gives $b_n \rightarrow \boxed{0}$. $\square$

For $n \geq 1$, the geometric sequence r^n tends to 0 when $|r| < 1$, and to 1 when $r = 1$. It diverges for all other real r . When $r = -1$, the even and odd subsequences have different limits.

27.2 Monotonicity and Boundedness

Definition 27.2

A sequence is **nondecreasing** if $a_{n+1} \geq a_n$, and **nonincreasing** if $a_{n+1} \leq a_n$. Either property makes it **monotone**. It is **bounded** if $m \leq a_n \leq M$ for some fixed real m, M and every n .

Theorem 27.1 (Monotone Convergence for Sequences)

A nondecreasing sequence bounded above converges. A nonincreasing sequence bounded below also converges.

Proof. In the nondecreasing case, let $L = \sup\{a_n\}$. For every $\varepsilon > 0$, some $a_N > L - \varepsilon$. Then $L - \varepsilon < a_N \leq a_n \leq L$ for $n \geq N$, so $a_n \rightarrow L$. Use an infimum in the nonincreasing case. $\square$

Example 27.3 (A Recursive Sequence). Let $a_1 = 1$ and $a_{n+1} = \sqrt{6 + a_n}$. Prove convergence and find the limit.

Solution. Induction gives $1 \leq a_n \leq 3$: the next term lies between $\sqrt{7}$ and 3 whenever the current term lies in $[1, 3]$. Also $6 + a_n - a_n^2 = (3 - a_n)(a_n + 2) \geq 0$, so $a_{n+1} \geq a_n$. The sequence converges by the theorem. Only now may we pass to the limit:

$$L = \sqrt{6 + L}, \quad (L - 3)(L + 2) = 0.$$

Since $L \geq 1$, $\boxed{L = 3}$. $\square$

Note

Every convergent sequence is bounded. The converse fails, as $\{(-1)^n\}$ shows. Solving a fixed-point equation for a recursive rule gives possible limits, but does not prove convergence.

Lesson 28: Infinite Series and Partial Sums

Goals of this lesson.

- Define an infinite sum as a limit of finite sums.
- Sum geometric and telescoping series.
- Use the Divergence Test and explain its limitation.

28.1 What Does an Infinite Sum Mean?

We cannot perform infinitely many additions in one step. Instead, consider the total after N steps and ask whether these totals approach a finite number.

Definition 28.1 (Series)

The N th **partial sum** of $\sum_{n=1}^{\infty} a_n$ is $s_N = \sum_{n=1}^N a_n$. The series **converges** to S if $\lim_{N \rightarrow \infty} s_N = S$, and we then write

$$\sum_{n=1}^{\infty} a_n = S.$$

Otherwise the series **diverges**. Other starting indices are allowed.

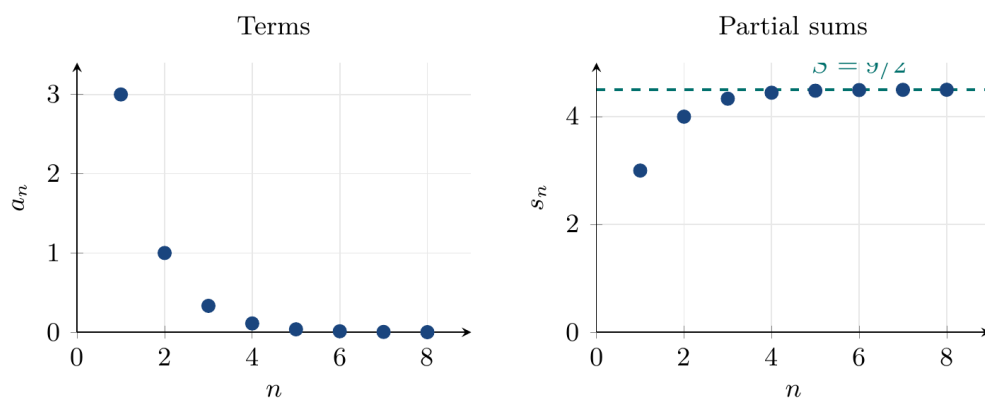


Figure 46: For $a_n = 3(1/3)^{n-1}$, the terms tend to 0 while the partial sums tend to $9/2$. These are two different sequences.

28.2 Geometric Series

For $s_N = a + ar + \cdots + ar^{N-1}$, subtracting rs_N gives $(1 - r)s_N = a(1 - r^N)$. This produces the sum formula.

Theorem 28.1 (Geometric Series)

For $a \neq 0$, $\sum_{n=0}^{\infty} ar^n$ converges exactly when $|r| < 1$, and then

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

When $a = 0$ the series is identically zero for every r .

Example 28.1. Evaluate $\sum_{n=2}^{\infty} 4 \left(-\frac{2}{5}\right)^n$.

Solution. The first term is $4(4/25) = 16/25$, and $r = -2/5$. Thus

$$S = \frac{16/25}{1 + 2/5} = \boxed{\frac{16}{35}}.$$

The first term must come from the actual starting index. □

Example 28.2 (A Repeating Decimal). Write $0.\overline{27}$ as a fraction.

Solution.

$$0.\overline{27} = \frac{27}{100} + \frac{27}{100^2} + \cdots = \frac{27/100}{1 - 1/100} = \boxed{\frac{3}{11}}.$$

□

28.3 Telescoping Series

Example 28.3. Evaluate $\sum_{n=1}^{\infty} \frac{2}{n(n+2)}$.

Solution. Partial fractions give $2/[n(n+2)] = 1/n - 1/(n+2)$. Work with a finite sum first:

$$\begin{aligned} s_N &= \sum_{n=1}^N \left(\frac{1}{n} - \frac{1}{n+2} \right) \\ &= 1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2}. \end{aligned}$$

Taking $N \rightarrow \infty$ gives $\boxed{S = 3/2}$. We did not subtract two divergent harmonic series. □

28.4 A Necessary Condition**Theorem 28.2 (Divergence Test)**

If $\sum a_n$ converges, then $a_n \rightarrow 0$. Consequently, if a_n does not tend to 0, the series diverges.

Proof. If $s_N \rightarrow S$, then $a_N = s_N - s_{N-1} \rightarrow S - S = 0$. □

Example 28.4. Test $\sum_{n=1}^{\infty} \frac{2n+1}{n+4}$.

Solution. The terms tend to $2 \neq 0$, so the series diverges. □

Zero Terms in the Limit Are Not Enough

The harmonic series $\sum_{n=1}^{\infty} 1/n$ diverges even though $1/n \rightarrow 0$. We will prove this next. When $a_n \rightarrow 0$, the Divergence Test is inconclusive.

Rules for Convergent Series

If $\sum a_n$ and $\sum b_n$ converge, their sums can be added, subtracted, and multiplied by constants term by term. Adding, removing, or changing finitely many terms does not change convergence, though it can change the sum. Do not separate a convergent series into divergent pieces and subtract their supposed sums.

Lesson 29: The Integral Test and p -Series

Goals of this lesson.

- Relate a positive series to an improper integral.
- Check continuity, positivity, and eventual monotonicity.
- Apply the p -Series Test and logarithmic benchmarks.

29.1 Rectangles Compared with Area

If f is nonnegative and nonincreasing, rectangles of height $f(n)$ bound areas under its graph:

$$\int_1^{N+1} f(x) dx \leq \sum_{n=1}^N f(n) \leq f(1) + \int_1^N f(x) dx.$$

Thus the integrals and partial sums become unbounded together, or remain bounded together.

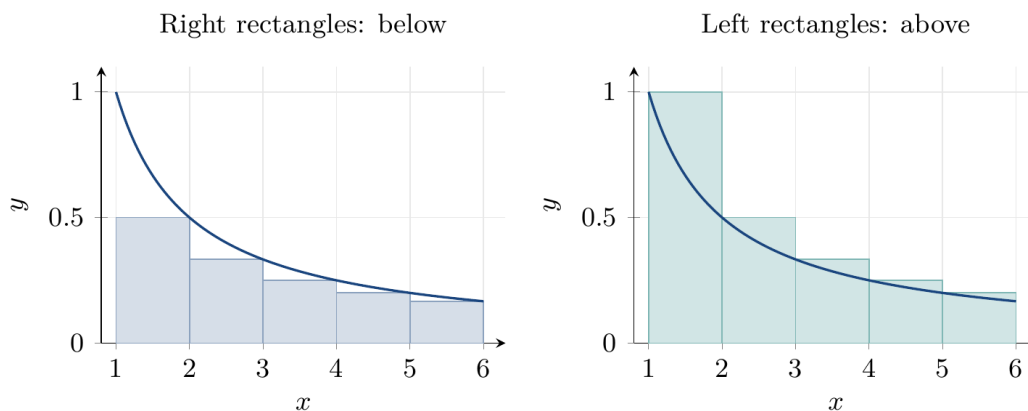


Figure 47: For a decreasing function, right-endpoint rectangles lie below the curve and left-endpoint rectangles lie above it. These inequalities control convergence.

Theorem 29.1 (Integral Test)

Suppose $a_n = f(n)$ for $n \geq N_0$, where f is continuous, positive, and nonincreasing on $[N_0, \infty)$. Then

$$\sum_{n=N_0}^{\infty} a_n \text{ converges} \iff \int_{N_0}^{\infty} f(x) dx \text{ converges.}$$

Note

The theorem compares convergence, not values. The series sum usually differs from the integral. The hypotheses need only hold eventually, since finitely many initial terms do not affect convergence.

Theorem 29.2 (p -Series Test)

For real p , $\sum_{n=1}^{\infty} n^{-p}$ converges exactly when $p > 1$.

Proof. For $p > 0$, apply the Integral Test and the power-integral criterion in [Lesson 20](#). For $p \leq 0$, the terms do not tend to 0. $\square$

In particular, the harmonic series ($p = 1$) diverges.

Example 29.1. Test $\sum_{n=1}^{\infty} n^{-9/4}$ and $\sum_{n=1}^{\infty} n^{-4/5}$.

Solution. The first converges because $9/4 > 1$. The second diverges because $4/5 \leq 1$. Both have terms tending to 0. $\square$

Example 29.2 (Logarithm and a Power). Test $\sum_{n=2}^{\infty} \frac{\ln n}{n^3}$.

Solution. Take $f(x) = \ln x/x^3$. It is positive and continuous for $x \geq 2$, and

$$f'(x) = \frac{1 - 3 \ln x}{x^4} < 0 \quad (x \geq 2).$$

Integration by parts gives

$$\begin{aligned} \int_2^{\infty} \frac{\ln x}{x^3} dx &= \lim_{b \rightarrow \infty} \left[-\frac{\ln x}{2x^2} - \frac{1}{4x^2} \right]_2^b \\ &= \frac{\ln 2}{8} + \frac{1}{16} < \infty. \end{aligned}$$

Thus the series converges by the Integral Test. $\square$

Example 29.3 (A Logarithmic Threshold). For which real q does $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^q}$ converge?

Solution. The function is positive and continuous for $x > 1$. Its derivative has sign opposite to $1 + q/\ln x$, so it is eventually decreasing for every fixed q . With $u = \ln x$,

$$\int_2^{\infty} \frac{dx}{x(\ln x)^q} = \int_{\ln 2}^{\infty} u^{-q} du.$$

This converges exactly when $q > 1$. The same is true of the series. $\square$

Optional: Estimating a Positive Tail

If the hypotheses hold and the series converges, its remainder $R_N = \sum_{n=N+1}^{\infty} f(n)$ satisfies

$$\int_{N+1}^{\infty} f(x) dx \leq R_N \leq \int_N^{\infty} f(x) dx.$$

This extension is useful for approximation, but the main task here is deciding convergence.

Lesson 30: Comparison Tests for Series

Goals of this lesson.

- Compare nonnegative series with geometric or p -series.
- Use limit comparison when direct inequalities are awkward.
- Explain why a proposed comparison does or does not establish a conclusion.

30.1 Direct Comparison

The partial sums of a nonnegative series form a nondecreasing sequence. If those sums have a finite upper bound, they converge.

Theorem 30.1 (Comparison Test)

Suppose $0 \leq a_n \leq b_n$ for all sufficiently large n .

- If $\sum b_n$ converges, then $\sum a_n$ converges.
- If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Example 30.1 (A Bounded Numerator). Test $\sum_{n=1}^{\infty} \frac{\sin^2 n}{n^{7/4}}$.

Solution. Since $0 \leq \sin^2 n \leq 1$,

$$0 \leq \frac{\sin^2 n}{n^{7/4}} \leq \frac{1}{n^{7/4}}.$$

The upper-bound p -series converges. Hence the given series converges by comparison. □

Example 30.2 (An Oscillating Positive Numerator). Test $\sum_{n=1}^{\infty} \frac{2 + \cos n}{n + 1}$.

Solution. Because $2 + \cos n \geq 1$, $\frac{2 + \cos n}{n + 1} \geq \frac{1}{n + 1} > 0$. The lower-bound series is a tail of the divergent harmonic series, so the given series diverges. □

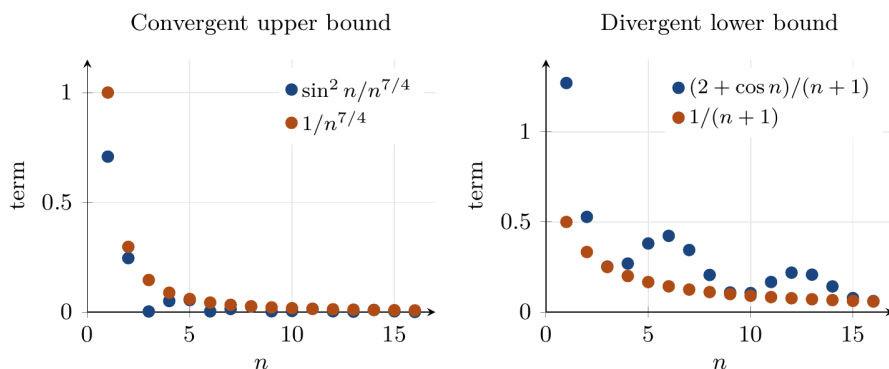


Figure 48: Compare in the direction needed: a convergent upper bound proves convergence, and a divergent lower bound proves divergence.

30.2 Limit Comparison

If a_n/b_n approaches a positive finite number, the two sequences eventually differ only by bounded positive factors. Their series must have the same convergence behavior.

Theorem 30.2 (Limit Comparison Test)

Suppose $a_n, b_n > 0$ for all sufficiently large n . If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c, \quad 0 < c < \infty,$$

then $\sum a_n$ and $\sum b_n$ either both converge or both diverge.

Proof. Eventually $c/2 < a_n/b_n < 3c/2$. Apply direct comparison in both directions. $\square$

Example 30.3 (Leading Powers). Test $\sum_{n=1}^{\infty} \frac{4n^2 + n}{n^4 + 7}$.

Solution. The leading powers suggest $b_n = 1/n^2$. Then

$$\frac{a_n}{b_n} = \frac{4n^4 + n^3}{n^4 + 7} \rightarrow 4.$$

Since $\sum 1/n^2$ converges and 4 is positive and finite, the given series converges by limit comparison. $\square$

Example 30.4 (A Radical Denominator). Test $\sum_{n=1}^{\infty} \frac{3n+2}{\sqrt{n^3+4}}$.

Solution. Use $b_n = n^{-1/2}$. Then

$$\frac{a_n}{b_n} = \frac{3 + 2/n}{\sqrt{1 + 4/n^3}} \rightarrow 3.$$

The benchmark p -series has $p = 1/2$ and diverges. Hence the given series diverges. $\square$

What if the Ratio Limit Is Zero?

The two-way theorem above does not apply when $c = 0$. If $a_n/b_n \rightarrow 0$ and $\sum b_n$ converges, then $a_n \leq b_n$ eventually, so $\sum a_n$ converges by direct comparison. If $\sum b_n$ diverges, the limit 0 alone gives no conclusion.

Problem 30.5 (Quick Check). Is $\sum_{n=1}^{\infty} \frac{1}{n5^n}$ smaller than a harmonic series? Does that prove convergence?

Solution. It is smaller, but that comparison is inconclusive. Instead use $1/(n5^n) \leq 1/5^n$, a convergent geometric upper bound. $\square$

Lesson 31: Alternating Series

Goals of this lesson.

- Check all hypotheses of the Alternating Series Test.
- Explain convergence through the even and odd partial sums.
- Distinguish a failed test from a proof of divergence.

31.1 Cancellation Can Help

An alternating series has the form $\sum(-1)^{n-1}b_n$ or $\sum(-1)^nb_n$, where $b_n \geq 0$. As successive terms shrink, the partial sums move back and forth by smaller amounts.

Theorem 31.1 (Alternating Series Test)

If $b_n \geq 0$, $b_{n+1} \leq b_n$ for all sufficiently large n , and $b_n \rightarrow 0$, then $\sum(-1)^{n-1}b_n$ converges.

Proof. After removing any initial terms, assume the hypotheses hold from $n = 1$. The even partial sums increase, the odd partial sums decrease, and every even partial sum is at most every later odd partial sum. Both subsequences converge by monotone convergence. Their gap is $s_{2N+1} - s_{2N} = b_{2N+1} \rightarrow 0$, so their limits agree. $\square$

Example 31.1 (A Shifted Power). Test $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n+3}}$.

Solution. $b_n = 1/\sqrt{n+3}$ is positive, decreasing, and tends to 0. Thus the series converges by the Alternating Series Test. Its absolute-value series diverges by limit comparison with $1/\sqrt{n}$. We will call this *conditional convergence* in the next lesson. $\square$

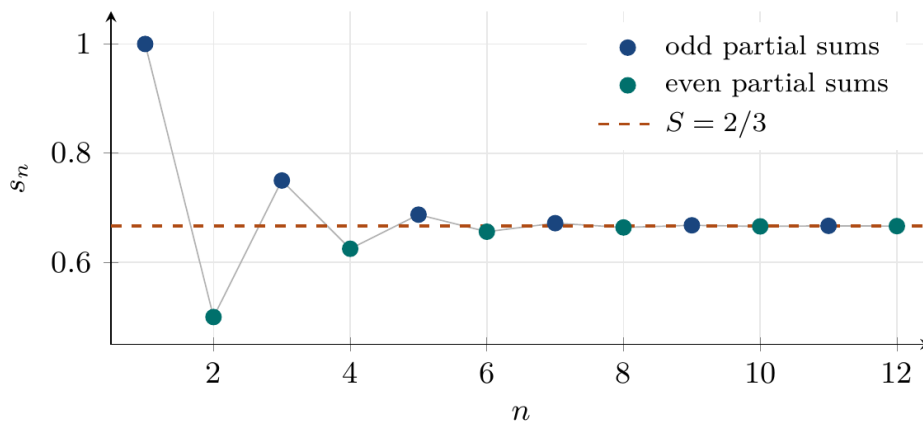


Figure 49: For the alternating geometric series with terms $(-1)^{n-1}2^{-(n-1)}$, even partial sums approach $2/3$ from below and odd partial sums approach it from above.

Example 31.2 (Eventually Decreasing). Test $\sum_{n=1}^{\infty} (-1)^n \frac{n}{n^2 + 4}$.

Solution. For $f(x) = x/(x^2 + 4)$,

$$f'(x) = \frac{4 - x^2}{(x^2 + 4)^2} < 0 \quad (x > 2).$$

The magnitudes are eventually decreasing, positive, and tend to 0. Hence the series converges. The first few terms do not affect this conclusion. □

Example 31.3 (Alternation Alone Is Not Enough). Test $\sum_{n=1}^{\infty} (-1)^n \frac{n+2}{4n+1}$.

Solution. The magnitudes tend to $1/4$, so the terms do not tend to 0. The series diverges by the Divergence Test. □

Optional: Error of an Alternating Approximation

If the magnitudes are nonincreasing from the start and tend to 0, the sum S lies between successive partial sums. Consequently,

$$|S - s_N| \leq b_{N+1}.$$

This also holds for a tail once the hypotheses hold. The next omitted magnitude bounds the error. We will use this fact to justify a Taylor approximation.

Use the Right Conclusion

If the magnitudes fail to decrease, the Alternating Series Test is inconclusive. If the terms fail to approach 0, use the Divergence Test. A convergence theorem does not become a divergence theorem merely because one of its hypotheses fails.

Problem 31.4 (Quick Check). Test $\sum_{n=2}^{\infty} (-1)^n \frac{\ln n}{n}$.

Solution. The magnitudes are positive and tend to 0. For $f(x) = \ln x/x$, $f'(x) = (1 - \ln x)/x^2 < 0$ when $x \geq 3$, so they are eventually decreasing. The series converges by the Alternating Series Test. Its absolute-value series diverges since $\ln n/n \geq 1/n$ for $n \geq 3$. □

Lesson 32: Absolute Convergence, Ratio Test, and Root Test

Goals of this lesson.

- Classify series as absolutely convergent, conditionally convergent, or divergent.
- Apply the Ratio and Root Tests with absolute values.
- Recognize when a test is inconclusive.

Definition 32.1 (Absolute and Conditional Convergence)

A series $\sum a_n$ **converges absolutely** if $\sum |a_n|$ converges. It **converges conditionally** if $\sum a_n$ converges but $\sum |a_n|$ diverges.

Theorem 32.1 (Absolute Convergence Test)

If $\sum |a_n|$ converges, then $\sum a_n$ converges.

Proof. The two nonnegative series with terms $(|a_n| + a_n)/2$ and $(|a_n| - a_n)/2$ are each bounded term by term by $|a_n|$, so both converge by comparison. Their difference is $\sum a_n$, which therefore converges. $\square$

Example 32.1 (Absolute Convergence without Alternation). Test $\sum_{n=1}^{\infty} \frac{\cos(2n)}{n^3}$.

Solution. $|\cos(2n)/n^3| \leq 1/n^3$, and the p -series converges. Thus the series converges absolutely. No predictable sign pattern is needed. $\square$

32.1 Ratio Test: Compare Consecutive Terms

Theorem 32.2 (Ratio Test)

Suppose $a_n \neq 0$ eventually and the limit $L = \lim_{n \rightarrow \infty} |a_{n+1}/a_n|$ exists in $[0, \infty]$.

- If $L < 1$, the series converges absolutely.
- If $L > 1$ (including $L = \infty$), the series diverges.
- If $L = 1$, the test is inconclusive.

When $L < 1$, choose $L < q < 1$. Eventually $|a_{n+1}| \leq q|a_n|$, giving a geometric bound on the tail. This explains the test.

Example 32.2 (Factorial in the Denominator). Test $\sum_{n=0}^{\infty} \frac{(-4)^n}{n!}$.

Solution.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{4^{n+1}}{(n+1)!} \frac{n!}{4^n} = \frac{4}{n+1} \rightarrow 0 < 1.$$

The series converges absolutely. $\square$

Example 32.3 (Factorials in Both Places). Test $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2}$.

Solution. $|a_{n+1}/a_n| = (2n+2)(2n+1)/(n+1)^2 \rightarrow 4 > 1$. The series diverges. □

32.2 Root Test: Remove an n th Power

Theorem 32.3 (Root Test)

Suppose $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ exists in $[0, \infty]$. If $L < 1$, the series converges absolutely. If $L > 1$, it diverges. If $L = 1$, the test is inconclusive.

Example 32.4. Test $\sum_{n=1}^{\infty} \left(\frac{2n+1}{5n-2} \right)^n$.

Solution. $\sqrt[n]{|a_n|} = (2n+1)/(5n-2) \rightarrow 2/5 < 1$. Thus the series converges absolutely. □

Example 32.5 (The Limit 1 Gives No Decision). Apply the Ratio Test to $\sum_{n=1}^{\infty} n^{-3/2}$ and

$$\sum_{n=1}^{\infty} n^{-1/2}.$$

Solution. For $a_n = n^{-p}$, the ratio is $(n/(n+1))^p \rightarrow 1$. Yet the first series converges and the second diverges by the p -Series Test. A ratio limit of 1 does not mean divergence. □

Which Test Fits?

Factorials or products suggest the Ratio Test. An expression raised to the n th power suggests the Root Test. Pure powers of n usually call for a p -series or comparison. If the absolute-value series diverges, the signed series still needs its own test.

Lesson 33: Choosing a Convergence Test

Goals of this lesson.

- Choose a test from the structure of a series.
- Give a complete justification, including hypotheses.
- Keep convergence, absolute convergence, and the exact sum separate.

33.1 Start with the Terms

First check for a geometric series, a telescoping sum, or a p -series. Next check whether $a_n \rightarrow 0$. If the terms pass that necessary check, inspect their structure and signs.

The Ten Tests

Test	When it fits	Conclusion
Geometric	ar^n , $a \neq 0$	Converges for $ r < 1$, diverges otherwise.
Divergence	Any series	Diverges if $a_n \not\rightarrow 0$. No conclusion if $a_n \rightarrow 0$.
Integral	$a_n = f(n)$, with f eventually positive, continuous, nonincreasing	Same convergence as $\int f$.
p -series	n^{-p}	Converges exactly when $p > 1$.
Comparison	Nonnegative terms with a usable inequality	Convergent upper bound, or divergent lower bound.
Limit comparison	Eventually positive a_n, b_n with $a_n/b_n \rightarrow c \in (0, \infty)$	Same convergence as $\sum b_n$.
Alternating	Eventually nonincreasing magnitudes $b_n \rightarrow 0$	Alternating series converges.
Absolute convergence	$\sum a_n $ converges	$\sum a_n$ converges absolutely.
Ratio	Eventually nonzero terms, $L = \lim a_{n+1}/a_n $	$L < 1$: absolute convergence. $L > 1$: divergence. $L = 1$: inconclusive.
Root	$L = \lim \sqrt[n]{ a_n }$	Same three conclusions as the Ratio Test.

Write a Complete Argument

Identify the terms, name the test, check its hypotheses, perform the calculation or comparison, and state the conclusion. A test that establishes convergence usually does not supply the sum.

Example 33.1 (Mixed Practice). Classify each series and choose a suitable test:

(a) $\sum_{n=1}^{\infty} \frac{n}{n+5}.$

(b) $\sum_{n=1}^{\infty} \frac{n^2+3}{n^4+2n}.$

(c) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^{3/4}}.$

(d) $\sum_{n=0}^{\infty} \frac{5^n}{(2n)!}.$

(e) $\sum_{n=1}^{\infty} \left(\frac{n+1}{3n+2} \right)^{2n}.$

Solution. (a) The terms tend to 1, so the series diverges by the Divergence Test.

(b) Limit comparison with $1/n^2$ gives ratio limit 1. It converges.

(c) The Alternating Series Test gives convergence. The absolute-value p -series has $p = 3/4$ and diverges, so convergence is conditional.

(d) The consecutive-term ratio is $5/[(2n+2)(2n+1)] \rightarrow 0$. The series converges absolutely.

(e) The n th root is $((n+1)/(3n+2))^2 \rightarrow 1/9 < 1$. The series converges absolutely. □

Problem 33.2 (Why This Argument Fails). A student observes $(1 + \sin^2 n)/n^2 \leq 2/n$ and concludes that $\sum_{n=1}^{\infty} (1 + \sin^2 n)/n^2$ diverges. Correct the argument.

Solution. Being smaller than a divergent series is inconclusive. Instead, $0 < (1 + \sin^2 n)/n^2 \leq 2/n^2$, and the upper-bound p -series converges. Therefore the given series converges. □

Beyond Convergence Tests

We now know how to decide whether many infinite sums exist. The next goal is to use infinite sums as functions, and eventually to recover and approximate familiar functions by their Taylor series.

Lesson 34: Power Series and Their Intervals of Convergence

Goals of this lesson.

- Identify the center and coefficients of a power series.
- Compute the radius of convergence.
- Test each finite endpoint separately.

Definition 34.1 (Power Series)

A **power series** centered at a has the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n.$$

For each fixed x , this is a numerical series. At $x = a$, its value is c_0 , because all positive-degree terms vanish.

Theorem 34.1 (Radius of Convergence)

Every power series has a radius $R \in [0, \infty]$ such that it converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$.

- $R = 0$: only the center gives convergence.
- $0 < R < \infty$: test $a - R$ and $a + R$ separately.
- $R = \infty$: the series converges for every real x .

The set of real x giving convergence is its **interval of convergence**.

Example 34.1 (One Endpoint Included). Find the radius and interval for $\sum_{n=1}^{\infty} \frac{(x+1)^n}{n3^n}$.

Solution. At $x = -1$ the series is zero. For $x \neq -1$, the ratio of absolute consecutive terms is

$$\frac{|x+1|}{3} \frac{n}{n+1} \rightarrow \frac{|x+1|}{3}.$$

Thus $R = 3$, with interior $-4 < x < 2$. At $x = -4$, the series is $\sum (-1)^n/n$, which converges by the Alternating Series Test. At $x = 2$, it is the divergent harmonic series. Hence

$$R = 3, \quad \text{interval } [-4, 2).$$

□

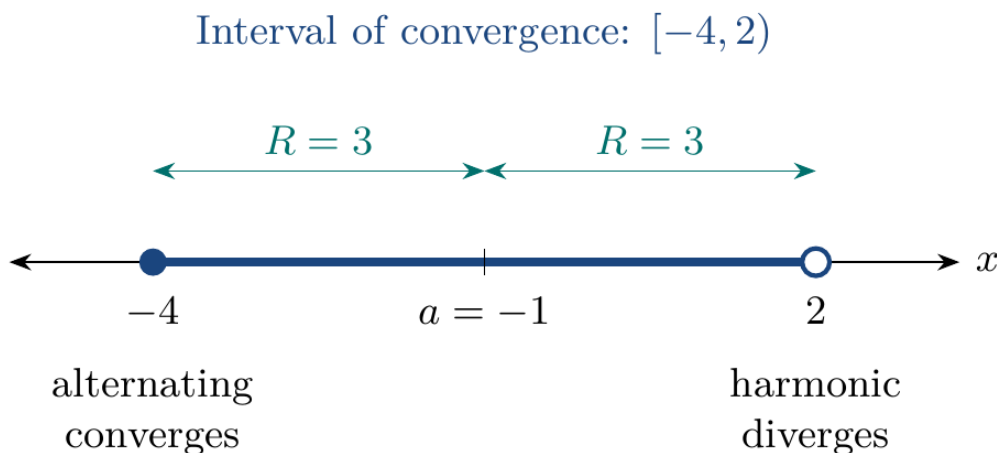


Figure 50: Center -1 and radius 3 determine the open interval $(-4, 2)$. Separate tests include the left endpoint and exclude the right endpoint.

Example 34.2 (Both Endpoints Included). Find the interval for $\sum_{n=1}^{\infty} \frac{(x-4)^n}{n^2 2^n}$.

Solution. The ratio limit is $|x-4|/2$, so $R = 2$. At $x = 2$ and $x = 6$, the absolute-value series is $\sum 1/n^2$, which converges. Therefore the interval is $[2, 6]$. $\square$

Example 34.3 (Infinite Radius). Find the convergence set of $\sum_{n=0}^{\infty} \frac{(2x+1)^n}{(n+1)!}$.

Solution. At the center $x = -1/2$ the sum is 1. Elsewhere the ratio is $|2x+1|/(n+2) \rightarrow 0$. Thus $R = \infty$ and the interval is $\mathbb{R}$. $\square$

Example 34.4 (Zero Radius). Find the convergence set of $\sum_{n=0}^{\infty} n!(x-3)^n$.

Solution. For $x \neq 3$, the ratio $(n+1)|x-3|$ tends to infinity, so the series diverges. At $x = 3$, the sum is 1. Hence $R = 0$ and the convergence set is $\{3\}$. $\square$

The Three Steps

Find the center and the inequality from a Ratio or Root Test. Convert it to $|x-a| < R$. Then substitute each finite endpoint into the original series and use an appropriate numerical-series test. The radius alone does not decide the endpoints.

Lesson 35: Representing Functions by Power Series

Goals of this lesson.

- Obtain new series from the geometric series.
- Differentiate and integrate inside the radius of convergence.
- Track the center, interval, and constant of integration.

35.1 One Identity, Many Functions

The geometric series gives the exact identity

$$\frac{1}{1-u} = \sum_{n=0}^{\infty} u^n, \quad |u| < 1.$$

Substitution and multiplication by a polynomial turn it into new power series.

Example 35.1 (Substitution). Expand $\frac{2x}{1+9x^2}$ about 0 and find its interval of convergence.

Solution. Set $u = -9x^2$:

$$\frac{2x}{1+9x^2} = 2x \sum_{n=0}^{\infty} (-9x^2)^n = \boxed{\sum_{n=0}^{\infty} 2(-9)^n x^{2n+1}}.$$

The condition is $|x| < 1/3$. At $x = \pm 1/3$, the terms fail to tend to 0, so the interval is $\boxed{(-1/3, 1/3)}$. □

Example 35.2 (A Nonzero Center). Expand $1/(x+2)$ about $a = 3$.

Solution. Write $x+2 = 5 + (x-3)$, so

$$\frac{1}{x+2} = \frac{1}{5} \frac{1}{1+(x-3)/5} = \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n (x-3)^n}{5^{n+1}}}.$$

This converges for $|x-3| < 5$. At both endpoints the terms do not tend to 0, so the interval is $\boxed{(-2, 8)}$. □

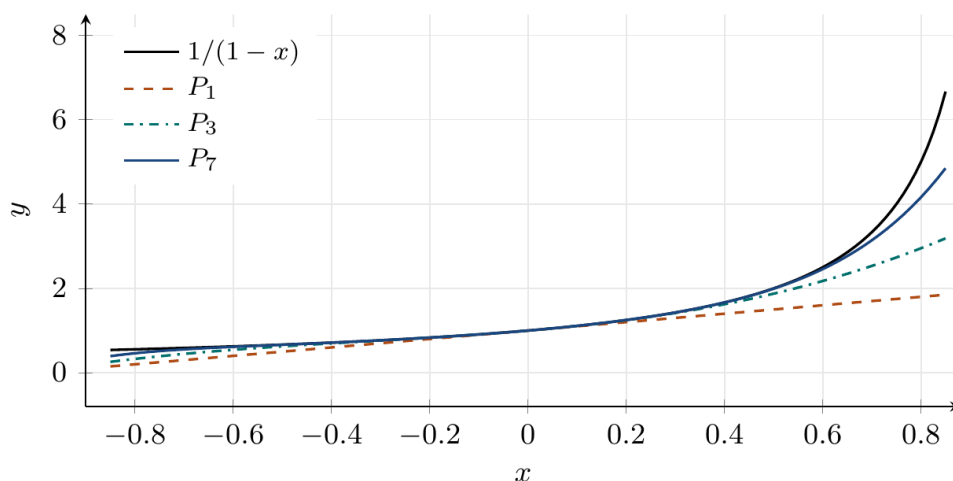


Figure 51: Partial sums of the geometric series approximate $1/(1-x)$ inside $|x| < 1$. They need not approximate the function outside that interval.

35.2 Differentiation and Integration

Theorem 35.1 (Term-by-Term Operations)

If $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n$ has radius $R > 0$, then for $|x-a| < R$,

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1},$$

$$\int_a^x f(t) dt = \sum_{n=0}^{\infty} \frac{c_n}{n+1} (x-a)^{n+1}.$$

Both resulting power series have radius R . Their endpoint behavior may differ from the original series. Integration over any closed interval strictly inside the radius is valid term by term.

Example 35.3 (Differentiation and an Exact Sum). Use the geometric series to find $\sum_{n=1}^{\infty} \frac{n}{4^n}$.

Solution. Differentiating $1/(1-x) = \sum x^n$ gives

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1}, \quad |x| < 1.$$

Multiply by x and substitute $x = 1/4$:

$$\sum_{n=1}^{\infty} \frac{n}{4^n} = \frac{1/4}{(1-1/4)^2} = \boxed{\frac{4}{9}}.$$

□

Example 35.4 (Integration and a Logarithm). Find a power series for $\ln(1 + x/3)$.

Solution. The derivative has the geometric expansion

$$\frac{1}{3+x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^{n+1}}, \quad |x| < 3.$$

Integrating from 0 to x gives

$$\boxed{\ln(1 + x/3) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n 3^n}, \quad |x| < 3.}$$

There is no undetermined constant because both sides vanish at 0. The series also converges at $x = 3$ and diverges at $x = -3$, so its interval of convergence is $(-3, 3]$. Equality at an endpoint needs a separate argument. $\square$

Two Useful Standard Identities

Integrating geometric expansions of $1/(1+x)$ and $1/(1+x^2)$ yields

$$\begin{aligned} \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, & |x| < 1, \\ \arctan x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, & |x| < 1. \end{aligned}$$

These are exact identities inside the indicated intervals.

Why the Endpoint Values Are Also Correct

For $0 \leq t \leq 1$, the finite geometric identity gives

$$\frac{1}{1+t} = \sum_{k=0}^{N-1} (-t)^k + \frac{(-t)^N}{1+t}.$$

After integrating from 0 to 1, the remainder has magnitude at most $1/(N+1)$, so the logarithm series at 1 sums to $\ln 2$. For arctangent, use the finite identity

$$\frac{1}{1+t^2} = \sum_{k=0}^N (-1)^k t^{2k} + \frac{(-1)^{N+1} t^{2N+2}}{1+t^2}.$$

Its integrated remainder on $[0, 1]$ has magnitude at most $1/(2N+3)$, proving the arctangent endpoint value $\pi/4$. Odd symmetry gives the value at -1 for arctangent. These endpoint identities do not follow merely from term-by-term integration inside the radius.

Lesson 36: Taylor and Maclaurin Series

Goals of this lesson.

- Find the only possible coefficients of a power-series representation.
- Compute Taylor series using derivatives or known expansions.
- Distinguish convergence of a Taylor series from equality with its function.

36.1 The Coefficients Are Forced

Suppose $f(x) = \sum c_n(x-a)^n$ in an open interval about a . Differentiating k times and evaluating at a gives

$$f^{(k)}(a) = k!c_k.$$

Thus any power series representing f near a must have Taylor coefficients.

Definition 36.1 (Taylor and Maclaurin Series)

If f has derivatives of every order near a , its **Taylor series** at a is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n.$$

The series at $a = 0$ is its **Maclaurin series**. Its partial sums are the Taylor polynomials from [Lesson 26](#).

A Necessary Distinction

The Taylor series exists formally whenever the derivatives exist. It may diverge away from the center. Even where it converges, its sum need not equal $f(x)$. To claim equality, justify that $f(x) - T_n(x) \rightarrow 0$, or obtain the expansion from an already justified power-series identity.

Example 36.1 (An Exponential). Find the Maclaurin series of e^{2x} and its radius.

Solution. $f^{(n)}(0) = 2^n$, so the Taylor series is

$$\sum_{n=0}^{\infty} \frac{2^n x^n}{n!}.$$

For $x \neq 0$, its ratio limit is $\lim 2|x|/(n+1) = 0$, so $R = \infty$. The remainder estimate in the next lesson proves equality with e^{2x} for every real x . Hence

$$e^{2x} = \sum_{n=0}^{\infty} \frac{(2x)^n}{n!}.$$

□

Example 36.2 (A Center Away from Zero). Find the Taylor series of $\ln x$ centered at 2 and its interval of convergence.

Solution. For $n \geq 1$, $f^{(n)}(x) = (-1)^{n-1}(n-1)!/x^n$. Thus

$$\ln x = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}(x-2)^n}{n2^n}, \quad |x-2| < 2.$$

Equality also follows by writing $\ln x = \ln 2 + \ln(1 + (x-2)/2)$ and using the identity from [Lesson 35](#). At $x = 0$, the series is $\ln 2 - \sum 1/n$ and diverges. At $x = 4$, it converges by alternation and sums to $\ln 4$ by the endpoint argument in that lesson. The interval is $(0, 4]$. $\square$

Example 36.3 (Substitution Is Faster than Many Derivatives). Find the Maclaurin series of $x^2 \sin(3x)$.

Solution. Using the standard sine expansion,

$$x^2 \sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{2n+3}}{(2n+1)!}.$$

The first terms are $3x^3 - \frac{9}{2}x^5 + \frac{81}{40}x^7 - \dots$. The radius is infinite. $\square$

Standard Series

$$\begin{aligned} \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n, & R &= 1, \\ e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & R &= \infty, \\ \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, & R &= \infty, \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, & R &= \infty, \\ \ln(1+x) &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, & R &= 1, \\ \arctan x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}, & R &= 1. \end{aligned}$$

The logarithm interval is $(-1, 1]$, the arctangent interval is $[-1, 1]$, and the geometric interval is $(-1, 1)$. Exponential, sine, and cosine identities hold for every real x .

Problem 36.4 (An Exact Series Value). Evaluate $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!}$.

Solution. This is $\cos(\pi/3) = \boxed{1/2}$. $\square$

Lesson 37: Taylor Series: Remainders, Applications, and Binomial Series

Goals of this lesson.

- Justify a Taylor representation with a remainder estimate.
- Use series to approximate integrals and compute limits.
- Expand fractional powers using the binomial series.

37.1 When Does the Series Equal the Function?

Theorem 37.1 (Taylor's Theorem with Remainder)

If f has continuous derivatives through order $n + 1$ on the interval between a and x , then

$$f(x) = T_n(x) + R_n(x), \quad R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$$

for some ξ between a and x . In particular, if $|f^{(n+1)}(t)| \leq M$ throughout that interval, then

$$|R_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}.$$

The Taylor series represents $f(x)$ exactly when $R_n(x) \rightarrow 0$.

This theorem supplies the extra justification for the Taylor identities we use. The coordinator's main emphasis is computation and application, so the remainder discussion can be treated as an extension.

Example 37.1 (Justifying the Exponential and Trigonometric Series). Why do the Taylor series of e^{2x} , $\sin x$, and $\cos x$ equal their functions for every real x ?

Solution. For a fixed x , derivatives of e^{2t} on the interval from 0 to x have magnitude at most $2^{n+1}e^{2|x|}$. Thus

$$|R_n(x)| \leq e^{2|x|} \frac{(2|x|)^{n+1}}{(n+1)!} \rightarrow 0.$$

For sine and cosine, all derivative magnitudes are at most 1, so $|R_n(x)| \leq |x|^{n+1}/(n+1)! \rightarrow 0$. Factorials eventually dominate any fixed exponential. $\square$

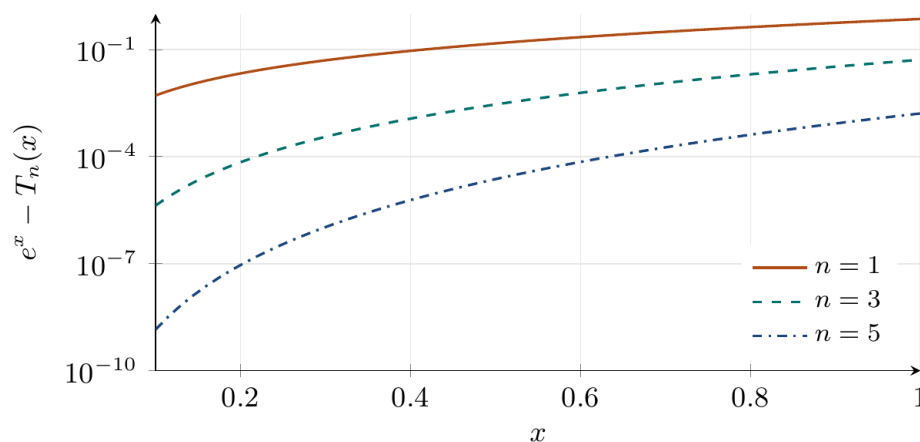


Figure 52: The error $e^x - T_n(x)$ for $0.1 \leq x \leq 1$ and orders 1, 3, and 5. The vertical axis is logarithmic. Remainder bounds make the improvement quantitative.

Example 37.2 (A Numerical Error Bound). Use T_3 for e^x at 0 to approximate $e^{0.3}$ and bound the error.

Solution.

$$T_3(0.3) = 1 + 0.3 + \frac{0.3^2}{2} + \frac{0.3^3}{6} = 1.3495.$$

On $[0, 0.3]$, $e^t \leq 1/(1-t) \leq 10/7$, using the nonnegative exponential series and $1/k! \leq 1$. Hence

$$|e^{0.3} - 1.3495| \leq \frac{10}{7} \frac{0.3^4}{4!} < 0.000483.$$

□

37.2 Integrals and Limits

Example 37.3 (Returning to the Integral from Lesson 26). Approximate $I = \int_0^{1/2} e^{-x^2} dx$ with a certified error.

Solution. Term-by-term integration is valid on $[0, 1/2]$ since the exponential series has infinite radius:

$$I = \sum_{n=0}^{\infty} \frac{(-1)^n (1/2)^{2n+1}}{(2n+1)n!}.$$

The term magnitudes decrease to 0, so the alternating error estimate gives

$$I \approx \frac{1}{2} - \frac{1}{24} + \frac{1}{320} = \frac{443}{960}, \quad \left| I - \frac{443}{960} \right| \leq \frac{1}{5376} < 0.000187.$$

This is the approximation from [Lesson 26](#), now justified.

□

Example 37.4 (A Limit from the First Nonzero Term). Compute $\lim_{x \rightarrow 0} \frac{\sin(2x) - 2x}{x^3}$.

Solution. Taylor's theorem gives $\sin(2x) = 2x - (2x)^3/6 + O(x^5)$ near 0. Thus

$$\frac{\sin(2x) - 2x}{x^3} = -\frac{4}{3} + O(x^2) \rightarrow \boxed{-\frac{4}{3}}.$$

Here $O(x^k)$ means an error bounded in magnitude by $C|x|^k$ near 0, for a fixed constant C . □

37.3 Binomial Series

Theorem 37.2 (Binomial Series)

For a real exponent α and $|x| < 1$,

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n, \quad \binom{\alpha}{0} = 1, \quad \binom{\alpha}{n} = \frac{\alpha(\alpha-1) \cdots (\alpha-n+1)}{n!}.$$

If α is a nonnegative integer, the series terminates and is a polynomial valid for every x , with radius ∞ . Otherwise its radius is 1 and its endpoints need separate tests.

Justification on $|x| < 1$. For a nonterminating series, the Ratio Test gives radius 1. Its coefficients satisfy $(n+1)c_{n+1} = (\alpha-n)c_n$, so its sum B satisfies $(1+x)B' = \alpha B$ and $B(0) = 1$. Differentiating $B(x)(1+x)^{-\alpha}$ shows it is constant on $(-1, 1)$, giving $B(x) = (1+x)^\alpha$. The terminating case is the usual binomial theorem. □

Example 37.5 (A Fractional Power). Find the first four nonzero terms of $(1+2x)^{-1/2}$.

Solution. Substitute $2x$ and $\alpha = -1/2$:

$$\boxed{(1+2x)^{-1/2} = 1 - x + \frac{3}{2}x^2 - \frac{5}{2}x^3 + \cdots}, \quad |x| < \frac{1}{2}.$$

At $x = -1/2$, the terms are positive. If their sum were a finite number C , then $\sum b_n u^n \leq C$ for $0 < u < 1$. But the binomial identity gives $\sum b_n u^n = (1-u)^{-1/2} \rightarrow \infty$ as $u \rightarrow 1^-$, a contradiction. At $x = 1/2$, the magnitudes satisfy $b_{n+1}/b_n = (n+1/2)/(n+1) < 1$ and tend to 0: indeed $b_n = \prod_{k=1}^n (1 - 1/(2k)) \leq \exp(-\frac{1}{2} \sum_{k=1}^n 1/k) \rightarrow 0$. The series converges there by alternation. Its interval of convergence is $\boxed{(-1/2, 1/2]}$. □

Optional: Smooth Does Not Mean Analytic

The function $f(x) = e^{-1/x^2}$ for $x \neq 0$, with $f(0) = 0$, is infinitely differentiable and every derivative at 0 is zero. Its Taylor series at 0 is therefore zero, yet $f(x) > 0$ for $x \neq 0$. Derivatives away from 0 are a polynomial in $1/x$ times e^{-1/x^2} , and this product tends to 0 at 0. This explains why remainder control is needed.

Problem 37.6 (Quick Check). Find the coefficient of x^4 in the Maclaurin series of $e^{x^2} \cos x$.

Solution. Multiply $(1 + x^2 + x^4/2 + \cdots)(1 - x^2/2 + x^4/24 + \cdots)$. The coefficient is $1/2 - 1/2 + 1/24 = 1/24$. $\square$